

BUSINESS

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UNIT 1: LINEAR EQUATIONS AND THEIR INTERPRETATIVE APPLICATIONS IN BUSINESS

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1.0 objectives

After reading the chapter students must be able to:

- define algebraic expression, equation & linear equation
- explain the different ways of formulating or developing equations of a line
- understand the breakeven point and its application
- define the cost output relation ship explain the
- different cost elements

1.1 INTRODUCTION

Mathematics, old and newly created, coupled with innovative applications of the rapidly evolving electronic computer and directed toward management problems, resulted in a new field of study called quantitative methods, which has become part of the curriculum of colleges of business. The importance of quantitative approaches to management problems is now widely accepted and a course in mathematics, with management applications is included in the core of subjects studied by almost all management students. This manual develops mathematics in the applied context required for an understanding of the quantitative approach to management problems.

1.2 Linear Equations

Equation: - A mathematical statement which indicates two algebraic expressions are equal

Example: $Y = 2X + 3$

Algebraic expressions: - A mathematical statement indicating that numerical quantities are linked by mathematical operations.

Example: $X + 2$

Linear equations: - are equations with a variable & a constant with degree one.

- Are equations whose terms (the parts separated by +, -, = signs)
- Are a constant, or a constant times one variable to the first power

Example: $2X - 3Y = 7$

- the degree (the power) of the variables is 1
- the constant or the fixed value is 7
- the terms of the equation are $2X$ and $3Y$ separated by – sign

However $2X + 3XY = 7$ isn't a linear equation, because $3XY$ is a constant times the product of 2 variables.

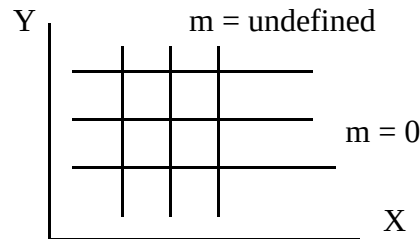
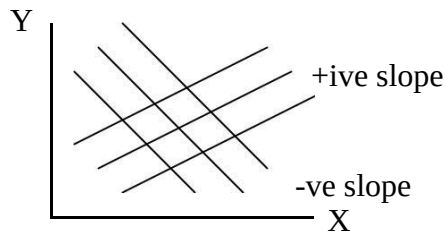
* No X^2 terms, No X/Y terms, and no XY terms are allowed.

- Linear equations are equations whose slope is constant throughout the line.
- The general notion of a linear equation is expressed in a form $Y = mx + b$ where m = slope, b = the Y- intercept, Y = dependent variable and X = independent variable.

If Y represents Total Cost, the cost is increased by the rate of the amount of the slope m .

$$\text{Slope (m)} = \frac{\Delta Y}{\Delta X} = \frac{\text{rise}}{\text{run}} = \frac{Y_2 - Y_1}{X_2 - X_1} \text{ if } X_1 \neq X_2$$

Slope measures the steepness of a line. The larger the slope the more steep (steeper) the line is, both in value and in absolute value.



- A line that is parallel to the X-axis is the gentlest of all lines i.e. $m = 0$
- A line that is parallel to the Y-axis is the steepest of all lines i.e. $m = \text{undefined}$ or infinite.

The slope of a line is defined as the change-taking place along the vertical axis relative to the corresponding change taking place along the horizontal axis, or the change in the value of Y relative to a one-unit change in the value of X.

1.2.1 Developing equation of a line

There are at least three ways of developing the equation of a line. These are:

1. The slope-intercept form
2. The slope-point form
3. Two-point form

1. The slope-intercept form

This way of developing the equation of a line involves the use of the slope & the intercept to formulate the equation.

Often the slope & the Y-intercept for a specific linear function are obtained directly from the description of the situation we wish to model.

Example # 1

Given Slope = 10

Y-intercept = +20, then

Slope-intercept form: the equation of a line with slope = m and Y-intercept b is

$$Y = mx + b$$

$$Y = 10X + 20$$

Interpretative Exercises

#2 Suppose the Fixed cost (setup cost) for producing product X be br. 2000. After setup it costs br. 10 per X produced. If the total cost is represented by Y:

1. Write the equation of this relationship in slope-intercept form.
2. State the slope of the line & interpret the number
3. State the Y-intercept of the line & interpret the number

#3. A sales man has a fixed salary of br. 200 a week In addition; he receives a sales commission that is 20% of his total volume of sales. State the relationship between the sales man's total weekly salary & his sales for the week.

Answer $Y = 0.20X + 200$

2. The slope point form

The equation of a non-vertical line, L, of slope, m, that passes through the point (X_1, Y_1) is : defined by the formula $Y - Y_1 = m (X - X_1)$

$$Y - Y_1 = m (X - X_1)$$

Example #1

$$\Rightarrow Y - 2 = 4 (X - 1)$$

Given, slope = 4 and

$$Y - 2 = 4X - 4$$

Point = (1, 2)

$$Y = 4X - 2$$

#2 A sales man earns a weekly basic salary plus a sales commission of 20% of his total sales. When his total weekly sales total br. 1000, his total salary for the week is 400. derive the formula describing the relationship between total salary and sales.

Answer $Y = 0.2X + 200$

#3 If the relationship between Total Cost and the number of units made is linear, & if costs increases by br. 7.00 for each additional unit made, and if the Total Cost of 10 units is br. 180.00. Find the equation of the relationship between Total Cost (Y) & number of units made (X)

Answer: $Y = 7X + 110$

3. Two-point form

Two points completely determine a straight line & of course, they determine the slope of the line. Hence we can first compute the slope, then use this value of m together with

either point in the point-slope form $Y - Y_1 = m(X - X_1)$ to generate the equation of a line.

⇒ By having two coordinate of a line we can determine the equation of the line.

$$\text{Two point form of linear equation: } (Y - Y_1) = \left(\frac{Y_2 - Y_1}{X_2 - X_1} \right) (X - X_1)$$

Example #1 given (1, 10) & (6, 0)

$$\text{First slope} = \frac{0 - 10}{6 - 1} = \frac{-10}{5} = \underline{\underline{-2}}, \text{ then}$$

$$Y - Y_1 = m(X - X_1) \Rightarrow Y - 10 = -2(X - 1)$$

$$Y - 10 = -2X + 2$$

$$\underline{\underline{Y = -2X + 12}}$$

#2 A salesman has a basic salary &, in addition, receives a commission which is a fixed percentage of his sales volume. When his weekly sales are Br. 1000, his total salary is br. 400. When his weekly sales are 500.00, his total salary is br. 300. Determine his basic salary & his commission percentage & express the relationship between sales & salary in equation form.

Answer: $Y = 0.2X + 200$

#3 A printer costs a price of birr 1,400 for printing 100 copies of a report & br. 3000 for printing 500 copies. Assuming a linear relationship what would be the price for printing 300 copies?

Answer: $Y = 4X + 1000$

$$\text{Cost} = 4.0(300) + 1000 = \text{br. } \underline{\underline{2200}}$$

1.2.2 Special formats

a) Horizontal & vertical lines

When the equation of a line is to be determined from two given points, it is a good idea to compare corresponding coordinates because if the Y values are the same the line is horizontal & if the X values are the same the line is vertical

Example: 1 Given the points (3, 6) & (8, 6) the line through them is horizontal because both Y-coordinates are the same i.e. 6

The equation of the line becomes $Y = 6$, which is different from the form $Y = mx + b$

If the X-coordinates of the two different points are equal

Example (5, 2) & (5, 12) the line through them is vertical, & its equation is $X = 5$ i.e. X is equal to a constant. If we proceed to apply the point slope procedure, we would obtain.

$$\text{Slope (M)} = \frac{12 - 2}{5 - 5} = \frac{10}{0} = \infty \text{ \& if } m = \infty \text{ infinite the line is vertical \& the form of the}$$

equation is: $X = \text{constant}$

b) Parallel & perpendicular lines (//and $\perp$)

Two lines are parallel if the two lines have the same slope, & two lines are perpendicular if the product of their slope is -1 or the slope of one is the negative reciprocal of the slope of the other. However, for vertical & horizontal lines. (They are perpendicular to each other), this rule of M_1 (the first slope) times M_2 (the second slope) equals -1 doesn't hold true. i.e. $M_1 \times M_2 \neq -1$

Example: $Y = 2X - 10$ & $Y = 2X + 14$ are parallel because their slope are equal i.e. 2

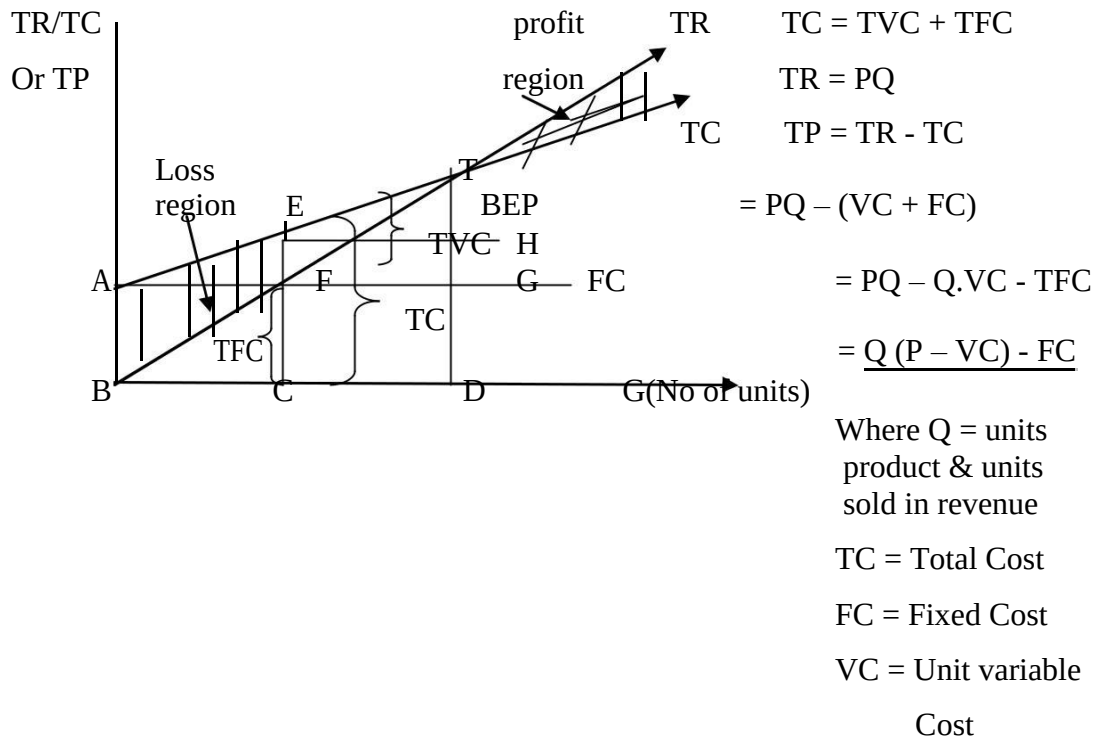
$Y = \frac{3}{2}X + 10$ & $Y = -\frac{2}{3}X + 100$ are perpendicular to each other because the multiplication result of the two slope are -1 i.e. $\frac{3}{2} \times \frac{-2}{3} = -1$

c) Lines through the origin

Any equation in the variables X & Y that has no constant term other than zero will have a graph that passes through the origin. Or, a line which passes through the origin has an X-intercept of (0, 0) i.e. both X and Y intercepts are zero.

1.3 Application of linear equations

1.3.1 linear cost output relationships – VC, FC, TC, AC, MC, TR, TP



Interpretation of the graph:

1. The vertical distance between AB, FC, GD is the same because Fixed Cost is the same at any levels of output.
2. There is no revenue without sales (because Total Revenue function passes through the origin), but there is cost without production (because of Fixed Cost) & the TC function starts from A & doesn't pass through the origin
3. Up to point T, Total Cost is greater than Total Revenue $\Rightarrow$ results in loss. While at point T, (Total Revenue = Total Cost) i.e. Breakeven. (0 profit), & above point T, $TR > TC \Rightarrow +ve$ profit.
4. TFC remains constant regardless of the number of units produced. Given that there is no any difference in scale of production.

5. As production increases, Total Variable Cost increases at the same rate and Marginal cost is equal with Unit Variable Cost (MC = VC) only in linear equations.
6. As production increases TC increases by the rate equal to the $AVC = MC$ (average cost equal to marginal cost)
7. AVC is the same through out any level of production, however Average Fixed Cost (AFC) decreases when Quantity increases & ultimately ATC decreases when Q increases because of the effect of the decrease in AFC.
8. As Quantity increases TR increases at a rate of P. and average revenue remains constant.

$$AR = \frac{TR}{Q} = \frac{P \cdot Q}{Q} = \cancel{P \cdot Q} = P \text{ in linear functions}$$

1.2.2 Breakeven Analysis

Break-even point is the point at which there is no loss or profit to the company. It can be expressed as either in terms of production quantity or revenue level depending on how the company states its cost equation.

Manufacturing companies usually state their cost equation in terms of quantity (because they produce and sell) where as retail business state their cost equation in terms of revenue (because they purchase and sell)

Case 1: Manufacturing Companies

Consider a Company with equation

$$TC = VC + FC / \text{Total cost} = \text{Variable cost} + \text{Fixed cost}$$

$$TR = PQ / \text{Total Revenue} = \text{Price} \times \text{Quantity}$$

$$\text{At Break-even point, } TR = TC \quad \text{i.e } TR - TC = 0$$

$$PQ = VC + FC$$

where Q_e = Breakeven Quantity

$$PQ - VC \cdot Q = FC$$

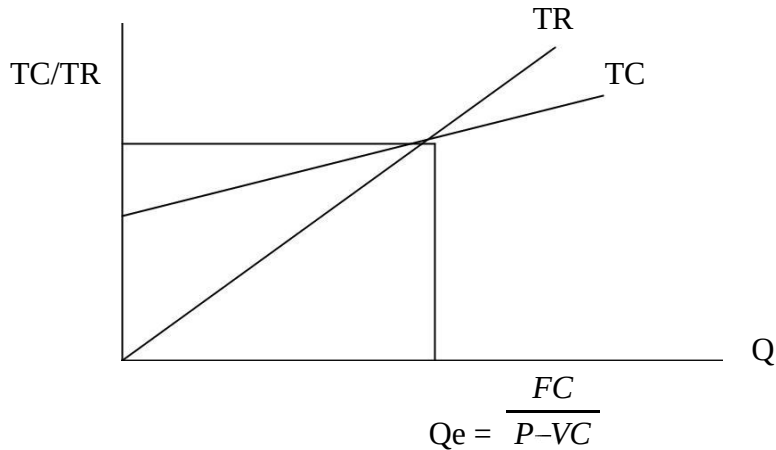
FC = Fixed cost

$$Q(P - VC) = FC$$

P = unit selling price

$$Q_e = \frac{FC}{P-VC}$$

VC = unit variable cost



Example #1 A manufacturing Co. has a Total Fixed Cost of Br. 10,000 & a Unit Variable Cost of Br. 5. if the co. can sell .What it produces at a price of Br. 10,

- Write the Revenue, cost & Profit functions
- Find the breakeven point in terms of quantity and sales volume
- Show diagrammatically the Total Revenue, Total Cost, Total Profit, Fixed Cost and Variable Costs.
- Interpret the results

Answer

a) TC = VC + FC	TR = PQ	Profit(Π) = TR – TC
TC = 5Q + 10,000	TR = 10Q	= 10Q – (5Q + 10,000)
		= 5Q – 10,000

- b) At break even point TR = TC

$$10Q = 5Q + 10,000$$

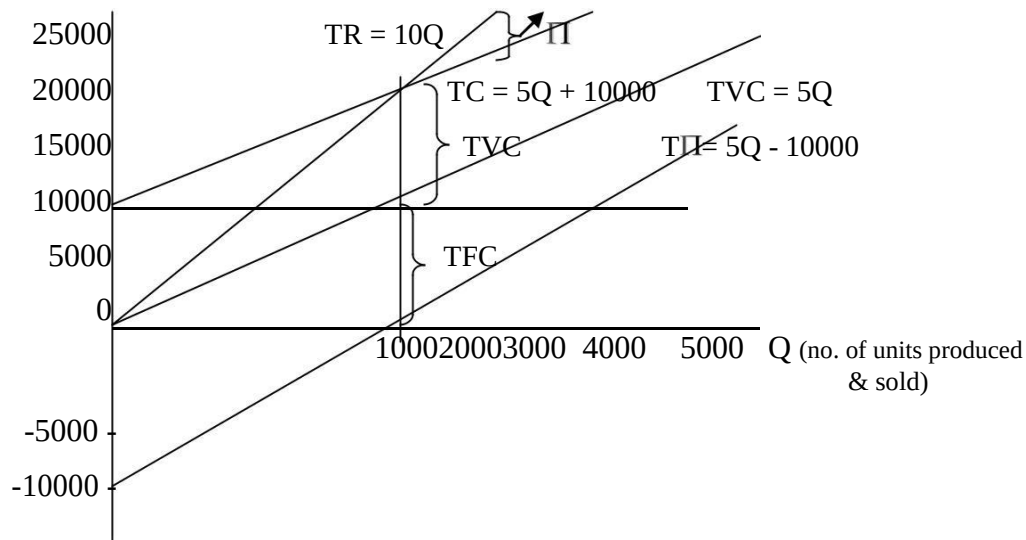
$$5Q - 10,000 = 0$$

$$Q_e = \frac{10,000}{5}$$

$$Q_e = 2000 \text{ units}$$

i.e. Breakeven Quantity is 2000 units

Sales volume = $2000 \times 10 = \underline{20,000 \text{ br.}}$



Interpretation:

When a co. produces & sells 2000 units of output, there will not be any loss or gain (no profit, no loss)

⇒ **The effect of changing one variable keeping other constant**

Case 1 - Fixed cost

Assume for the above problem FC is decreased by Br. 5000, Citrus Paribus (other things being constant)

$$TC = 5Q + 5000 \quad \Rightarrow Q_e = \frac{5000}{5} = 1000 \text{ units}$$

$$TR = 10Q$$

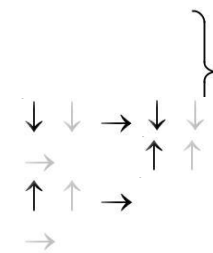
Therefore, $\begin{matrix} FC \downarrow \rightarrow Q_e \downarrow \\ FC \uparrow \rightarrow Q_e \uparrow \end{matrix} \left. \vphantom{\begin{matrix} FC \downarrow \rightarrow Q_e \downarrow \\ FC \uparrow \rightarrow Q_e \uparrow \end{matrix}} \right\} \text{FC \& } Q_e \text{ have direct relationship}$

Case 2-Unit variable cost

Assume for the above problem UVC decreased by 1 br. Citrus Paribus (keeping other thing constant)

$$TC = 4Q + 10000 \quad \Rightarrow Q_e = \frac{10000}{6} = \underline{\underline{1,667 \text{ units}}}$$

$$TR = 10Q$$

Therefore,  VC & Qe have direct relationship

Case 3- Selling price

Assume for the above problem selling price is decreased by br. 1, Citrus Paribus,

$$TC = 5Q + 10,000 \quad \Rightarrow Q_e = \frac{10000}{4} = \underline{\underline{2500units}}$$

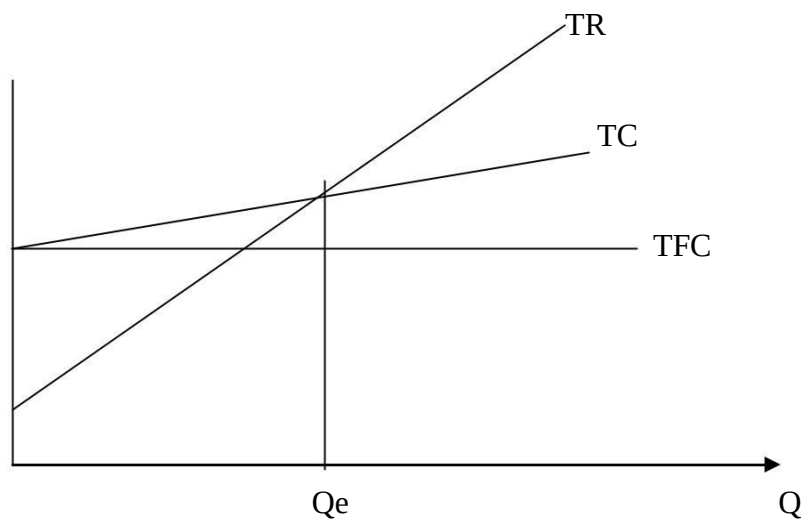
$$TR = 9Q$$

Therefore $P \downarrow \rightarrow Q_e \uparrow$ } Price and breakeven point have indirect relationship
 $P \uparrow \rightarrow Q_e \downarrow$ }

In the above example a company has the following options (to minimize its breakeven point and maximize profit).

- decreasing FC
- decreasing unit VC
- increasing the unit selling price

And if the organization is between option 2 & 3, it is preferable to decrease the unit variable cost because if we increase the selling price, the organization May loose its customers & also decreasing the FC is preferable.



Finding the quantity level which involves profit or loss

$$BEP = \frac{FC + 0}{P - V}, \text{ any } Q \text{ is related to the cost, profit}----$$

$$\Pi = TR - TC$$

Where: BEP = breakeven point

$$\Pi = PQ - (VC.Q + FC)$$

Π = Profit

$$\Pi = (P.Q - VC.Q) - FC$$

TR = Total revenue

$$\Pi = Q(P - VC) - FC$$

TC = Total cost

$$\frac{\Pi + FC}{P - VC} = \text{for any qty level } Q = \frac{FC + \Pi}{P - VC}$$

Q = Quantity

VC = Unit variable cost

Example #1 For the above manufacturing co. if it wants to make a profit of 25000 br.
What should be the quantity level?

$$TR = 10Q \quad Q = \frac{FC + \Pi}{P - V} \text{ when there is } \Pi \text{ the quantity produced \&}$$

sold have to be greater than the

Breakeven quantity

$$TC = 5Q + 10,000 \quad = \frac{10,000 + 25,000}{10 - 5}$$

$$\Pi = 25,000 \quad = \underline{7000 \text{ units}}$$

$$Q = ?$$

If it expects a loss of br. 5000 what will be the quantity level.

$$Q = \frac{FC \pm \overline{VC}}{P - VC} = \frac{10,000 - 5000}{10 - 5} \quad * \text{ when there is loss, the qty produced \& sold}$$

should be less than the BEQ

Case 2 Merchandising /Retail Business

$$\text{Breakeven Revenue} = BEQ \times P$$

Assume a bus. Firm with product A has the following cost & revenue items.

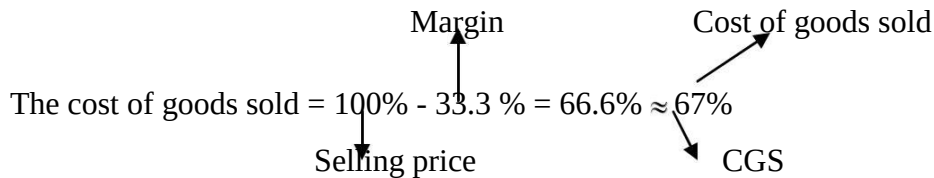
Variable cost of A = 100 br.

Selling price = 150 br.

Markup = Selling price – Variable cost = 150 – 100 = 50

i. as a function of cost, the markup is $50/100 = 50\%$

ii. as a function of retail price, the markup is $50/150 = 33.3\%$ it is also called margin.



Given other selling expense = 1% of the selling price i.e. $0.01X$

So, the TC equation becomes:

$$Y = 0.68X + FC$$

Where: X is sales revenue

Y is total cost

Out of 100% selling price 68% is the variable cost of goods purchased & sold

Example Suppose a retail business sale its commodities at a margin of 25% on all items purchased & sold. Moreover the company uses 5% commission as selling expense & br. 12000 as a Fixed Cost.

Find the Breakeven revenue for the retail business after developing the equation

<u>Solution</u>	Selling price	100%
	Margin	<u>25%</u>
	CGS	75%
	Comm. Exp.	<u>5%</u>
	Total VC	80%

Let X represents selling price

Y = total cost

FC = 12000

Xe = Breakeven revenue

$$\therefore Y = 0.8X + 12000$$

Break even revenue is obtained by making sales revenue & cost equals

At breakeven point TC = TR

i.e. Y = X then,

$$Y = mx + b$$

unit variable cost

$$0.8X + 12000 = X \quad \Rightarrow \frac{FC}{1-m} \text{ or } \frac{FC \pm \Pi}{1-m} \text{ where } m = \frac{VC}{P} = \frac{TVC}{TR}$$

$$-0.2X = -12000$$

$$\underline{X = 60,000 \text{ br.}} \quad \Rightarrow \text{When the co. receives br. 60,000 as sales revenue, there will be no loss or profit.}$$

The Breakeven revenue (BER = $\frac{FC \pm \Pi}{1-m}$) method is useful, because we can use a single

formula for different goods so far as the company uses the same amount of profit margin

for all goods. However, in Breakeven quantity method or BEQ = $\frac{FC \pm \Pi}{P-V}$ it is not

possible and hence we have to use different formula for different items.

Example #1 It is estimated that sales in the coming period will be br. 6000 & that FC will be br. 1000 & variable costs br. 3600, develop the total cost equation & the breakeven revenue.

Answer: $Y = \frac{3600}{6000} X + 1000 = 0.6X + 1000$

Where Y = Total Cost

X = Total revenue

$$BER = X_e = \frac{1000}{1-0.6} = \frac{1000}{0.4} = \underline{\underline{2500br.}}$$

At the sales volume of br. 2500, the company breaks even.

* When the breakeven revenue equation is for more than one item it is impossible to find the breakeven quantity. It is only possible for one item by $Q_e = X_e/P$

Where X_e = Break even revenue

P = selling price

Q_e = Breakeven quantity

To change the breakeven revenue equation in to Breakeven quantity . We have to multiple price by the coefficient of X. likewise, to change in to breakeven revenue from Break even quantity, we have to divide the unit VC by price.

1.4 Model examination questions

1. XYZ company's cost function for the next four months is

$$C = 500,000 + 5Q$$

- a) Find the BE dollar volume of sales if the selling price is br. 6 / unit
 - b) What would be the company's cost if it decides to shutdown operations for the next four months
 - c) If, because of strike, the most the company can produce is br. 100,000 units, should it shutdown? Why or why not?
2. In its first year, "Abol Buna Co" had the following experience
- | | |
|----------------------|-------------------------|
| Sales = 25,000 units | Selling price = br. 100 |
| TVC = br. 1,500,000 | TFC = br. 350,000 |

Required:

1. Develop Revenue, cost & profit functions for the co. in terms of quantity.
2. Find the Breakeven point in terms of quantity
3. Convert the cost equation in terms of quantity in to a cost equation in terms of revenue
4. Find the Breakeven revenue
5. If profit had been br. 500,000 what would have been the sales volume (revenue) & the quantity of sales
6. What would have been the profit if sales are br. 2,000,000.
3. A small home business set up with an investment of \$ 10,000 for equipment. The business manufactures a product at a cost of br. 0.64 per unit. If the product sales for Br. 1.20 per unit how many units must be sold before the business breaks even?
4. A retail co. plans to work on a margin of 44% of retail price & to incur other Variable Cost of 4%. If is expected Fixed cost of Br. 20,000.
 - i. Find the equation relating Total Cost to sales
 - ii. Find the profit if sales are Br. 60,000
 - iii. Find the breakeven revenue
 - iv. If profit is Br. 15,000 what should be the revenue level?
 - v. If you have any one item at a price of Br. 15/unit how do you convert the cost equation in terms of revenue in to a cost equation in terms of quantity?

UNIT 2 MATRIX ALGEBRA AND ITS APPLICATION

Contents

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- 2.2 Matrix algebra
 - 2.2.1 Types of matrices
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- 2.3 Matrix Application
 - 2.3.1 Solving systems of linear equations
 - 2.3.2 Word problems
 - 2.3.3 Markov Chains

2.0 AIMS AND OBJECTIVES

After reading this chapter students will be able to:

- explain what a matrix is
- define the different types of matrices
- perform matrix operations
- find inverse of a square matrix
- explain how to solve a systems of linear equations
- solve word problems applying matrices
- understand the concept of Markov chain

2.1 INTRODUCTION

Brevity in mathematical statements is achieved through the use of symbols. The price paid for brevity, of course, is the effort spent in learning the meaning of the symbol.

In this unit we shall learn the symbols for matrices, and apply them in the statement and solution of input-output problems and other problem involving linear systems

2.2 MATRIX ALGEBRA

Algebra is a part of mathematics, which deals with operations (+, -, x, ÷).

A matrix is a rectangular array of real numbers arranged in m rows & n columns. It is symbolized by a bold face capital letter enclosed by a bracket or parentheses.

$$\text{eg. } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \quad \text{in which } a_{ij} \text{ are real numbers}$$

Each number appearing in the array is said to be an element or component of the matrix. Element of a matrix are designated using a lower case form of the same letter used to symbolize the matrix itself. These letters are subscripted as a_{ij} , to give the row & column location of the element within the array. The first subscript always refers to the row location of the element; the second subscript always refers to its column location. Thus, component a_{ij} is the component located at the intersection of the i^{th} row and j^{th} column.

The number of rows (m) & the number of columns (n) of the array give its order or its dimension, M x n (reads “M” by “n”)

Eg. The following are examples of matrices

$$A = \begin{bmatrix} 1 & 7 \\ 5 & 3 \\ 4 & 2 \end{bmatrix} \quad \begin{array}{l} \text{this is 3 x 2 matrix} \\ \text{element } a_{12} = 7 \\ a_{21} = 5 \\ a_{32} = 2 \end{array}$$

$$X = \begin{bmatrix} 1 & 5 & 9 & 15 \\ 2 & 6 & 10 & 20 \\ 3 & 7 & 11 & 30 \\ 4 & 8 & 12 & 45 \end{bmatrix} \quad \begin{array}{l} \text{This is a 4 x 4 matrix} \\ \text{Element } X_{44} = 45 \\ X_{34} = 30 \\ X_{42} = 8 \\ X_{32} = 7 \end{array}$$

2.2.1 Types of Matrices

There are different types of matrices. These are

1. Vector matrix – is a matrix, which consists of just one row or just one column. It is an $m \times 1$ or $1 \times n$ matrix.

1.1 Row vector is a $1 \times n$ matrix i.e. a matrix with 1 row

eg. $W = \begin{bmatrix} -10 & 6 & 1 & 3 \end{bmatrix}$

1.2. Column vector: is an $m \times 1$ matrix i.e. a matrix with one column only

eg. $Z = \begin{bmatrix} 0 \\ 20 \\ 5 \end{bmatrix} 3 \times 1$

2. Square matrix: - a matrix that has the same number of rows & columns. It is also called n -th order matrix

eg. 2×2 , 3×3 , $n \times n$ $X = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} 2 \times 2$

3. Null or zero matrix: - is a matrix that has zero for every entry.

It's generally denoted by $O_{m \times n}$ eg. $Y = \begin{bmatrix} 0 & 0 & 0 \\ 0 \end{bmatrix}$

4. Identity (unit) matrix: - a square matrix in which all of the primary diagonal entries are ones & all of the off diagonal entries are zeros. Its denoted by I .

eg. $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} 2 \times 2$ $I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} 4 \times 4$

N.B. Each identity matrix is a square matrix

* Primary diagonal represents: $a_{11}, a_{22}, a_{33}, a_{44}, \dots, a_{nn}$ entries element

$A \times I = A$ & $I \times A = A$ that is, the product of any given matrix & the identity matrix is the given matrix itself. Thus, the identity matrix behaves in a matrix multiplication like number 1 in an ordinary arithmetic.

5. Scalar matrix: - is a square matrix where elements on the primary diagonal are the same.

“ An identity matrix is a scalar matrix but a scalar matrix may not be an identity matrix”.

2.2.2 Matrix operations (Addition, Subtraction, Multiplication)

Matrix Addition/ Subtraction

Two matrices of the same dimension are said to be CONFORMABLE FOR ADDITION. Adding corresponding elements from the two matrices & entering the result in the same row-column position of a new matrix perform the addition.

If A & B are two matrices, each of size $m \times n$, then the sum of A & B is the $m \times n$ matrix C whose elements are:

$$C_{ij} = a_{ij} + b_{ij} \quad \text{for } i = 1, 2, \dots, m \quad \begin{matrix} C_{11} = a_{11} + b_{11} \\ C_{22} = a_{22} + b_{22} \\ C_{12} = a_{12} + b_{12} \text{ etc} \end{matrix}$$

$$j = 1, 2, \dots, n$$

$$\text{eg. } \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 7 & 9 \\ 8 & -10 \end{bmatrix} = \begin{bmatrix} 8 & 12 \\ 10 & -6 \end{bmatrix}$$

$$\text{eg. } \begin{bmatrix} -2 & 7 & 2 \\ 4 & 6 & 9 \end{bmatrix} + \begin{bmatrix} 8 & 7 \\ 6 & 4 \end{bmatrix} = \text{These two matrices aren't conformable for addition because they aren't of the same dimension.}$$

Laws of matrix addition

The operation of adding two matrices that are conformable for addition has these two basic properties.

1. $A + B = B + A$ the commutative law of matrix addition
2. $(A + B) + C = A + (B + C)$ the associative law of matrix addition

→ The laws of matrix addition are applicable to laws of matrix subtraction, given that the two matrices are conformable for subtraction $A - B = A + (-B)$

$$\text{eg. } A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 \\ 2 & 5 \end{bmatrix}$$

$$A - B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Matrix Multiplication

a) By a constant (scalar multiplication)

A matrix can be multiplied by a constant by multiplying each component in the matrix by a constant. The result is a new matrix of the same dimension as the original matrix.

If K is any real number & A is an $M \times n$ matrix, then the product KA is defined to be the matrix whose components are given by K times the corresponding component of A ; i.e.

$$KA = \begin{bmatrix} K_{aij} \end{bmatrix} \quad (m \times n)$$

$$\text{eg. If } X = \begin{bmatrix} 6 & 5 & 7 \end{bmatrix} \text{ then } 2X = \begin{bmatrix} (2 \times 6) & (2 \times 5) & (2 \times 7) \end{bmatrix}$$

$$2X = \begin{bmatrix} 12 & 10 & 14 \end{bmatrix}$$

Laws of scalar multiplication

The operation of multiplying a matrix by a constant (a scalar) has the following basic properties. If X & Y are real numbers & A & B are $m \times n$ matrices, conformable for addition, then

1. $XA = AX$
2. $(X + Y)A = XA + YA$
3. $X(A + B) = XA + XB$
4. $X(YA) = XY(A)$

Laws of scalar multiplication

eg. Given matrices A & B and two real numbers X & Y

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 2 & 1 \\ 3 & 0 & 5 \end{bmatrix}$$

$$X = 2$$

$$Y = 4$$

1) $XA = AX$

Proof: $XA = 2 \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad AX = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} 2$

$$XA = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix} \quad AX = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$$

Therefore $XA = AX$

2) $(X + Y) A = (XA + YA)$

Proof:

→ $(X + Y) A$ means first add X with Y and then multiply the result by matrix A.

The result of $X + y$ is $(2 + 4) = 6$ then 6 will be multiplied by matrix A

$$\begin{bmatrix} 6 & 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \text{ becomes } \begin{bmatrix} 6 & 12 & 18 \\ 24 & 30 & 36 \end{bmatrix}$$

$$\text{Therefore } (X + Y) A = \begin{bmatrix} 6 & 12 & 18 \\ 24 & 30 & 36 \end{bmatrix}$$

→ $XA + YA$ means multiply the constant numbers X and Y with matrix A independently, then add the two results together

$$XA = 2 \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad YA = 4 \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$XA = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix} \quad YA = \begin{bmatrix} 4 & 8 & 12 \\ 16 & 20 & 24 \end{bmatrix}$$

Then add the result XA with YA

$$\begin{aligned} XA + YA &= \begin{bmatrix} 24 & 6 \\ 8 & 10 & 12 \end{bmatrix} + \begin{bmatrix} +4 & 8 & 12 \\ 16 & 20 & 24 \end{bmatrix} \\ &= \begin{bmatrix} 6 & 12 & 18 \\ 24 & 30 & 36 \end{bmatrix} \end{aligned}$$

Therefore it is true that $(X + Y)A$ is equal with $XA + YA$

$$3) X(A + B) = XA + XB$$

Proof:

→ $X(A + B)$ means add matrix A and B first and then multiply the result by a constant X

Given constant number $X = 2$ matrices A and B then the result of $X(A + B)$ will be

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 2 & 1 \\ 3 & 0 & 5 \end{bmatrix} \quad A + B = \begin{bmatrix} 5 & 4 & 4 \\ 7 & 5 & 11 \end{bmatrix}$$

$$\begin{aligned} X(A + B) &= 2 \begin{bmatrix} 5 & 4 & 4 \\ 7 & 5 & 11 \end{bmatrix} \\ &= \begin{bmatrix} 10 & 8 & 8 \\ 14 & 10 & 22 \end{bmatrix} \end{aligned}$$

→ $XA + XB$ means multiply matrices A and B by a constant number X independently then add the results

$$\begin{aligned} XA &= \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix} & XB &= \begin{bmatrix} 8 & 4 & 2 \\ 6 & 0 & 10 \end{bmatrix} \\ XA &= \begin{bmatrix} 21 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} & XB &= \begin{bmatrix} 2 & 42 & 1 \\ 3 & 0 & 5 \end{bmatrix} \end{aligned}$$

Then Add XA with XB

$$\begin{aligned} \text{i.e. } XA + XB &= \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix} + \begin{bmatrix} 8 & 4 & 2 \\ 6 & 0 & 10 \end{bmatrix} \\ &= \begin{bmatrix} 10 & 8 & 8 \\ 14 & 10 & 22 \end{bmatrix} \end{aligned}$$

Therefore it is true that $X(A + B)$ is equivalent with $XA + XB$

4) $X(YA) = XY(A)$

Proof:

→ $X(YA)$ means multiply the second constant number Y with matrix A first and then multiply the first constant number X with the result.

$$\begin{aligned} YA &= 4 \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} & X(YA) &= \begin{bmatrix} 24 & 8 & 12 \\ 16 & 20 & 24 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 8 & 12 \\ 16 & 20 & 24 \end{bmatrix} & &= \begin{bmatrix} 8 & 16 & 24 \\ 32 & 40 & 48 \end{bmatrix} \end{aligned}$$

→ $XY(A)$ means multiply the two constant real numbers X and Y first and multiply the result by matrix A.

$$\begin{aligned} XY &= 2 \times 4 \\ &= \underline{8} \end{aligned} \quad XY(A) = \begin{bmatrix} 8 & 16 & 24 \\ 32 & 40 & 48 \end{bmatrix}$$

$$\begin{bmatrix} 8 & 16 & 24 \\ 32 & 40 & 48 \end{bmatrix}$$

Therefore it is also true that $X(YA) = XY(A)$ of columns in B

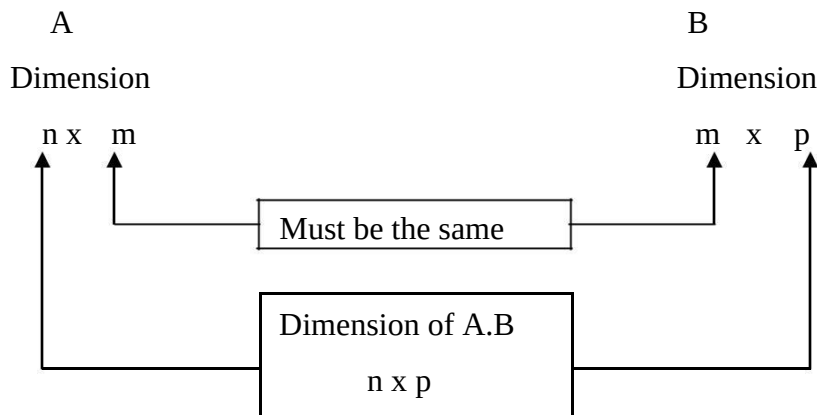
b) Matrix by matrix multiplication

If A & B are two matrices, the product AB is defined if and only if the number of Columns in A is equal to the number of rows in B, i.e. if A is an $m \times n$ matrix, B should be an $n \times b$.

If this requirement is met., A is said to be conformable to B for multiplication. The matrix resulting from the multiplication has dimension equivalent to the number of rows in A & the number columns in B

If A is a matrix of dimension $n \times m$ (which has m columns) & B is a matrix of dimension $p \times q$ (which has p rows) and if m and p aren't the same product A.B is not defined. That is, multiplication of matrices is possible only if the number of columns of the first equals the number of rows of the second.

If A is of dimension $n \times m$ & if B is of dimension $m \times p$, then the product A.B is of dimension $n \times p$



eg. $A = \begin{bmatrix} 2 & 3 & 4 \\ 6 & 9 & 7 \end{bmatrix} 2 \times 3$ $B = \begin{bmatrix} -1 & 7 \\ 0 & 8 \\ 5 & 1 \end{bmatrix} 3 \times 2$

$$AB = (2 \times -1) + (3 \times 0) + (4 \times 5) \\ = 18$$

$$(6 \times -1) + (9 \times 0) + (7 \times 5) \\ = 29$$

$$(2 \times 7) + (3 \times 8) + (4 \times 1) \\ = 42$$

$$(6 \times 7) + (9 \times 8) + (7 \times 1) \\ = 121$$

$$\therefore AB = \begin{bmatrix} 18 & 42 \\ 29 & 121 \end{bmatrix}$$

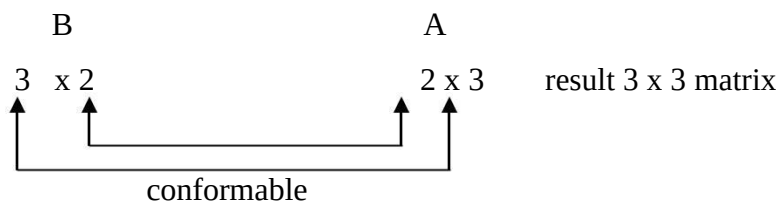
Find BA =

$$B = \begin{bmatrix} -1 & 7 \\ 0 & 8 \\ 5 & 1 \end{bmatrix}$$

3 x 2

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 6 & 9 & 7 \end{bmatrix}$$

2 x 3



$$BA = (-1 \times 2) + (7 \times 6) \quad (-1 \times 3) + (7 \times 9) \quad (-1 \times 4) + (7 \times 7)$$

$$= 40 \quad 60 \quad 45$$

$$(0 \times 2) + (8 \times 6) \quad (0 \times 3) + (8 \times 9) \quad (0 \times 4) + (8 \times 7)$$

$$= 48 \quad 72 \quad 56$$

$$(5 \times 2) + (1 \times 6) \quad (5 \times 3) + (1 \times 9) \quad (5 \times 4) + (1 \times 7)$$

$$16 \quad 24 \quad 27$$

$$BA = \begin{bmatrix} 40 & 60 & 45 \\ 48 & 72 & 56 \\ 16 & 24 & 27 \end{bmatrix}$$

Special properties of matrix application

1 The associative & distributive laws of ordinary algebra apply to matrix multiplication.

Given three matrices A, B & C which are conformable for multiplication,

- i. $A(BC) = AB(C)$ ~~As~~ Associative law, (not $C(AB)$)
- ii. $A(B+C) = AB + AC \rightarrow$ Distributive property

iii. $(A + B)C = AC + BC \rightarrow$ Distributive property

2 on the other hand, the commutative law of multiplication doesn't apply to matrix multiplication. For any two real numbers X & Y , the product XY is always identical to the product YX . But for two matrices A & B , it is not generally true that AB equals BA . (in the product AB , we say that B is pre multiplied by A & that A is post multiplied by B .)

3 In many instances for two matrices, A & B , the product AB may be defined while the product BA is not defined or vice versa.

In some special cases, AB does equal BA . In such special cases A & B are said to be Commute. $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}_{2 \times 2}$ $B = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}_{2 \times 2}$ $AB = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$
 $BA = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$

4 Another unusual property of matrix multiplication is that the product of two matrices can be zero even though neither of the two matrices themselves is zero: we can't conclude from the result $AB = 0$ that at least one of the matrices A or B is a zero matrix

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 & 0 \\ 7 & -10 & 4 \\ 8 & 3 & 2 \end{bmatrix} \quad AB = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

5 Also we can't, in matrix algebra, necessarily conclude from the result $AB = AC$ that $B = C$ even if $A \neq 0$. Thus the cancellation law doesn't hold, in general, in matrix multiplication

$$\text{eg. } A = \begin{bmatrix} 1 & 3 \\ -2 & -6 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$AB = AC = \begin{bmatrix} 10 & 14 \\ -20 & -28 \end{bmatrix} \quad \text{but } B \neq C$$

2.2.3 The multiplicative inverse of a matrix

If A is a square matrix of order n , then a square matrix of its inverse (A^{-1}) of the same order n is said to be the inverse of A , if and only if $A \times A^{-1} = I = A^{-1} \times A$

Two square matrices are inverse of each other, if their product is the identity matrix.

$$AA^{-1} = A^{-1}A = I$$

Not all matrices have an inverse. In order for a matrix to have an inverse, the matrix must, first of all, be a square matrix.

Still not all square matrices have inverse. If a matrix has an inverse, it is said to be INVERTIBLE OR NON-SINGULAR. A matrix that doesn't have an inverse is said to be singular. An invertible matrix will have only one inverse; that is, if a matrix does have an inverse, that inverse will be unique.

Note: i. Inverse of a matrix is defined only for square matrices

ii. If B is an inverse of A , then A is also an inverse of B

iii. Inverse of a matrix is unique

iv. If matrix A has an inverse, A is said to be invertible & not all. Square matrices are invertible.

Finding the inverse of a matrix

Lets begin by considering a tabular format where the square matrix A is AUGMENTED with an identity matrix of the same order as $[A \mid I]$ i.e. the two matrices separated by a vertical line

Now if the inverse matrix A^{-1} were known, we could multiply the matrices on each side of the vertical line by A^{-1} as

$$[AA^{-1} \mid A^{-1}I]$$

Then because $AA^{-1} = I$ & $A^{-1}I = A^{-1}$, we would have $[I \mid A^{-1}]$. We don't follow this procedure, because the inverse is not known at this juncture, we are trying to determine

the inverse. We instead employ a set of permissible row operations on the augmented matrix $[A \mid I]$ to transform A on the left of the vertical line in to an identity matrix (I). As the identity matrix is formed on the left of the vertical line, the inverse of A is formed on the right side. The allowable manipulations are called Elementary row operations. **ELEMENTARY ROW OPERATIONS:** are operations permitted on the rows of a matrix. In a matrix Algebra there are three types of row operations

Type 1: Any pair of rows in a matrix may be interchanged / Exchange operations

Type 2: a row can be multiplied by any non-zero real number / Multiple operation

Type 3: a multiple of any row can be added to any other row. / Add A-multiple operation

In short the operation can be expressed as

1. Interchanging rows
2. The multiplication of any row by a non-zero number.
3. The addition / subtraction of (a multiple of) one row to /from another row

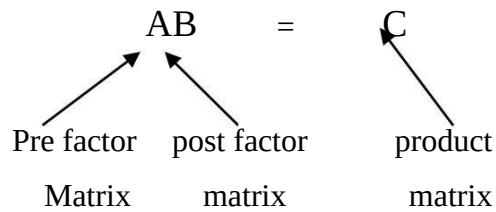
eg.1. $A = \begin{bmatrix} 4 & 3 & 2 \\ -2 & 6 & 7 \end{bmatrix}$ $B = \begin{bmatrix} -2 & 6 & 7 \\ 4 & 3 & 2 \end{bmatrix} \Rightarrow$ interchanging rows

2. $A = \begin{bmatrix} 4 & 3 & 2 \\ -2 & 6 & 7 \end{bmatrix}$ $B = \begin{bmatrix} 8 & 6 & 4 \\ -2 & 6 & 7 \end{bmatrix}$ Multiplying the first rows by 2

3. $A = \begin{bmatrix} 4 & 3 & 2 \\ -2 & 6 & 7 \end{bmatrix}$ $B = \begin{bmatrix} 4 & 3 & 2 \\ 6 & 12 & 11 \end{bmatrix}$ multiplying the 1st row by 2 & add to the 2nd row. This case there is no change to the first row.

Theorem on row operations

A row operation performed on product of two matrices is equivalent to row operation performed on the pre factor matrix.



eg. $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}_{2 \times 3}$ $B = \begin{bmatrix} 1 & 2 \\ 1 & 1 \\ 2 & 3 \end{bmatrix}_{3 \times 2}$ $C = \begin{bmatrix} 9 & 13 \\ 13 & 19 \end{bmatrix}_{2 \times 2}$

Interchange row1 (R₁) with row-2 (R₂)

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 1 & 1 \\ 2 & 3 \end{bmatrix} \quad C = \begin{bmatrix} 13 & 19 \\ 9 & 13 \end{bmatrix}$$

Basic procedures to find the inverse of a square matrix

1. To set ones first in a column & next zeros (with in a given column)
2. To set zeros first in a matrix & next ones.

Ones first method

Find the inverse of the following matrix

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

augment A
with the same dimension $\Rightarrow$
identity matrix first

$$\begin{array}{cc|cc}
 3 & 2 & 1 & 0 \\
 1 & 1 & 0 & 1
 \end{array}$$

Interchange rows (row 1 with row 2)

$$\begin{array}{cc|cc}
 1 & 1 & 0 & 1 \\
 3 & 2 & 1 & 0
 \end{array}$$

Multiply R₁ by -3 & add to R₂

$(-3R_1 + R_2)$ i.e. No change to R_1

$$\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & -3 \end{array}$$

Multiply R_2 by $-1 = (-R_2)$

$$\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 3 \end{array}$$

Multiply R_2 by -1 & add to R_1

$$\begin{array}{cc|cc} 1 & 0 & 1 & -2 \\ 0 & 1 & -1 & 3 \end{array}$$

Ones first: try to set ones first in a column and then zeros of the same column. Goes from left to right

Therefore inverse of A is

$$A^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$$

Zeros first method

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

Find inverse of A

$$\text{Augmentation} \rightarrow \begin{array}{cc|cc} 3 & 2 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}$$

$-2R_2 + R_1$

$$\begin{array}{cc|cc} 1 & 0 & 1 & -2 \\ 1 & 1 & 0 & 1 \end{array}$$

$-1R_1 + R_2$

$$\begin{array}{cc|cc} 1 & 0 & 1 & -2 \\ 0 & 1 & -1 & +3 \end{array}$$

Therefore, inverse of A

$$\text{i.e. } A^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$$

Exercise: Find the inverse for the following matrices (if exist)

$$1. \quad A = \begin{bmatrix} -2 & 2 & 3 \\ 1 & -1 & 0 \\ 0 & 1 & 4 \end{bmatrix}$$

$$A^{-1} = \frac{1}{3} \begin{bmatrix} -4 & -5 & 3 \\ -4 & -8 & 3 \\ 1 & 2 & 0 \end{bmatrix}$$

$$2. \quad B = \begin{bmatrix} 2 & -17 & 11 \\ -1 & 11 & -7 \\ 0 & 3 & -2 \end{bmatrix}$$

$$B^{-1} = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix}$$

$$3. \quad C = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix}$$

4. What do you conclude from question 2 and 3?

$$5. \quad D = \begin{bmatrix} 2 & 7 & 1 \\ -3 & -9 & 2 \end{bmatrix}$$

Answers for exercises

1) Finding inverse of matrix A

Uses ones first method. First
augment the matrix with

The same dimension identity matrix

$$\text{i.e. } \begin{array}{ccc|ccc} -2 & 2 & 3 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 4 & 0 & 0 & 1 \end{array}$$

Then our objective is trying to change
the given matrix into identity format
by applying elementary row operations
exchange row -1 with row -2

$$\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 1 & 0 \\ -2 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 0 & 1 \end{array}$$

next change the remaining number

With in the same column into zero.

the appropriate operation is multiply
row 2 by 1 and add the result to row
number 1.

i.e. $1R_2 + R_1$

$$\begin{array}{ccc|ccc} 1 & 0 & 4 & 0 & 1 & 1 \\ 0 & 1 & 4 & 0 & 0 & 1 \\ 0 & 0 & 3 & 1 & 2 & 0 \end{array}$$

now proceed to the third column and
change the column into its required
form. First change the primary
diagonal entries into one. By
multiplying the third row by $1/3$

Next multiply Row 1 by 2 and add the result to Row -2 i.e. $2R_1 + R_2$

$$\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 3 & 1 & 2 & 0 \\ 0 & 1 & 4 & 0 & 0 & 1 \end{array}$$

Now proceed to the 2nd column and change the primary diagonal entry into positive one by applying elementary row operation. The best operation is is exchanging row 2 with row 3

$$\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 4 & 0 & 0 & 1 \\ 0 & 0 & 3 & 1 & 2 & 0 \end{array}$$

i.e. $1/3 R_3$

$$\begin{array}{ccc|ccc} 1 & 0 & 4 & 0 & 1 & 1 \\ 0 & 1 & 4 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1/3 & 2/3 & 0 \end{array}$$

then change the remaining numbers into zero.

multiply Row 3 by -4 and add the result to row 2

$$\begin{array}{ccc|ccc} 1 & 0 & 4 & 0 & 1 & 1 \\ 0 & 1 & 0 & -4/3 & -8/3 & 1 \\ 0 & 0 & 1 & 1/3 & 2/3 & 0 \end{array}$$

now multiply the third row by -4 add the result to row 1 i.e. $-4R_3 + R_1$

$$\begin{array}{ccc|ccc} 1 & 0 & 0 & -4/3 & -5/3 & 1 \\ 0 & 1 & 0 & -4/3 & -8/3 & 1 \\ 0 & 0 & 1 & 1/3 & 2/3 & 0 \end{array}$$

Therefore the resulting matrix, that is a matrix consisting of the elements at the right side is assumed to be inverse of matrix A i.e. A^{-1}

$$A^{-1} = \begin{bmatrix} -4/3 & -5/3 & 1 \\ -4/3 & -8/3 & 1 \\ 1/3 & 2/3 & 0 \end{bmatrix}$$

2) Inverse of matrix B is

$$B^{-1} = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix}$$

3) Inverse of matrix C is

$$C^{-1} = \begin{bmatrix} 2 & -17 & 11 \\ -1 & 11 & -7 \\ 0 & 3 & -2 \end{bmatrix}$$

- 4) We can conclude (observe) that matrix B and C are inverse to each other.
 5) Matrix d doesn't have an inverse because it is not a square matrix.

2.3 MATRIX APPLICATIONS

2.3.1 Solving Systems of Linear Equations

I. n by n systems

Systems of linear equations can be solved using different methods. Some are:

- i. Estimation method – for two (2) variable problems (equation)
- ii. Matrix method
 - Inverse method
 - Gaussian method

Inverse method:

Steps 1. Change the system of linear equation into matrix form. The result will be 3 different matrices constructed using coefficient of the variables, unknown values and right hand side (constant) values

2. Find the inverse of the coefficient matrix

3. Multiply the inverse of coefficient matrix with the vector of constant, and the resulting values are the values of the unknown matrix.

eg. $2X + 3Y = 4$ Given this system of linear equation applying
 $X + 2Y = 2$ inverse method we can find the unknown values.

Step 1. Change it into matrix form

- Using coefficient construct one matrix i.e. coefficient matrix

$$\begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix} = \text{Coefficient matrix}$$

- Using the unknown variables construct unknown matrix & it is a column vector (a matrix which has one column)

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \text{vector of unknown}$$

-Using the constant values again construct vector of constant

$$2 \quad \begin{bmatrix} 4 \end{bmatrix} = \text{vector of constant}$$

Step 2. Find inverse of the coefficient matrix

Now we are familiar how to find an inverse for any square matrix. Assuming once first method find the inverse for matrix $2 \quad \begin{bmatrix} 3 \\ 1 \quad 2 \end{bmatrix}$

Its inverse become $\begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$

Step 3. Multiply the coefficient inverse with the vector of constant

$$\begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

Therefore the resulting matrix that is $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$ is

the value for the unknown variables i.e. $\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

Then $X = 2$ and $Y = 0$ that is unique solution

* The logic is this given three matrices, coefficient matrix, unknown matrix and vector of constant in the following order.

$$AX = B$$

A = coefficient matrix

Given this we can apply different

X = vector of unknown

Operations, say multiply both sides

B = vector of constant

Of the expression by A^{-1}

$$\underline{A^{-1}AX} = A^{-1}B$$

$$\underline{IX} = A^{-1}B$$

$X = A^{-1}B$ this implies that multiplying inverse of the coefficient matrix will gives us the value of the unknown matrix

Limitations of inverse method

- It is only used whenever the coefficient matrix is square matrix
- In addition to apply the method the coefficient matrix needs to have an inverse
- It doesn't differentiate between no solution and infinite solution cases.

Gaussian method

It is developed by a mathematician Karl F. Gauss (1777-1855). It helps to solve systems of linear equations with different solution approaches i.e. unique solution, No solution and infinite solution cases.

“n” by “n” systems

- Step: 1. Change the system of linear equation into a matrix form
2. Augment the coefficient matrix with the vector of constant.
 3. Change the coefficient matrix into identity form by applying elementary row operation and apply the same on the vector of constant.
 4. The resulting values of the vector of constant will be the solution or the value of the unknown

Example:
$$\begin{cases} 2X + 3Y = 4 \\ X + 2Y = 2 \end{cases}$$

Step 1. Change it into matrix form

$$\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

Coefficient	unknown	vector of
Matrix	matrix	constant

Step: 2. Augumentation

$$\begin{array}{cc|c} 2 & 3 & 4 \\ 1 & 2 & 2 \end{array}$$

Step: 3. Change the coefficient matrix into identity form by applying elementary row operation (use ones first method)

$$\begin{array}{cc|c} \textcircled{2} & 3 & 4 \\ 1 & 2 & 2 \end{array}$$

Change first the primary diagonal entry from the first row into positive one. Possible operation is exchange row one with row two.

$$\begin{array}{cc|c} 1 & 2 & 2 \\ \textcircled{2} & 3 & 4 \end{array}$$

Next change the remaining numbers in the first column into zero, this case number 2

Now multiply the 1st row by -2 & add the result to row -2

$$\begin{array}{cc|c} 1 & 2 & 2 \\ 0 & -1 & 0 \end{array}$$

Then proceed to column 2 and change the primary diagonal entry i.e. -1 into 1

Multiply the 2nd row by -1 (-1R₂)

$$\begin{array}{cc|c} 1 & \textcircled{2} & 2 \\ 0 & 1 & 0 \end{array}$$

Now change the remaining number with in the same column (column -2) into zero i.e. number 2

Multiply 2nd row by -2 and add the result to the 1st row

$$\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 0 \end{array}$$

Therefore X = 2 and Y = 0

Example 2. $\begin{cases} X + Y = 2 \\ 2X + 2Y = 4 \end{cases}$

Step-1

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

Step-2

$$\begin{array}{cc|c} 1 & 1 & 2 \\ \textcircled{2} & 2 & 4 \end{array}$$

Multiply Row-1 by -2 & add the result to row-1 ($-2R_1 + R_2$)

$$\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & \textcircled{0} & 0 \end{array}$$

The next step is changing the primary diagonal entry in the 2nd row to 1. But there is no possible operation that can enable you to change it in to number 1

Therefore the implication is that you can't go further but we can observe something from the result. And it is implying an infinite solution case

Example 3.
$$\begin{cases} X + Y = 5 \\ X + Y = 9 \end{cases}$$

Step 1.
$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$

Step 2
$$\begin{array}{cc|c} 1 & 1 & 5 \\ \textcircled{1} & 1 & 9 \end{array}$$

Change the encircled number above in to zero

Multiply the first row by -1 & add the result to the 2nd row.

$$\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & 0 & 4 \end{array}$$

$0 = 4$ no solution

There is no possible operation that we can apply in order to change the primary diagonal entry in the 2nd column without affecting the first column structure. Therefore stop there, but here we can observe something i.e. it is no solution case.

Therefore, Gaussian method makes a distinction between No solution & infinite solution. Unlike the inverse method.

* Summarizing our results for solving an “n” by “n” system, we start with the matrix. (A/B), & attempt to transform it into the matrix (I/C) one of the three things will result.

1. an “n” by “n” matrix with the unique solution.

eg.

$$\begin{array}{ccc|c} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & 3 \end{array}$$

2. A row that is all zeros except in the constant column, indicating that there are no solutions,

eg.

$$\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 0 & 7 \end{array}$$

3. A matrix in a form different from (1) & (2), indicating that there are an unlimited number of solutions. Note that for an n by n system, this case occurs when there is a row with all zeros, including the constant column.

Eg.

$$\begin{array}{ccc|c} 1 & 0 & 2 & 5 \\ 0 & 1 & 3 & -3 \\ 0 & 0 & 0 & 0 \end{array}$$

Reference Exercise

$$\begin{array}{lll} 1. \begin{cases} X + 2Y - 3Z = 11 \\ 3X + 2Y + Z = 1 \\ 2X + Y - 5Z = 11 \end{cases} & 2. \begin{cases} X + Y + Z = 4 \\ 5X - Y + 7Z = 25 \\ 2X - Y + 3Z = 8 \end{cases} & 3. \begin{cases} X + Y + Z = 4 \\ 5X - Y + 7Z = 20 \\ X - Y + 3Z = 8 \end{cases} \end{array}$$

Unique solution case

* No solution case

* Many solution case

$$\text{i.e } X = -1$$

$$Y = 3$$

$$Z = 2$$

$$4. \begin{cases} 2X + 6Y - Z = 18 \\ Y + 3Z = 9 \\ 3X - 5Y + 8Z = 4 \end{cases} \quad X = 1, Y = 3, Z = 2$$

II M by n linear systems

The $m \times n$ linear systems are those systems where the number of rows (m) and number of columns (n) are unequal or it is the case where the number of equations (m) & the number of variables (n) are unequal. And it may appear as $m > n$ or $m < n$.

Linear equation where $m > n$

To solve an m by n system of equations with $m > n$, we start with the matrix (A/B) and attempt to transform it into the matrix (I/C) .

One of the three things will result:

1. An m by n identifying matrix above $m - n$ bottom rows that are all zeros, giving the unique solution:

$$\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 3X_1 + 2X_2 + X_3 = 23 \\ 0 & 1 & 0 & -5 & X_1 + 3X_2 + 2X_3 = 26 \\ 0 & 0 & 1 & 4 & 2X_1 + X_2 + 2X_3 = 10 \\ 0 & 0 & 0 & 0 & 4X_1 + 5X_2 + 3X_3 = 49 \end{array} \quad \{3, 5, 4\}$$

2. A row that is $m - n$ bottom row is all zeros except in the constant column, indicating that there are no solutions

eg.

$$\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 2X_1 + X_2 = 30 \\ 0 & 1 & 0 & -5 & X_1 + 2X_2 = 24 \\ 0 & 0 & 1 & 7 & 4X_1 + 5X_2 = 72 \\ 0 & 0 & 0 & 1 & \end{array}$$

3. A matrix in a form different from (1) & (2), indicating that there are an unlimited number of solutions

eg.1	0	2	-4	$3X_1 + 2X_2 + X_3 = 6$
	0	1	8	$6X_1 + 4X_2 + 3X_3 = 12$
	0	0	0	$9X_1 + 6X_2 + 3X_3 = 18$
	0	0	0	$15X_1 + 10X_2 + 5X_3 = 30$

Linear Equations where $m < n$

Our attempts to transform (A/B) into (I/C) in the case where $m < n$ will result in:

1. A row which is all zeros except in the constant columns, indicating that there are no solutions, or
2. A matrix in a form different from number one above indicating that there are an unlimited number of solutions.

“Every system of linear equations has either No solution, Exactly one solution or infinitely many solutions.”

Example Solve the following systems of linear equations

1) $4X_1 + 6X_2 - 3X_3 = 12$

$6X_1 + 9X_2 - 9/2X_3 =$

20 No solution

$X_1 + 3X_2 + X_3 = 6$

$-X_1 + X_2 + X_3 = 2$

Unlimited solution

$2X_1 + 3X_2 + 4X_3 + X_4 = 37$

$X_1 + 2X_2 + 3X_3 + 2X_4 = 24$

$3X_1 + X_2 + X_3 + 3X_4 = 33$

Unlimited solution

Solution for an “n” by “n” system

$$\begin{array}{ccc|c} 1.1 & 2 & -3 & 11 \\ 3 & 2 & 1 & 1 \\ 2 & 1 & -5 & 11 \end{array}$$

$-3R_1 + R_2 / -2R_1 + R_3$

$$\begin{array}{ccc|c} 1 & 2 & -3 & 11 \\ 0 & -4 & 10 & -32 \\ 0 & -5 & 4 & -22 \end{array}$$

$-1/4 R_2 / -2 R_2 + R_1 / 3 R_2 + R_3$

$$\begin{array}{ccc|c} 1 & 0 & 2 & -5 \\ 0 & 1 & -5/2 & 8 \\ 0 & 0 & -13/2 & 13 \end{array}$$

$-2/13 R_3 / 5/2 R_3 + R_2 / -2 R_3 + R_1$

$$\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{array}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix}$$

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix}$$

$\therefore \underline{X = -1} \quad \underline{Y = 3} \quad \underline{Z = -2}$

$$\begin{array}{ccc|c} 2.1 & 1 & 1 & 4 \\ 5 & -1 & 7 & 25 \\ 2 & -1 & 3 & 8 \end{array}$$

$-5R_1 + R_2 / -2R_1 + R_3$

$$\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -6 & 2 & 5 \\ 0 & -3 & 1 & 0 \end{array}$$

$-1/6 R_2 / -R_2 + R_1 / 3 R_2 + R_3$

$$\begin{array}{ccc|c} 1 & 0 & 4/3 & 13/2 \\ 0 & 1 & -1/3 & -5/6 \\ 0 & 0 & 0 & -5/2 \end{array}$$

$\therefore$ this implies No solution case

$$\begin{array}{ccc|c} 3) & 1 & 1 & 4 \\ 5 & -1 & 7 & 20 \\ 2 & -1 & 3 & 8 \end{array}$$

$-5R_1 + R_2 / -2R_1 + R_3$

$$\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -6 & 2 & 0 \\ 0 & -3 & 1 & 0 \end{array}$$

$-1/6 R_2 / 3 R_2 + R_3 / -R_2 + R_1$

$$\begin{array}{ccc|c} 1 & 0 & 4/3 & 4 \\ 0 & 1 & -1/3 & 0 \\ 0 & 0 & 0 & 0 \end{array}$$

This implies that there are so many or infinite solution

“m” by “n” systems

$$1) \quad \begin{array}{ccc|c} 3 & 2 & 1 & 23 \\ 1 & 3 & 2 & 26 \\ 2 & 1 & 2 & 19 \\ 4 & 5 & 3 & 49 \end{array}$$

$$R_1 \leftrightarrow R_2 / -3R_1 + R_2 / -2R_1 + R_3 / -4R_1 + R_4$$

$$\begin{array}{ccc|c} 1 & 3 & 2 & 26 \\ 0 & -7 & -5 & -55 \\ 0 & -5 & -2 & -33 \\ 0 & -7 & -5 & -55 \end{array}$$

$$-1/7 R_2 / -3R_2 + R_1 / 5R_2 + R_3 / 7R_2 + R_4$$

$$\begin{array}{ccc|c} 1 & 0 & -1/7 & 17/7 \\ 0 & 1 & 5/7 & 55/7 \\ 0 & 0 & 11/7 & 44/7 \end{array}$$

$$7/11R_3 / -5/7R_3 + R_2 / 1/7R_3 + R_1$$

$$\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array}$$

Unique solution case

$$\therefore \underline{X_1 = 3} \quad \underline{X_2 = 5} \quad \underline{X_3 = 4}$$

$$2) \quad \begin{array}{cc|c} 2 & 1 & 30 \\ 1 & 2 & 24 \\ 4 & 5 & 72 \end{array}$$

$$R_1 \leftrightarrow R_2 / -2R_1 + R_2 / -4R_1 + R_3$$

$$\begin{array}{cc|c} 1 & 2 & 24 \\ 0 & -3 & -18 \\ 0 & -3 & -24 \end{array}$$

$$-1/3 R_2 / -2R_2 + R_1 / 3R_2 + R_3$$

$$\begin{array}{cc|c} 1 & 0 & 14 \\ 0 & 1 & 6 \\ 0 & 0 & -6 \end{array}$$

this implies that there is no solution which uniquely satisfy the system

$$3) \quad \begin{array}{ccc|c} 3 & 2 & 1 & 6 \\ 6 & 4 & 3 & 12 \\ 9 & 6 & 3 & 18 \\ 15 & 10 & 5 & 30 \end{array}$$

$$1/3R_1 / -6R_1 + R_2 / -9R_1 + R_3 /$$

$$-15R_1 + R_4$$

$$\begin{array}{ccc|c} 1 & 2/3 & 1/3 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}$$

$\therefore$ This implies that there are unlimited numbers of solutions

m by n system where m < n

i.e. number of equations are less than # of variables

1) $4X_1 + 6X_2 - 3X_3 = 12$

$$6X_1 + 9X_2 - 9/2X_3 = 20$$

$$\begin{bmatrix} 4 & 6 & -3 \\ 6 & 9 & -9/2 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 20 \end{bmatrix}$$

$$\begin{array}{ccc|c} 4 & 6 & -3 & 12 \\ 6 & 9 & -9/2 & 20 \end{array}$$

$$1/4R_1 / -6R_1 + R_2$$

$$\begin{array}{ccc|c} 1 & 3/2 & -3/4 & 3 \\ 0 & 0 & 0 & 2 \end{array}$$

No solution

2) $X_1 + 3X_2 + X_3 = 6$

$$-X_1 + X_2 + X_3 = 2$$

$$\begin{bmatrix} 1 & 3 & 1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

$$\begin{array}{ccc|c} 1 & 3 & 1 & 6 \\ -1 & 1 & 1 & 2 \end{array}$$

$$1 \times R_1 + R_2$$

$$\begin{array}{ccc|c} 1 & 3 & 1 & 6 \\ 0 & 4 & 2 & 8 \end{array}$$

$$1/4R_2$$

$$\begin{array}{ccc|c} 1 & 3 & 1 & 6 \\ 0 & 1 & 1/2 & 1/2 \end{array}$$

Infinite solution

2.3.2 Word problems

Steps

1. Represent one of the unknown quantities by a letter usually X & express other unknown quantities if there is any in terms of the same letter like X_1 X_2 etc
2. Translate the quantities from the statement of the problem in to algebraic form & set up an equation
3. Solve the equation (s) for the unknown that is represented by the letter & find other unknowns from the solution
4. check the findings according to the statement in the problem

Example:

- 1) A Manufacturing firm which manufactures office furniture finds that it has the following variable costs per unit in dollar/unit

	Desks	Chairs	Tables	Cabinet
Material	50	20	15	25
Labor	30	15	12	15
Overhead	30	15	8	20

Assume that an order of 5 desks, 6 chairs, & 4 tables & 12 cabinets has just been received. What is the total material, labor & overhead costs associated with the production of ordered items?

Answer:

Material cost = \$ 750

Labor cost = \$ 918

Overhead cost = \$ 512

2. Kebede carpet co. has an inventory of 1,500 square yards of wool & 1,800 square yards of nylon to manufacture carpeting. Two grades of carpeting are produced. Each roll of superior grade carpeting requires 20 sq. yards of wool & 40sq. yards of nylon. Each roll of quality-grade carpeting requires 30 square yards of wool & 30 square yard of

nylon. If Kebede would like to use all the material in inventory, how many rolls of superior & how many rolls of quality carpeting should be manufactured?

15 & 40

3. Getahun invested a total of br. 10000 in three different saving accounts. The accounts paid simple interest at an annual rate of 8%, 9% & 7.5% respectively. Total interest earned for the year was br. 845. The amount in the 9% account was twice the amount invested in the 7.5% account. How much did Getahun invest in each account?

1000, 6000, 3000

4. A certain manufacturer produces two products P & Q. Each unit of product P requires (in its production) 20 units of raw material A & 10 units of raw material B. Each unit of product Q requires 30 units of raw material A & 50 units of raw material B. There is a limited supply of 1200 units of raw material A & 950 units of raw material B. How many units of P & Q can be produced if we want to exhaust the supply of raw materials?

Answer: 45 units of P and

10 units of Q

5. Attendance records indicate that 80,000 South Koreans attended the 2002 world cup at its opening ceremony. Total ticket receipts were Birr 3,500,000. Admission prices were Birr 37.5 for the second-class and Birr 62.50 for the first class. Determine the number of South Koreans who attended the football game at first class and second class.

Solutions / word problems

1)	D	Ch	T	Cb		
Mt	$\left[\begin{array}{ccccc} 50 & & 20 & 15 & 25 \\ 30 & & 15 & 12 & 15 \\ 30 & & 15 & 8 & 20 \end{array} \right]$				5	Desk
Lab.					6	Chair
FOH					4	Tables
					12	Cabinet

$$\begin{aligned} \text{Material cost} &= (50 \times 5) + (20 \times 6) + (15 \times 4) + (25 \times 12) \\ &= \underline{730 \text{ br.}} \end{aligned}$$

$$\begin{aligned} \text{Labor cost} &= (30 \times 5) + (15 \times 6) + (12 \times 4) + (15 \times 12) \\ &= \underline{\$ 468.00} \end{aligned}$$

$$\text{FOH cost} = (30 \times 5) + (15 \times 6) + (8 \times 4) + (20 \times 12) \\ = \underline{\$ 1710.00}$$

2) Given: wool inventory 1500 yards
nylon inventory 1800 yards

	Grade one carpet	The other grade carpet
Wool requirement		
per roll	20 Sq.	30 Sq.
Nylon requirement		
per roll	40 Sq.	30 Sq.

Let X be rolls of superior grade carpeting &
Y be rolls of quality grade carpeting

$$\begin{bmatrix} 20 & 30 \\ 40 & 30 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} 1500 \\ 1800 \end{bmatrix}$$

$$\therefore 20X + 30Y = 1500$$

$$\underline{40X + 30Y = 1800}$$

$$-20X = -300$$

$$\underline{X = 15}$$

$$20X + 30Y = 1500$$

$$20(15) + 30Y = 1500$$

$$30Y = 1500 - 300$$

$$Y = \underline{1200}$$

$$30$$

$$\underline{Y = 40}$$

$$\begin{array}{cc|c} 20 & 30 & 1500 \\ 40 & 30 & 1800 \end{array}$$

$$\begin{array}{cc|c} 2 & 3 & 150 \\ 4 & 3 & 180 \end{array}$$

$$1/2 R_1 / -4R_1 + R_2$$

$$\begin{array}{cc|c} 1 & 3/2 & 75 \\ 0 & -3 & -120 \end{array}$$

$$-1/3 R_2 / -3/2 R_2 + R_1$$

$$\begin{array}{cc|c} 1 & 0 & 15 \\ 0 & 1 & 40 \end{array}$$

$$\therefore \underline{X = 15 \text{ \& } Y = 40}$$

3) Total investment 10000.00

interest earned from the three accounts 8%, 9% and 7.5%

total interest earned for the year was 845.00

amount in 9% = 2 (amount in 7.5%)

Let amount invested in 8% account be X_1

amount invested in 9% account be X_2

amount invested in 7.5% account be X_3

$$X_1 + X_2 + X_3 = 10,000$$

$$0.08X_1 + 0.09X_2 + 0.075X_3 = 845$$

$$X_2 = 2(X_3)$$

or

$$X_1 + X_2 + X_3 = 10,000$$

$$0.08X_1 + 0.09X_2 + 0.075X_3 = 845$$

$$X_2 - 2X_3 = 0$$

$$\begin{array}{ccc|c} 1 & 1 & 1 & 10,000 \\ 80 & 90 & 75 & 845,000 \\ 0 & 1 & -2 & 0 \end{array}$$

$$-80R_1 + R_2$$

$$\begin{array}{ccc|c} 1 & 1 & 1 & 10,000 \\ 0 & 10 & -5 & 45,000 \\ 0 & 1 & -2 & 0 \end{array}$$

$$1/10R_2 \quad -R_2 + R_1 \quad -R_2 + R_3$$

$$\begin{array}{ccc|c} 1 & 0 & 3/2 & 5500 \\ 0 & 1 & -1/2 & 4500 \\ 0 & 0 & -3/2 & -4500 \end{array}$$

$$-2/3 R_3 \quad 1/2 R_3 \quad 1/2 R_3 + R_2 \quad -3/2 R_3 + R_1$$

$$\begin{array}{ccc|c}
 1 & 0 & 0 & 1000 \\
 0 & 1 & 0 & 6000 \\
 0 & 0 & 1 & 3000
 \end{array}$$

$\therefore \underline{X_1 = 1000}, \quad \underline{X_2 = 6000}, \quad \underline{X_3 = 3000}$

2.3.3 Markov Chains

This model is a forecasting model. It is probabilistic (stochastic) model. A Russian Mathematician called Andrew Markov around 1907 develops this model.

Markov chains are models, which are useful in studying the evolution of certain systems over repeated trials. These repeated trials are often successive time periods where the state (outcome condition) of the systems in any particular time period can't be determined with certainty. Therefore, a set of transition probabilities is used to describe the manner in which the system makes transition from one period to the next. Hence,. We can predict the probabilities of the system being in a particular state at a given time period. We can also talk about the long run or equilibrium or steady state.

The necessary assumptions of the chain:

1. The system condition (outcome) state in any given period depends on its state in the Preceding period & on the transition probabilities
2. The transition probabilities are constant overtime
3. Change in the system will occur once & only once each period
eg. If it's a week, its only once in a week
4. The transition period occurs with regularities
* if we start with days, we use the day until we reach our end.

Information flow in the analysis

The Markov model is based on two sets of input data

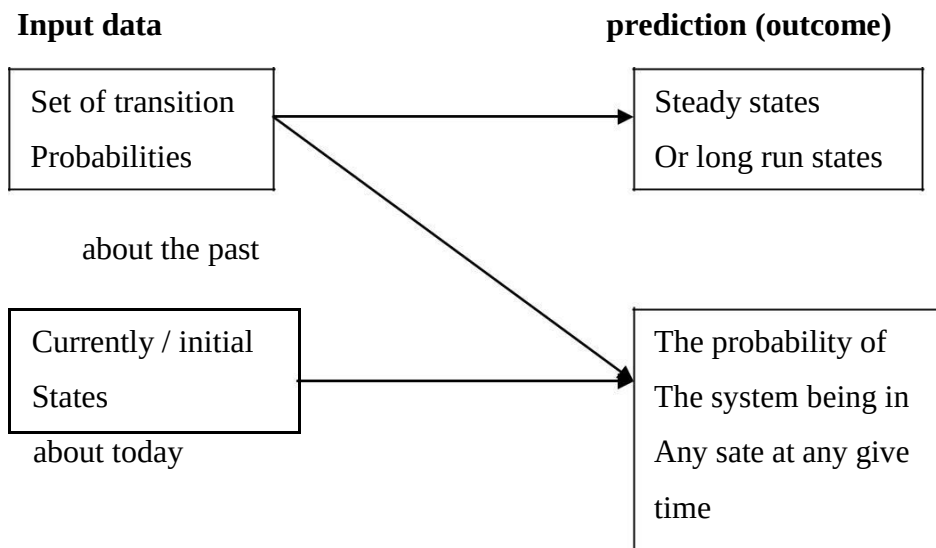
- The set of transition probabilities
- The existing or initial or current conditions or states

The Markov process, therefore, describes the movement of a system from a certain state in the current state/time period to one of n possible states in the next stage. The system

makes in an uncertain environment, all that is known is the probability associated with any possible move or transition. This probability is known as transition probability, symbolized by P_{ij} . It is the likelihood that the system which is currently in state i will move to state j in the next period. From these inputs the model makes two predictions usually expressed as vectors.

1. The probabilities of the system being in any state at any given future time period
2. The long run (equilibrium) or steady state probabilities.

The set of transition probabilities are necessary for both prediction (time period n , & steady state), but the initial state is needed for only the first prediction.



Example

Currently its known that 80% of customers shop at store 1 & 20% shop at store 2. In reviewing a past data suppose we find that out of all customer who shopped at store 1 in a given week 90% remain loyal for the next week (store one again), 10% switch to store 2. On the other hand, out of all customers who shopped at store 2, in a given week 80% remains loyal for the next week (store 2 again), 20% switch to store 1. What will be the proportion of customers shopping at store 1 & 2 in each of the next two weeks.

Solution

Let S_1 be the proportion of store 1

S₂ be the proportion of store 2

- Initial state/current state probability matrix for store one and two will be:

$$V_{12} = (0.8 \quad 0.2)$$

To the next weekly shopping period

from one		S ₁	S ₂
week	S ₁	0.9	0.1
shopping	S ₂	0.2	0.8
period			

Transition probability matrix

is a square matrix such that each entry indicates the prob. Of the system moving from a given state to another state.

- The sum of rows in the transition matrix should be 1
- We have to be consistent in writing the elements

Markov Chain Formula

a) $V_{ij}(n) = [V_{ij}(n-1)] \times P$

where P = Transition matrix

$V_{ij}(n)$ = Vector for period n

$V_{ij}(n-1)$ = Vector for period n - 1

$V_{12}(0) = (0.8 \ 0.2) \longrightarrow$ current share

$V_{12}(1) = V_{12}(0) \times P$

$= V_{12}(0) \times P$

$(0.8 \ 0.2) \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$

$= (0.8 \times 0.9) + (0.2 \times 0.2)$

$= 0.72 + 0.04$

$= \underline{0.76}$

$(0.8 \times 0.1) + (0.2 \times 0.8)$

$0.08 + 0.16$

$\underline{0.24}$

$V_{12}(1) = (0.76 \quad 0.24)$

$V_{12}(2) = V_{12}(1) \times P$

$= V_{12}(1) \times P$

$(0.76 \ 0.24) \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$

$$\begin{array}{ccc}
 & 0.2 & 0.8 \\
 (0.76 \times 0.9) + (0.24 \times 0.2) & & 0.76(0.1) + (0.24 \times 0.8) \\
 (0.732 & & 0.268)
 \end{array}$$

b) Long run market share

Assumption

In the long run the share of the systems is assumed to be constant.

Let - the share of store 1 in the long run be V_1

- the share of store 2 in the long run be V_2

$$\begin{array}{ccc}
 n & p & n+1 \\
 (V_1 \ V_2) & \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} & (V_1 \ V_2)
 \end{array}$$

$$0.9V_1 + 0.2V_2 = V_1$$

$$0.1V_1 + 0.8V_2 = V_2$$

$$-V_1 + 0.9V_1 + 0.2V_2 = 0$$

$$0.1V_1 + 0.8V_2 - V_2 = 0$$

$$\left. \begin{array}{l} -0.1V_1 + 0.2V_2 = 0 \\ 0.1V_1 + (0.8V_2 - V_2) = 0 \end{array} \right\}$$

$$0.9V_1 + 0.2(V_2 - V_1) = V_1$$

$$0.9V_1 + 0.2 - 0.2V_1 = V_1$$

$$0.7V_1 + 0.2 = V_1$$

$$0.2 = 0.3V_1$$

$$\underline{V_1 = 2/3}$$

$$V_2 = 1 - V_1$$

$$= 1 - 2/3$$

$$\underline{V_2 = 1/3}$$

In the long run 67% of the customers will shop in store 1 & 33% in shop 2.

Prediction:

Long run: only the transition matrix

= **At specified time:-** the transition matrix & state vector. Hence unless the transition matrix is affected, the long run state will not be affected. Moreover, we can't know the number of years, weeks to attain the long run state / point but we can know the share

Exercises

1. A division of the ministry of public health has conducted a simple survey on the public attitude to wards smoking . From the results of the survey the department concluded that currently only 20% of the population smokes cigarette & every month 10% of non-smokers become smokers where as 5% of smokers discontinue smoking.

Required:

1. Write the current & transition matrices
2. What will be the proportion of the non-users (non-smokers) & users (smokers) in the long run

Solution

- 1) Let U – stands for Smokers N
– stands for non-Smokers

$$\text{Initial state } V_{UN}^{(0)} = (0.2 \ 0.8)$$

To the next month

		<u>Smokers</u>	<u>Non smokers</u>	
From	{	Smokers	0.95	$P = \begin{bmatrix} .95 & .05 \\ .10 & .90 \end{bmatrix}$
one month		Non-smokers	0.10	

$$\begin{aligned}
 2. \ V_{UN}^{(1)} &= V_{UN}^{(0)} \times P \\
 &= (0.2 \ 0.8) \begin{bmatrix} .95 & .05 \\ .10 & .90 \end{bmatrix} \\
 &= \underline{(0.27 \quad 0.73)}
 \end{aligned}$$

$$\begin{aligned}
 V_{UN}^{(2)} &= V_{UN}^{(1)} \times P \\
 &= (.27 \ .73) \begin{bmatrix} .95 & .05 \\ .10 & .90 \end{bmatrix} \\
 &= \underline{(0.3295 \quad 0.6705)}
 \end{aligned}$$

2. A population of 100,000 consumers make the following purchases during a particular week: 20,000 purchase Brand A, 35,000 Brand B & 45,000 purchase neither Brand. From a market study, it is estimated that of those who purchase Brand A, 80% will purchase it again next week, 15% will purchase brand B next week, & 5% will purchase neither brand. Of those who purchase B, 85% will purchase it again next week, 12% will purchase brand A next week, & 3% will purchase neither brand. Of those who purchase neither brand, 20% will purchase A next week, 15% will purchase Brand B next week, & 65% will purchase neither brand next week. If this purchasing pattern continues, will the market stabilize? What will the stable distribution be?

Yes. The share of A, B and C is = (0.4 0.5 0.1) respectively

3. A vigorous television advertising campaign is conducted during the football season to promote a well-known brand X shaving cream. For each of several weeks, a survey is made & it is found that each week 80% of those using brand X continue to use it & 20% switch. It is also found that those not using brand X, 20% switch to brand X while the other 80% continue using another brand.

a) Write the transition matrix, assuming the transition percentage continues to hold for succeeding weeks.

b) If 20% of the people are using brand X at the start of the advertising campaign, what percentage will be brand X 1 week later? Two weeks later?

Solutions for question number 2, 3 & 4

2) Given: 20,000 purchase brand A

35,000 purchase brand B

45,000 purchase neither brand

Total consumers = 100,000 (20,000 + 35,000 + 45,000)

Let V_A represents the share of brand A purchasers

V_B represents the share of Brand B purchasers

V_N represent the share of neither brand purchasers

The system is arranged in a weekly basis

		To the next week shopping period		
From		A	B	N
One week	A	0.80	0.15	0.05
Shopping	B	0.12	0.85	0.03
Period	N	0.20	0.15	0.65

= P

The stable market means the long run or steady state market because it is noted that in the long run the share will be stable.

And in the long run we have said that the share at n period is equal with the share at n + 1 period. Therefore

$$(\text{The share at n period}) \times (\text{the transition probabilities}) = (\text{the share at n + 1 period})$$

Let the share of brand A purchasers be V_1 in the long run

the share of brand B purchasers be V_2 in the long run

the share of neither purchasers be V_3 in the long run

$$\text{Then } (V_1 \ V_2 \ V_3) \begin{bmatrix} 0.80 & 0.15 & 0.05 \\ 0.12 & 0.85 & 0.03 \\ 0.20 & 0.15 & 0.65 \end{bmatrix} = (V_1 \ V_2 \ V_3)$$

$$0.8V_1 + 0.12V_2 + 0.2V_3 = V_1$$

$$0.15V_1 + 0.85V_2 + 0.15V_3 = V_2$$

$$0.05V_1 + 0.03V_2 + 0.65V_3 = V_3$$

$$V_1 + V_2 + V_3 = 1$$

$$\text{Therefore } V_1 = 1 - V_2 - V_3$$

Substitute $1 - V_2 - V_3$ in place of V_1 to reduce your system in to two variable case Take the first equation

$$0.8V_1 + 0.12V_2 + 0.2V_3 = V_1$$

$$0.8(1 - V_2 - V_3) + 0.12V_2 + 0.2V_3 = 1 - V_2 - V_3$$

$$0.8 - 0.8V_2 - 0.8V_3 + 0.12V_2 + 0.2V_3 + V_2 + V_3 = 1$$

add similar variables together

$$0.32V_2 + 0.4V_3 = 1 - 0.8$$

$$\underline{0.32V_2 + 0.4V_3 = 0.2} \text{ this is a summarized equation from the first equation. And}$$

do the same for the second equation.

$$0.15V_1 + 0.85V_2 + 0.15V_3 = V_2$$

$$0.15(1 - V_2 - V_3) + 0.85V_2 + 0.15V_3 = V_2$$

$$0.15 - 0.15V_2 - 0.15V_3 + 0.85V_2 + 0.15V_3 = V_2$$

add together similar variables and take the constant values to the right side.

$$-0.3V_2 = -0.15$$

$$V_2 = \underline{-0.15}$$

$$-0.30$$

$V_2 = 0.5$ this implies the long run share for the purchasers of brand B will be 50%. Then using this value and the first summarized equation ($0.32V_2 + 0.4V_3 = 0.2$)

Find the value of V_3 in the long run

$$\text{That is, } 0.32(0.5) + 0.4V_3 = 0.2$$

$$0.16 + 0.4V_3 = 0.2$$

$$0.4V_3 = 0.20 - 0.16$$

$$V_3 = \underline{0.04}$$

$$0.04$$

$$\underline{\underline{V_3 = 0.1}}$$

This implies the share of neither brand purchases in the long run will be 10%. Hence the share of Brand A purchasers will be $0.4(1 - 0.5 - 0.1)$

$$\text{Because we said above } V_1 = 1 - V_2 - V_3$$

$$\text{Therefore } V_1 = 1 - 0.5 - 0.1$$

$$\underline{V_1 = 0.4}$$

That is the long run share of Brand A purchasers will be 40% of the total population

$$\text{i.e. } V_A \quad V_B \quad V_N = (0.4 \quad 0.5 \quad 0.1)$$

3) A) let X be the proportion of brand X users X^1

be the proportion of other brand users

To the next week
Shopping period

$$\begin{array}{lcl} \text{From this} & & \\ \text{Week} & X & \begin{bmatrix} X & X^1 \\ 0.80 & 0.20 \end{bmatrix} P \\ \text{Shopping} & X^1 & \begin{bmatrix} 0.20 & 0.80 \end{bmatrix} \\ \text{period} & & \end{array}$$

B) Current users of brand X = 20% then, the users of other brand will be 80%

$$V_{XX}^1(0) = (0.20 \quad 0.80)$$

$$\begin{aligned} V_{XX}^1(1) &= V_{XX}^1(0) [P] \\ &= (0.20 \quad 0.80) \begin{bmatrix} 0.80 & 0.20 \\ 0.20 & 0.80 \end{bmatrix} \\ &= \underline{(0.32 \quad 0.68)} \end{aligned}$$

The proportion of brand X and other brand users after one week period is expected to be 32% and 68% respectively. Then the expected users in the 2nd week will be

$$\begin{aligned} V_{XX}^1(2) &= V_{XX}^1(2-1) (P) \\ &= \begin{bmatrix} V_{XX}^1(1) & P \end{bmatrix} \begin{bmatrix} 0.80 & 0.20 \\ 0.20 & 0.80 \end{bmatrix} \\ &= \underline{\underline{(0.392 \quad 0.608)}} \end{aligned}$$

The expected share of brand X and other brand users is 39.2% and 60.8% in the second week.

UNIT 3: INTRODUCTION TO LINEAR PROGRAMMING

Contents

- 3.0 Aims and Objectives
- 3.1 Introduction
- 3.2 Linear Programming Models
- 3.3 Formulating Lp Models

3.0 OBJECTIVES

At the end of this chapter students will be able to apply the knowledge of linear programming to solve managerial problems, they will be able to use linear programming models to solve managerial problems having quantitative application.

3.1 INTRODUCTION

Linear programming- is an optimization method which shows how to allocate scarce resources in the best possible way subject to more than one limiting condition expressed in the form of inequalities and /or equations. It

- enables users to find optional solution to certain problems in which the solution must satisfy a given set of requirements or constraints.
- Optimization in linear programming implies either maximization (max) Profit, revenue, sales, market share or minimization (min) Cost, time, distance, or a certain objective function.
- Involves linearly related multi-variety functions i.e. functions with more than one independent variables.
- The goal in linear programming is to find the best solution given the constraints imposed by the problem, hence the term constrained optimization.

We can't
max/min two
quantities in one
model.

3.2 LINEAR PROGRAMMING MODELS

LP models are mathematical representation of LP problems. Some models have a specialized format where as others have a more generalized format. Despite this, LPMs

have certain characteristics in common knowledge of these characteristics enables us to recognize problems that are amenable to a solution using LP models and to correctly formulate an LP model. The characteristics can be grouped into two categories: Components and assumptions. The components relate to the structure of a model, where as the assumptions describe the conditions under which the model is valid.

Components

1. Objective function
 2. Decision variables
 3. Constraints
 4. Parameters and Right.
Hand Side Values
- } Model
} Structure

Assumptions

1. Linearity
 2. Divisibility
 3. Certainty
 4. Non-negativity
- } Model
} Validity

3.2.1 Components of LP Model

- a) The Objective function: is the mathematical/ quantitative expression of the objective of the company/ model. The objective in problem solving is the criterion by which all decisions are evaluated. In LPMs a single quantifiable objective must be specified by the decision maker. For example, the objective might relate to profits, or costs or market share, best to only one of these. Moreover, because we are dealing with optimization, the objective will be either maximization or minimization, but not both at a time
- b) The Decision Variables: represent unknown quantities to be resolved for. These decision variables may represent such things as the:
 - number of units of different products to be sold
 - the # of dollars to invest in various projects
 - the # of ads to place with different media

Since the decision maker has freedom of choice among actions, these decision variables are controllable variables.

- c) The constraints: are restrictions which define or limit the attainability (achievability) feasibility of a proposed course of action. They limit the degree to which the objective can be pursued.

A typical restriction embodies scarce resources (such as labor supply, RMs, production capacity, machine time, storage space), legal or contractual requirements (leg. Product standards, work standards), or they may reflect other limits based on forecasts, customer orders, company policies etc.

- d) Parameters- are fixed values that specify the impact that one unit of each decision variable will have on the objective and on any constraint it pertains to as well as to the numerical value of each constraint.

The components are the building blocks of an LP model. We can better understand their meaning by examining a simple LP model as follows.

Example:

Maximize:

Maximize: $4X_1 + 7X_2 + 5X_3$ (profit)... objective function subject to

$$\left. \begin{array}{lcl} 2X_1 + 3X_2 + 6X_3 & 300 \text{ labor hrs} \\ 5X_1 & + 4X_3 & 200 \text{ raw mata.} \\ 3X_1 + 5X_2 + 2X_3 & 360 \end{array} \right\} \text{System Constraints}$$

$$\left. \begin{array}{lcl} X_1 & & = 30 \\ X_1 - \text{qty of product 1} & X_2 & \geq 40 \end{array} \right\} \text{Individual Constraints}$$

$$\begin{array}{lcl} X_2 & \text{qty of product 2} & X_1, X_2, X_3 \geq 0 \\ X_3 & \text{qty of product 3} & \end{array} \longrightarrow \text{Non negativity constructs}$$

Variables
Decision

System constraints- involve more than one decision variables

Individual constraint- involve only one decision variable.

None-negativity constrains- specify that no variable will be allowed to take on a negative value. The non negativity constraints typically apply in an LP model, whether they are explicitly stated or not.

3.2.2 Assumption of LP models

a) Linearity

The linearity requirement is that each decision variable has a linear impact on the objective function and in each constraint in which it appears. Taking the above example,

producing one more unit of products add br 4 to the total profit. This is true over the entire range of possible values of x_1 . The same applies (true) to each of the constraints.

b) Divisibility: The divisibility requirement pertains to potential values of decision variables. If it is assumed that non-integer values are acceptable. For example: 3.5 TV sets/hr would be acceptable $\rightarrow 7$ TV sets/ 2hr.

c) Certainty: The parameters are known and constant. The certainty requirement involves two aspects of LP models. The constraint equations do not change.

- (1) With respect to model parameters (i.e. the numerical values) –If it is assumed that these values are known and constant. Eg. In the above example each unit of product 1 requires 2 labor hours is known and remain constant, and also the 300 labor available is deemed to be known and constant.
- (2) All the relevant constraints identified and represented in the model are as they are.

d) Non-negativity- The non-negativity constraint is that negative values of variables are unrealistic and, therefore, will not be considered in any potential solutions, only positive values and zero will be allowed.

3.3 FORMULATING LP MODELS

Once a problem has been defined, the attention of the analyst shifts to formulating a model. Just as it is important to carefully formulate the model that will be used to solve the problem. If the LP model is ill formulated, ill-structured, it can easily lead to poor decisions.

Formulating linear programming models involves the following steps:

- 1) Define the problem/ problems definition: to determine the no. of type 1 and type 2 products to be produced per month so as to maximize the monetary profit given the restriction.
- 2) Identity the decision variables or represent unknown quantities.
* Let X_1 and X_2 be the monthly quantities of type 1 and type 2 products.
- 3) Determine the objective function: Once the variables have been identified, the objective function can be specified. It is necessary to decide if the problem is a

maximization or a minimization problem and the coefficients of each decision variable.

4) Identify the constraints

- system constraints- more than one variable
- individual constraints- one variable
- non-negativity constraints.

Q(1) Check Your Progress Question

1. A firm that assembles computers computer equipment is about to start production of two new micro computers. Each type of microcomputer will require assembly time, inspection time, and storage space. The amount of each of there resources that can be devoted to the production of micro computers is limited. The mgnager. Of the firm would like to determine the quantity of each micro computer to produce in order to maximize the profit generated by sales of these micro computers.

Additional information

In order to develop a suitable model of the problem, the manager has met with the design and manufacturing personnel. As a result of these meetings, the manager has obtained the following information:

	<u>Type 1</u>	<u>Type 2</u>
Profit per unit	\$ 60	\$ 50
Assembly time per unit	4hrs	10hrs
Inspection time per unit	2hrs	1hr
Storage space per unit	3 cubic ft	3 cubic ft

The manager also has acquired information on the available company resources. These (weekly) amounts are:

<u>Resource</u>	<u>Resource available</u>
Assembly time	100hrs
Inspection time	22hrs
Storage space	39cubic feet

The manager has also met with the firm's marketing manager and learned that demand for the micro computers was such that whatever combination of these two types of micro computers is produced, all of the output can be sold.

Answer for the above check your progress question

Step 1: Problem definition

- To determine the no. of two types of microcomputers to be produced (and sold) per week so as to maximize the weekly profit given the restrictions.

Step 2: Variable representation

- Let X_1 and X_2 be the weekly quantities of type 1 and type 2 microcomputers respectively.

Step 3: Develop the objective function

$$\text{Maximize or } Z \text{ max} = 60X_1 + 50X_2$$

Step 4: Constraint identification

- System constraints: $4X_1 + 10X_2 \leq 100\text{hrs}$ Assembly
 $2X_1 + X_2 \leq 22\text{hrs}$ Inspective
 $3X_1 + 3X_2 \leq 39 \text{ cub Feet}$ Storage
- Individual constraintsNo
- Non-negativity constraint $X_1, X_2 \geq 0$

In summary, the mathematical model for the microcomputer problem is:

$$Z \text{ max} = 60X_1 + 50X_2$$

Subject to

$$4X_1 + 10X_2 \leq 100$$

$$2X_1 + X_2 \leq 22$$

$$3X_1 + 3X_2 \leq 39$$

$$X_1, X_2 \geq 0$$

Q(2) Check your progress

2. An electronics firm produces three types of switching devices. Each type involves a two-step assembly operation. The assembly times are shown in the following table:

	Assembly time per unit (minutes)	
	<u>Station 1</u>	<u>Station 2</u>
Model A	2.5	3.0
Model B	1.8	1.6
Model C	<u>2.0</u>	<u>2.2</u>
	7.5hr	7.5hr

Each workstation has a daily working time of 7.5 hrs. The manager want to obtain the greatest possible profit during the next five working days. Model A yields a profit of br. 8.25 per unit, Model B a profit of br 7.5 per unit and model C a profit of Br 7.8 per unit. Assume that the firm can sell all it produces. During this time, but it must fill outstanding orders for 20 units of each model type.

Required. Formulate the linear programming model of this problem.

Solution for check your progress question number 2.

Step 1: Problem definition: to determine the number of three types of searching devices to be produced to be produced and sold for the next 5 days (working) so as to maximize the 5 days profit.

2. Variable representation

Let X_1 , X_2 , and X_3 be the number of model A, B and C sketching devices to be produced and sold.

3. Develop objective function

$$Z \text{ max} = 8.25X_1 + 7.50X_2 + 7.80X_3$$

4.Constraint identification

$$\begin{array}{rcl}
 2.5X_1 + 1.8X_2 + 2.0X_3 & \leq & 450 \text{minutes...Assembly time station 1} \\
 3.0X_1 + 1.6X_2 + 2.2X_3 & \leq & 450 \text{minutes.... Assembly time station2} \\
 \left. \begin{array}{l} X_1 \geq 20 \text{ Model A} \\ X_2 \geq 20 \text{Model B} \\ X_3 \geq 20 \text{....Model C} \end{array} \right\} & & \text{Individual Constraints} \\
 X_1, X_2, X_3 \geq 0 & & \text{....no negativity}
 \end{array}
 \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{System Constraints}$$

Summary

$$Z \text{ max} = 8.25X_1 + 7.50X_2 + 7.8X_3$$

$$\text{Subject to: } 2.5X_1 + 1.8X_2 + 2X_3 \leq 450$$

$$3X_1 + 1.6X_2 + 2.2X_3 \leq 450$$

$$X_1 \geq 20$$

$$X_2 \geq 20$$

$$X_3 \geq 20$$

$$X_1, X_2, X_3 \geq 0$$

3.4 SOLUTION APPROACHES TO LINEAR PROGRAMMING PROBLEMS

There are two approaches to solve linear programming problems.

1. The graphic solution method
2. The algebraic solution/ simplex algorithm

3.4.1 The Graphic solution Method

It's a relatively straight forward method for determining the optimal solution to certain linear programming problems.

It gives us a clear picture.

This method can be used only to solve problems that involve two decision variables. However, most linear programming applications involve situations that have more than two decision variables, so the graphic approach is not used to solve these.

Example 1. Solving the micro-computer problem with graphic approach.

$$Z \text{ max: } 60X_1 + 50X_2$$

$$\text{S.t: } AX_1 + 10X_2 \leq 100$$

$$2X_1 + X_2 \leq 22$$

$$3X_1 + 3X_2 \leq 39$$

$$X_1, X_2 \geq 0$$

Steps

1. Plot each of the constraints and identify its region.
2. Identify the common region, which is all area that contains all of the points that satisfy the entire set of constraints.

3. Determine the optional solution-identify the point which lead to maximum benefit or minimum cost.

$$4X_1 + 10X_2 = 100$$

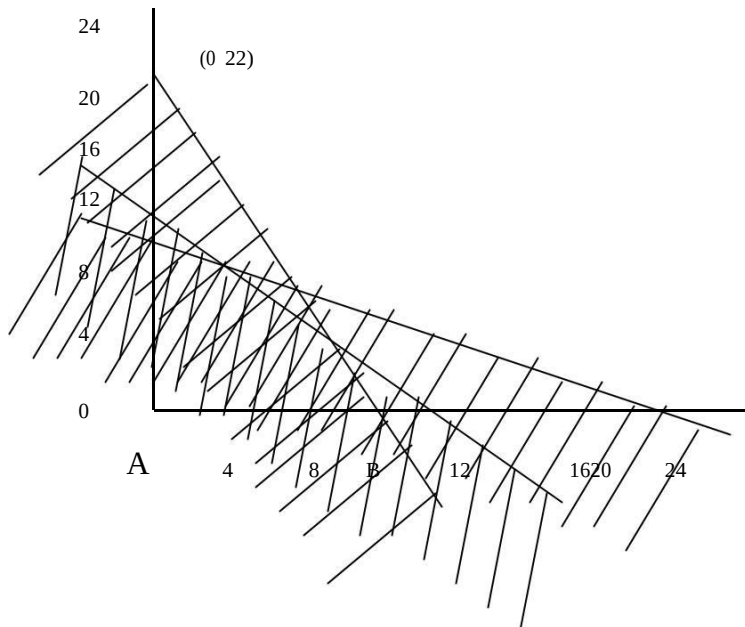
X ₁	0	25
X ₂	10	0

$$2X_1 + X_2 = 22$$

X ₁	0	11
X ₂	22	0

$$3X_1 + 3X_2 = 39$$

X ₁	0	13
X ₂	13	0



Region ABCDE is called feasible region
point C = ?

$$\begin{aligned} \begin{cases} 2x_1 + x_2 = 22 \\ 3x_1 + 3x_2 = 39 \end{cases} \quad x-3 \\ -6x_1 + 3x_2 = -66 \\ \underline{3x_1 + 3x_2 = 39} \\ -3x_1 = -27 \\ x_1 = -27/-3 = 9 \end{aligned}$$

$$\begin{aligned} 2(9) + x_2 &= 22 \\ x_2 &= 22 - 18 = 4 \end{aligned}$$

$$\begin{aligned} \text{Point D} = \begin{cases} 3x_1 + 3x_2 = 39 \\ 4x_1 + 10x_2 = 100 \end{cases} \quad x-4 \\ -12x_1 + 12x_2 = -156 \\ \underline{12x_1 + 30x_2 = 300} \\ 18x_2 &= 144 \\ x_2 &= 8 \\ 3x_1 + 3(8) &= 39 \\ 3x_1 &= 39 - 24 \\ x_1 &= 15/3 = 5 \end{aligned}$$

To identify the maximum (minimum) value we use the corner point approach or the extreme point approach. The corner point/ extreme point approach has one theorem. It states that:

For problems that have optional solutions, a solution will occur at an extreme, or corner point. Thus if a problem has a single optional solution, it will occur at a corner point. If it has multiple optional solutions, at least one will occur at a corner point consequently, in searching for an optional solution to a problem, we need any consider the extreme points because one of those must be optional. Further, determining the value of the objective function at each corner point, we could identify the optional solution by selecting the corner point that has the best value (i.e. maximum or minimum, depending on the optimization case) of the objective function. Extreme points represent interactions of constraints.

Determine the values of the decision variables at each corner point. Some times, this can be done by inspection (observation) and sometimes by simultaneous equation.

Substitute the value of the decision variables at each corner point into the objective function to obtain its value at each corner point.

After all corner points have been so evaluated, select the one with the highest or lowest value depending on the optimization case.

Corner Points	Coordinates		How determined?	Value of the objective function $Z = 60X_1 + 50X_2$
	X_1	X_2		
A	0	0	observation	0 br
B	11	0	observation	660 br
C	9	4	Simultaneous equation	740 br
D	5	8	Simultaneous equation	700 br
E	0	10	Observation	500 br

BASIC SOLUTION

$$X_1 = 9 \quad X_2 = 4 \quad Z = 740 \text{ Br.}$$

After we have got the optimal solution, we have to substitute the value of the decision variables into the constraints and check whether all the resources available are used or not. If there is any unused resources we can use it for any other purpose. The amount of unused resource is known as **slack**- the amount of a scarce resource that is unused by a given solution. The slack can range from zero, for a case in which all of a particular resource is used, to the original amount of the resource that was available (i.e. none of it is used.)

Computing the amount of slack

	<u>Amount used</u>	<u>Originally available</u>	<u>unused Amount of slack (Available-used)</u>
Constraint	$X_1 + 9X_2 = 4$		
Assembly	$4(9) + 10(4) = 76$	100	$100 - 76 = 24 \text{ hrs}$

Impective	$2(9) 9+ 1(4) = 22$	22	$22 - 22 = 0 \text{ hr}$
Storage	$3(9) + 3(4) = 39$	39	$39 - 39 = 0 \text{ hr}$

Constraints that have no slack are sometimes referred to as binding constraints since they limit or bind the solution. In the above cases, inspection time and storage space are binding constraints, while assembly time has slack.

Knowledge of unused capacity can be useful for planning. A manager may be able to use the remaining assembly time for other products, or, perhaps to schedule equipment maintenance, safety seminars, training sermons or other activities

Interpretation: The company is advised to produce 9 units of type 1 micro computer and 4 units of type 2 micro computers per week to maximize its early profit to Br. 740, and in doing so the company would be left with unused resource of 24 assembly hrs which can be used for other purposes.

Example 2: Solving the diet problem with graphic approach.

$$C \text{ min} = 5X_1 + 8X_2$$

$$10X_1 + 30X_2 \geq 140$$

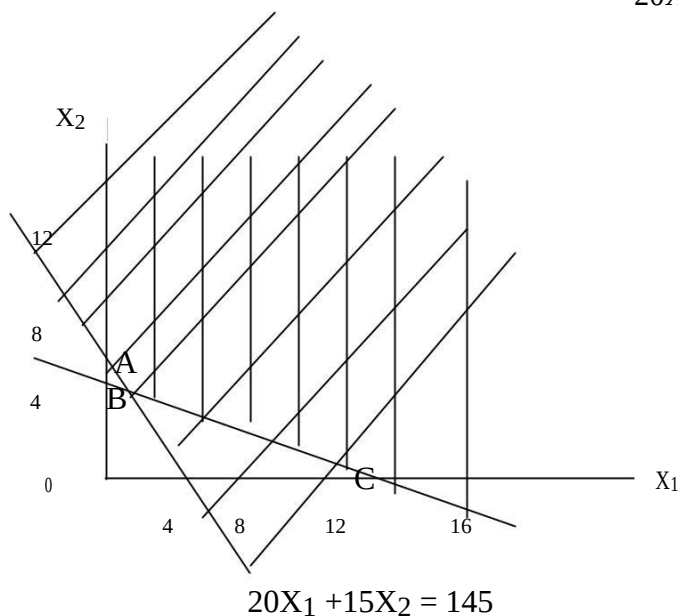
$$20X_1 + 15X_2 \geq 145$$

$$X_1, X_2 \geq 0$$

$$10X_1 + 30X_2 = 140$$

$$\begin{array}{c|cc} X_1 & 0 & 14 \\ \hline X_2 & 14/3 & 0 \end{array}$$

$$20X_1 + 15X_2 = 145 \quad \begin{array}{c|cc} X_1 & 0 & +25 \\ \hline X_2 & 9.67 & 0 \end{array}$$



Value of obj

Points	coordinates		How determined	Function $c_{min} = 5X_1 + 8X_2$
	X_1	X_2		
A	0	9.67	Observation	77.3 br.
B	5	3	Simul. equ ⁿ .	49 br.
C	14	0	Observation	70 br.

Basic solution $X_1 = 5$ pounds $X_2 = 3$ pounds $C = 49$ br.

Interpretation to make the diet the minimum cost of br 49 we have to purchase 5 pounds of type 1 food and 3 pounds type 2 food.

If there is a difference between the minimum required amount and the optimal solution, we call the difference surplus; that is: surplus is the amount by which the optimal solution causes a $\geq$ constraint to exceed the required minimum amount. It can be determined in the same way that slack can: substitute the optimum values of the decision variables into the left side of the constraint and solve. The difference between the resulting value and the original right-hand side amount is the amount of surplus.

Surplus can potentially occur in a $\geq$ constraint.

3.4.2 The Simplex Algorithm/ Algebraic Solution Method

The simplex method is an iterative technique that begins with a feasible solution that is not optimal, but serves as a starting point. Through algebraic manipulation, the solution is improved until no further improvement is possible (i.e. until the optional solution has been identified.) Each iteration moves one step closer to the optional solution.

The optimal solution to a linear programming model will occur at an extreme point of the feasible solution space. This is true even if a model involves more than two variables; optimal solutions will occur at these point of the feasible solution space; some will be outside of the feasible solution space. Hence, not every solution will be a feasible solution. Solutions which represent infasseetwim of constraints are called basic solutions; those which also satisfy all of the constraints, including the non-negativity constraints, are called basic feasible solutions. The simplex method is an algebraic procedure for systematically examining basic feasible solutions. If an optimal solution exists, the

simplex method will identify it. # of basic solution $n + mCm \Rightarrow$ not all basic solutions are feasible.

The simplex procedure for a maximization problem with all $\leq$ constraints consists of the following steps.

1. Write the LPM in a Standard form: When all of the constraints are written as equalities, the LP program is said to be in a standard form. We convert the LPM in to a standard form by applying the slack variables, s , which carries a subscript that denotes which constraint it applies to. For example, s_1 refers to the amount of slack in the first constraint, S_2 to the amount of slack in the second constraint, and so on. When slack variables are introduced to the constraints, they are no longer inequalities b/c the slack variable accounts, they become equalities. Further more, every variable in a model must be represented in the objective function. However, since slack does not provide any real contribution to the objective, each slack variable is assigned a coefficient of zero in the objective function.

Slack = Requirement – Production, scruples – Production – Requirement

Taking the microcomputer problem, its standard form is as follows

$\begin{aligned} Z \max &= 60X_1 + 50X_2 \\ 4X_1 + 10X_2 &\leq 100 \\ 2X_1 + X_2 &\leq 22 \Rightarrow \\ 3X_1 + 3X_2 &\leq 39 \\ X_1, X_2 &\geq 0 \end{aligned}$	$\begin{aligned} Z \max &= 60X_1 + 50X_2 + 0S_1 + 0S_2 + 0S_3 \\ 4X_1 + 10X_2 + S_1 &= 100 \\ 2X_1 + X_2 + S_2 &= 22 \\ 3X_1 + 3X_2 + S_3 &= 39 \\ \text{All variables} &\geq 0 \\ (X_1, X_2, S_1, S_2, S_3) &\geq 0 \end{aligned}$
--	---

2. Develop the initial tableau
 - a. List the variables across the top of the table and write the objective function coefficient of each variables just above it.
 - b. There should be one row in the body of the table for each constraint. List the slack variables in the basis column, one per row.
 - c. In the C_j column, enter the objective function coefficient of zero for each slack variable.
 - d. Compute values for row Z_j .
 - e. Computer values for $C_j - Z_j$.

C_j = Coeff of variable J in the obj function.
b_j = RHSV of constraint i.
A_{ij} – coefficient of variable j in constraint i

		Pivot column					Pivot element	
Solution	Cj	60	50	0	0	0		
Basis		X ₁	X ₂	S ₁	S ₂	S ₃	RHSV	
S1	0	4	10	1	0	0	100	100/4 = 25
S2	0	2	1	0	1	0	22	leaving 22/2 = 11
S3	0	3	3	0	0	1	39	39/3 = 13
Zj		0	0	0	0	0	0	
Cj - Zj		60	50	0	0	0		

Initial feasible solution
 $S_1 = 100$ Obtained by
 $S_2 = 22$ equating to
 $S_3 = 39$ Variables to
 $X_1 = 0$ Zero
 $X_2 = 0$ Decision
 $Z = 0$ Variable

Entering variable

3. Develop subsequent tables

3.1 Identify the entry variable –variable that has a largest positive value in the $C_j - Z_j$ row.

3.2 Identify the leaving variable –using the constraint coefficient or substitution rates in the entering variable column divide each one into the corresponding quantity value. However do not divide by a zero or negative value. The smallest non negative ratio that results indicate which variable will leave the solution

4. Find unique vectors for the new basic variable using row operations on the pivot element.

$-1/2 R_{21} - 4R_{2new} + R_{1old}$, $-3R_{2new} + R_{3old}$

Solution	Cj	60	50	0	0	0	
Basics		X ₁	X ₂	S ₁	S ₂	S ₃	RHSV
S1	0	0	8	1	-2	0	56
X1	60	1	1/2	0	1/2	0	11
S3	0	0	3/2	0	-3/2	1	6
Zs		60	30	0	30	0	660
Cj - Zj		0	20	0	-30	0	

Cj = b_i/x_j (a_{ij})
 $56/8 = 7$
 $11/1/2 = 22$
 $6/3/2 = 4$

Incoming Variable

Solution basis	Cj	60	50	0	0	0	RHSV
		X ₁	X ₂	S ₁	S ₂	S ₃	
S ₁	0	0	0	1	6	-16/3	24
X ₁	60	1	0	0	1	-1/3	9
X ₂	50	0	1	0	-1	2/3	4
Zj		60	50	0	10	40/3	740
Cj - Zj		0	0	0	-10	-40/3	

Opportunity cost

5. Compute $C_j - Z_j$ row
6. If all $C_j - Z_j$ Values are zeros and negatives, you have reached optimality
7. If this is not the case (step 6), repeat 2 to 5 until you get optional solution.

“A simplex solution in a maximization problem is optional if the $C_j - Z_j$ row consists entirely of zeros and negative numbers (i.e. there are no positive values in the bottom row.)”

Note: The variables in solution all have unit vectors in their respective columns for the constraint equations. Further, note that a zero appears in row $C - Z$ in every column whose variable is in solution, in row $C - Z$ in every column whose variable is in solution, indicating that its maximum contribution to the objective function has been realized.

Example 2

A manufacture of lawn and garden equipment makes two basic types of lawn mowers; a push type and a self propelled model. The push type require 9 minutes to assemble and 2 minutes to package; the self-propelled mover requires 12 minute to assemble and 6 minutes to package. Each type has an engine. The company has 12hrs of assembly time available, 75 engines, and 5hrs of packing time profits are Birr 70 for the self propelled model and br 45 for the push type mower per unit.

Required:

1. a. To determine how many units of each type of mower to produce so as to maximize profit.
- b. Let X_1 be push type mower
 X_2 be self propelled mower

c. Determine the objective function

$$Z \max = 45X_1 + 70X_2$$

d. Identify constraints

$$9X_1 + 12X_2 \leq 720 \text{ minutesassembly time}$$

$$2X_1 + 6X_2 \leq 300 \text{ minutespacking time}$$

$$X_1 + X_2 \leq 75 \text{ engineEngines}$$

$$X_1, X_2 \geq 0$$

In summary

$$Z \max = 45X_1 + 70X_2$$

$$\text{S.t. } 9X_1 + 12X_2 \leq 720$$

$$2X_1 + 6X_2 \leq 300$$

$$X_1 + X_2 \leq 75$$

$$X_1, X_2 \geq 0$$

2. a. Write the Lpm in a standard form

$$Z \max = 45X_1 + 70X_2 + 0S_1 + 0S_2 + 0S_3$$

$$\text{S.T. } 9X_1 + 12X_2 + S_1 = 720$$

$$2X_1 + 6X_2 + S_2 = 300$$

$$X_1 + X_2 + S_3 = 75$$

$$X_1, X_2, S_1, S_2, S_3 \geq 0$$

b. Develop the initial tableau

Solution basis	C	45	70	0	0	0	RHSV
		X1	X2	S1	S2	S3	
S1	0	9	12	1	0	0	720
S2	0	2	6	0	1	0	300
S3	0	1	1	0	0	1	75
Z		0	0	0	0	0	0
C - Z		45	70	0	0	0	

$$Q_{ij} = B_i/a_{ij}$$

$$720/12 = 60$$

$$300/6 = 50 - \text{Leaving}$$

$$75/1 = 75$$

entering

c. Develop the subsequent tableaus 1/6R2,

$$-1R_2\text{new} + R_3, -12R_2\text{new} + R_1$$

	C	45	70	0	0	0	
Solution Basis		X1	X2	S1	S2	S3	RHSV
S1	0	5	0	1	-2	0	120
X2	70	1/3	1	0	1/6	0	50
S3	0	2/3	0	0	-1/6	1	25
Z		70/30	70	0	70/6	0	3500
C - Z		65/3	0	0	-70/6	0	

Entering

$1/5R_1, -1/3 R_{1\text{new}} + R_{2\text{old}}, -2/3R, \text{New} + R_3 \text{ old}$

	C	45	70	0	0	0		Optimal solution
Solution Basis		X1	X2	S1	S2	S3	RHSV	
X1	45	1	0	1/5	-2/5	0	24	X1 = 24units
X2	70	0	1	-1/15	3/10	0	42	X2 = 42units
S3	0	0	0	-2/15	1/10	1	9	S1 = 0
Z		45	70	13/3	3	0	4020	S2 = 0
C - Z		0	0	-13/3	-3	0		S3 = 9 engine
								Z = <u>4020</u>

$$9X_1 + 12X_2 = 720$$

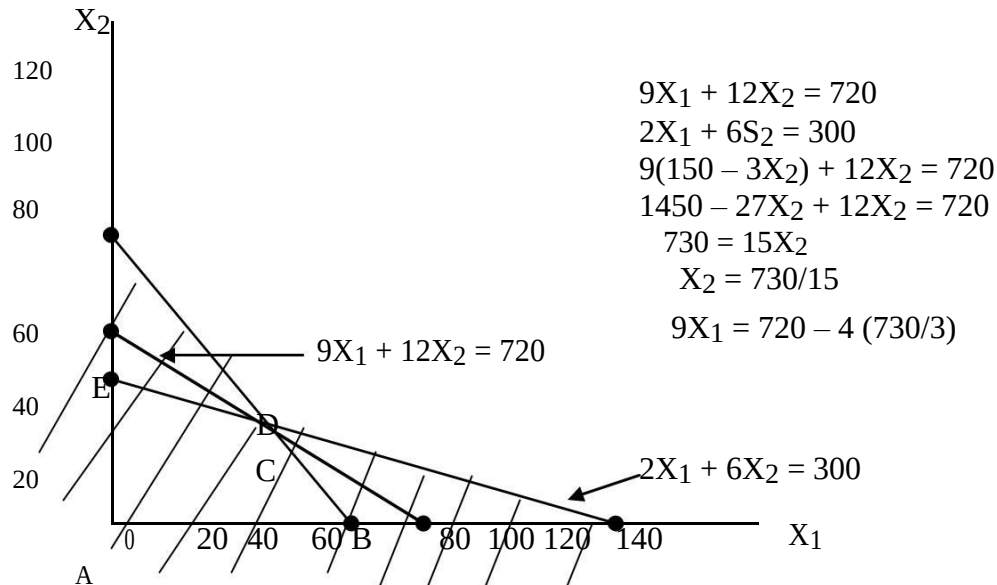
X1	0	80
X2	60	0

$$2X_1 + 6X_2 = 300$$

X1	0	150
X2	50	0

$$X_1 + X_2 = 75$$

X1	0	75
X2	75	0



Points	Coordinates		How determinable?	Profit
	X_1	X_2		
A	0	0	observation	$45(0) + 70(0) = 0$
B	75	0	observation	$45(75) + 70(0) = 3375$
C	60	15	simultaneous	$45(60) + 70(15) = 3750$
D	24	42	simultaneous	$45(24) + 70(42) = 4020$
E	0	50	observation	$45(0) + 70(50) = 3500$

$$X_1 = 24$$

$$b) 2(24) + 6(42) = 300$$

$$X_2 = 42$$

$$Z = \text{br. } 4020$$

$$c) 24 + 42 = 75$$

$$\underline{\underline{66}}$$

Constraints

$$a) (24) + 12(42) = 720$$

$$75 - 66 = 9$$

$$\underline{\underline{S_3 = 9}}$$

Interpretation. The company is advised to produce 24 units of push type mower and 40 units of self-propelled mowers so as to realize a profit of Br. 4020. in doing so, the company would be left with unused resource of engine which can be used for other purposes.

EXERCISES

1. A firm manufactures three products which must be processed through some of or all four departments. The table below indicates the number of hours a unit of each product requires in the different departments and the number of pounds of raw materials required. Also listed are the cost per unit, selling price, and weekly capacities of both work-hours and raw materials. If the objective is to maximize total weekly profit, formulate the linear programming model. (Only the model)

	Product A	Product B	Product C	Weekly Availability
Department 1	3	4	2	120 hours
Department 2		2	2	100 hours
Department 3	4	1		80 hours
Department 4	2	3	6	150 hours
Pounds of raw Material per unit	5	4	3	250 pounds
Selling price	\$ 50	\$ 60	\$ 65	
Cost / unit	41	40	43	

2. Solve the following linear programming problem

(a) (Use simplex method)

$$Z \text{ max. } 40X_1 + 32X_2$$

Subject to

$$40X_1 + 20X_2 \leq 600$$

$$4X_1 + 10X_2 \leq 100$$

$$2X_1 + 3X_2 \leq 38$$

$$X_1, X_2 \geq 0$$

(b) C min. : $50X_1 + 80X_2$

Subject to

$$20X_1 + 30X_2 \geq 1400$$

$$10X_1 + 40X_2 \geq 1200$$

$$X_1, X_2 \geq 0$$

(Use the graphic method)

(c) Z max: $X_1 + 1.2X_2 +$

$2X_3$ Subject to

$$X_1 + 2X_2 \leq 150$$

$$X_1 + 2X_3 \leq 150$$

$$2X_1 + X_2 \leq 80$$

$$2X_1 + 3X_2 + X_3 \leq 225$$

$$X_1, X_2, X_3 \geq 0$$

(Use the simplex method)

Solutions

(1) * Problem: Determine the number of three products (A, B, & C) that must be produced and sold in order to maximize the total weekly profit of the firm, given the different limiting factors.

* Let, X_1 = the number of product A, that must be produced & sold

X_2 = the number of product B, that must be produced & sold X_3

= the number of product C, that must be produced & sold

* Objective function: profit maximization

$$Z \text{ max} = 9X_1 + 20X_2 + 22X_3$$

* Constraints:

$$\rightarrow \text{Production time constraints: } 3X_1 + 4X_2 + 2X_3 \leq 120$$

$$\rightarrow \text{Production time constraints } 2X_2 + 2X_3 \leq 100$$

$$\rightarrow \text{Production time constraints } 4X_1 + X_2 \leq 80$$

$$\rightarrow \text{Production time constraints } 2X_1 + 3X_2 + 6X_3 \leq 150$$

→ Raw material constraints: $5X_1 + 4X_2 + 3X_3 \leq 250$

* Non-negative restrictions: $X_1, X_2, X_3 \geq 0$

⇒ Therefore, Z max: $9X_1 + 20X_2 + 22X_3$

Subject to:

$$3X_1 + 4X_2 + 2X_3 \leq 120$$

$$2X_2 + 2X_3 \leq 100$$

$$4X_1 + X_2 \leq 80$$

$$2X_1 + 3X_2 + 6X_3 \leq 150$$

$$5X_1 + 4X_2 + 3X_3 \leq 250$$

$$X_1, X_2, X_3 \geq 0$$

(2) a. * Problem: determining the number of bentwood rocking chairs and bentwood coffee tables that must be manufactured in order to maximize the total profit, given the different limiting factors.

* Let, X_1 = the number of bentwood rocking chairs that must be manufactured & sold

X_2 = the number of bentwood coffee tables that must be manufactured & sold

* Objective function: profit maximization

$$Z \text{ max} = 40X_1 + 32X_2$$

* Constraints

→ Raw material const. (Rosewood): $40X_1 + 20X_2 \leq 600$

→ Wood working hour const. $4X_1 + 10X_2 \leq 100$

→ Finishing hour constraint $2X_1 + 3X_2 \leq 38$

* Non-negative restrictions : $X_1, X_2 \geq 0$

⇒ Therefore, Z max = $40X_1 + 32X_2$

Subject to:

$$40X_1 + 20X_2 \leq 600$$

$$4X_1 + 10X_2 \leq 100$$

$$2X_1 + 3X_2 \leq 38$$

$$X_1, X_2 \geq 0$$

Standard form

$$Z \text{ max} = 40X_1 + 32X_2 + 0S_1 + 0S_2 +$$

$0S_3$ Subject to:

$$40X_1 + 20X_2 + S_1 = 600$$

$$4X_1 + 10X_2 + S_2 = 100$$

$$2X_1 + 3X_2 + S_3 = 38$$

$$X_1, X_2, S_1, S_2, S_3 \geq 0$$

Initial Table

Soln.	Cj	40	32	0	0	0	Quantity	
bases		X ₁	X ₂	S ₁	S ₂	S ₃	(RHSV)	Q/X ₁
S ₁	0	40	20	1	0	0	600	15
S ₂	0	4	10	0	1	0	100	25
S ₃	0	2	3	0	0	1	38	19
Zj		0	0	0	0	0	0	
Δj = Cj – Zj		40	32	0	0	0		

$$1/40 \text{ R}_1 / -4\text{R}_1 \text{ new} + \text{R}_2 \text{ old} // -2\text{R}_1 \text{ new} + \text{R}_3 \text{ old}$$

Soln.	Cj	40	32	0	0	0	quantity	
bases		X ₁	X ₂	S ₁	S ₂	S ₃	(RHSV)	Q/X ₂
X ₁	40	1	1/2	1/40	0	0	15	30
S ₂	0	0	8	-1/10	1	0	40	5
S ₃	0	0	2	-1/20	0	1	8	4
Zj		40	20	1	0	0	600	
Δj = Cj- Zj		0	12	-1	0	0		

$$1/2 \text{ R}_3 / -1/2\text{R}_3 \text{ new} + \text{R}_1 \text{ old} / -8\text{R}_3 \text{ new} + \text{R}_3 \text{ old}$$

Soln.	Cj	40	32	0	0	0	(Quantiy)
bases		X ₁	X ₂	S ₁	S ₂	S ₃	(RHSV)
X ₁	40	1	0	3/80	0	-1/4	13
S ₂	0	0	0	1/10	1	-4	8
X ₂	32	0	1	-1/40	0	1/2	4
Zj		40	32	7/10	0	6	648
Δj = cj - zj		0	0	-7/10	0	-6	

⇒ Since, the numbers in $\Delta j = c_j - z_j$ row are all 0 & Negative it shows that this solution is optimal solution.

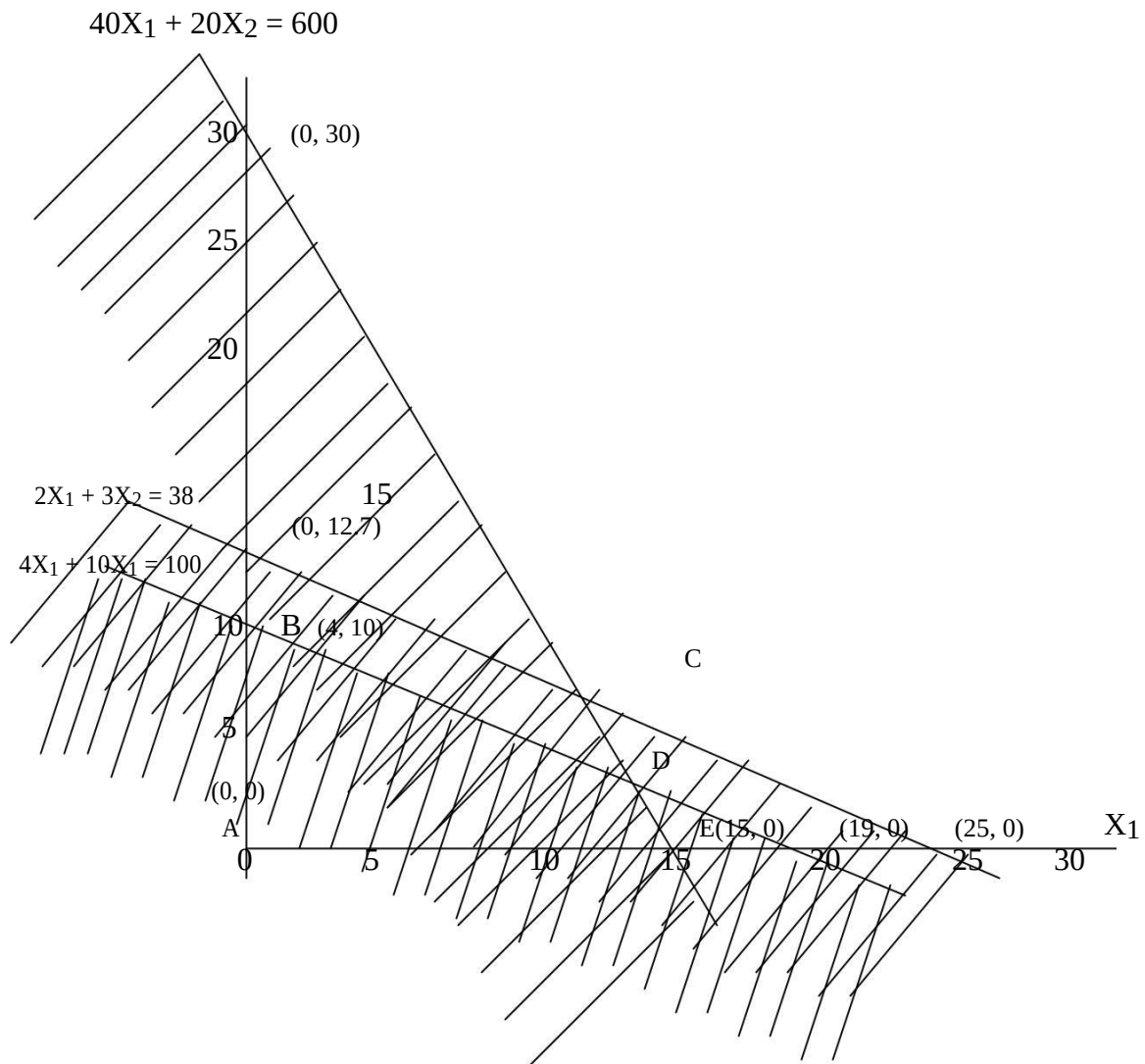
$$\underline{\underline{X_1 = 13}}, \quad \underline{\underline{X_2 = 4}}, \quad \underline{\underline{S_2 = 8}} \text{ (unused) hours} \quad \therefore Z \text{ max} = \$ 648$$

⇒ Proof for Question No. 2 by Graphic Method

$$* 40X_1 + 20X_2 = 600 \longrightarrow \begin{array}{c|c|c} X_1 & 0 & 15 \\ \hline X_2 & 30 & 0 \end{array}$$

$$* 4X_1 + 10X_2 = 100 \longrightarrow \begin{array}{c|c|c} X_1 & 0 & 25 \\ \hline X_2 & 10 & 0 \end{array}$$

$$* 2X_1 + 3X_2 = 38 \longrightarrow \begin{array}{c|c|c} X_1 & 0 & 19 \\ \hline X_2 & 12.7 & 0 \end{array}$$



C.P	X ₁	X ₂		Value of O.F 40X ₁ + 32X ₂		
A	0	0	Observation	0		
B	6	10	Observation	320		
C	10	6	Elimination	592		
	D	13	4	Elimination	648	*
	E	15	0	Observation	600	

* Point C

$$\begin{array}{rcl}
 & \left\{ \begin{array}{l} 4X_1 + 10X_2 = 100 \\ 2X_1 + 3X_2 = 38 \end{array} \right. & \\
 -2 & & \\
 \hline
 & 4X_1 + 10X_2 = 100 & \\
 & -4X_1 - 6X_2 = -76 & \\
 \hline
 & 4X_2 = 24 & \\
 & 4 \quad \quad 4 & \\
 & \underline{X_2 = 6} & \\
 & 4X_1 + 10(6) = 100 & \\
 & 4X_1 + 60 = 100 & \\
 & 4X_1 = 100 - 60 & \\
 & \underline{4X_1 - 40} & \\
 & 4 \quad \quad 4 \quad \quad \underline{\underline{X_1 = 10}} &
 \end{array}$$

Point D

$$\begin{array}{rcl}
 & \left\{ \begin{array}{l} 40X_1 + 20X_2 = 600 \\ 2X_1 + 3X_2 = 38 \end{array} \right. & \\
 -20 & & \\
 \hline
 & 40X_1 + 20X_2 = 600 & \\
 & -40X_1 - 60X_2 = -760 & \\
 \hline
 & -40X_2 = -160 & \\
 & -40 \quad \quad -40 & \\
 & \underline{\underline{X_2 = 4}} & \\
 & & \\
 & 40X_1 + 20(4) = 600 & \\
 & 40X_1 + 80 = 600 & \\
 & 40X_1 = 600 - 80 & \\
 & \underline{40X_1 = 520} & \\
 & 40 \quad \quad 40 & \\
 & \underline{\underline{X_1 = 13}} &
 \end{array}$$

Interpretation

It is advisable for the company to produce 13 units of bentwood rocking chairs and 4 units coffee tables and maximize its profit to birr 648, by doing so the company may be left with 8 unused wood working hours, therefore, the company can use these unused hours for other purposes.

- (b) * Problem: determine the number of two types of that must be produced by the two machines (M I & M II) in order to minimize costs of operating the machines.

* Let X_1 = the number of tiers that must be produced by M I

X_2 = the number of tiers that must be produced by M II

* Objective function : cost Minimization

$$Z \text{ min} = 50X_1 + 80X_2$$

* Constraints:

→ Grade A tiers constraints: $20X_1 + 30X_2 \geq 1,400$

→ Grade B tiers constraints : $10X_1 + 40X_2 \geq 1,200$

* Non-negative restrictions $X_1, X_2 \geq 0$

⇒ Therefore, $Z \text{ min} = 50X_1 + 80X_2$

Subject to:

$$20X_1 + 30X_2 \geq 1,400$$

$$10X_1 + 40X_2 \geq 1,200$$

$$X_1, X_2 \geq 0$$

⇒ Solution for Question No. 3 Graphic

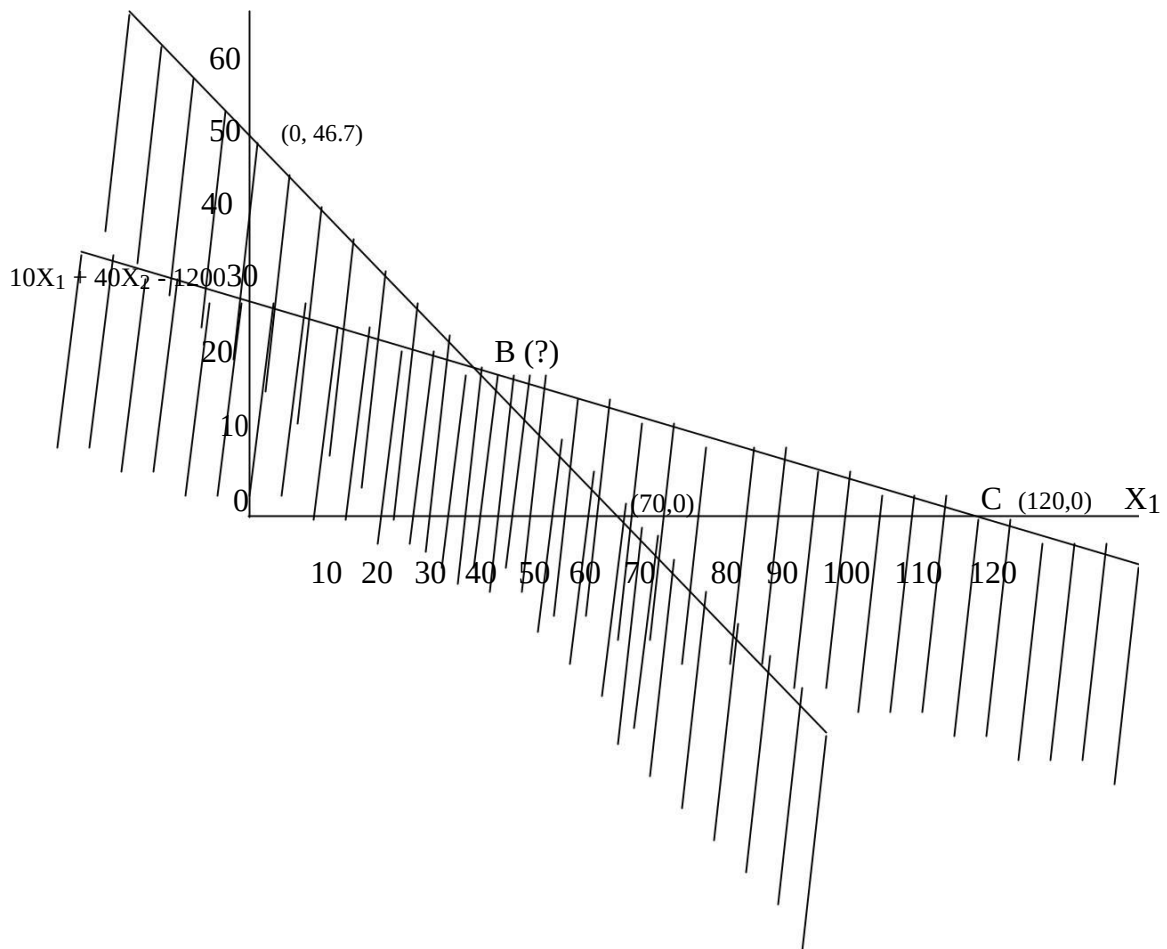
Method * $20X_1 + 30X_2 = 1,400$

X_1	0	70
X_2	$46 - 7$	0

* $10X_1 + 40X_2 = 1,200$

X_1	0	120
X_2	30	0

$$20X_1 + 30X_2 = 1400 \quad X_2$$



Corner points	X_1	X_2	How	Value of objective $F: 50X_1 + 80X_2$	
A	0	46.7	Obser.	3736	
B	40	20	Elimin.	3600	*
C	120	0	Obser.	6000	

* Point B

$$20X_1 + 30X_2 = 1400$$

$$-2 \quad 10X_1 + 40X_2 = 1200$$

$$20X_1 + 30X_2 = 1400$$

$$-20X_1 + 80X_2 = -2400$$

$$-50X_2 = -1000$$

$$50 \quad 50$$

$$\underline{\underline{X_2 = 20}}$$

$$10X_1 + 40(20) = 1200$$

$$10X_1 + 800 = 1200$$

$$10X_1 = 1200 - 800$$

$$10 \quad \frac{1}{2} R_2 / -R_2 \text{new} + R_4 \text{ old}$$

⇒ in order to Minimize

costs of operating the Machines

M I has to produce 40 tires

M II has to produce 20 tires

$$Z \text{ min} = \underline{\underline{3600}}$$

Sol ⁿ bases	C _j	1 X ₁	1.2 X ₂	2 X ₃	0 S ₁	0 S ₂	0 S ₃	0 S ₄	Quantity (RHSV)	Q/x ₂
S ₁	0	1	2	0	1	0	0	0	150	∞ 75
X ₃	2	1/2	0	1	0		1/2	0	75	75 ∞
S ₃	0	2	1	0	0	0	1	0	80	∞ 80
S ₄	0	3/2	3	0	0	-1/2	0	1	150	225 50 →
Z _j		1	0	2	0	1	0	0	150	
Δ j = C _j - Z _j		0	1.2	0	0	-1	0	0		

Sol ⁿ bases	C _j	1/3 R4 / -R4 New + R3 old / -2R4 new + R1 old							Quantity (RHSV)	
		1 X1	1.2 X2	2 X3	0 S1	0 S2	0 S3	0 S4		
S1	0	0	0	0	1	1/3	0	-2/3	150	
X3	2	1/2	0	1	0	1/2	0	0	75	
S3	0	1	0	0	0	1/6	1	-1/3	30	
X2	1.2	1/2	1	0	0	-1/6	0	1/3	50	
Z _j		1.6	1.2	2	0	0.8	0	0.4	210	
Δ _j = C _j - Z _j		-0.6	0	0	0	-0.8	0	-0.4		

⇒ since there is no positive number in $A_j = C_j - Z$ row we can suggest that we reach the optimum solution therefore, $X_2 = 50$

$$X_3 = 75$$

$$S_1 = 50$$

$$S_3 = 30$$

$$\text{and } Z_j = 210$$

UNIT 4: MATHEMATICS OF FINANCE

Contents

- 4.0 Aims and Objectives
- 4.1 Introduction
- 4.2 Interests
 - 4.2.1 Simple Interest
 - 4.2.2 Compound Interest
- 4.3 Effective Rate
- 4.4 Annuities
 - 4.4.1 Ordinary Annuity
 - 4.4.2 Sinking of Fund
 - 4.4.3 Amortization
 - 4.4.4 Mortgage
- 4.5 Exercise/Problems
- 4.6 Solution to the Exercise / Problem

4.0 OBJECTIVES

After studying this unit, you should be able to:

- tell how simple interest differ from compound interest;
- use annuity formula to solve business and economics problems which involve finance;
- analyze and solve mortgage problems;
- analyze future and present value of money (sum of money and/or an annuity) and
- solve sinking fund and amortization problems.

4.1 INTRODUCTION

The basic concept of mathematics of finance is that money has time value which is described either as present value or future. **Present value** is the value of money today;

future value is the value of money at some point in the future. The difference between money now and the same money in the future is called interest. Interest has a wide spread influence over decisions made by businesses and every one of us in our personal lives. Therefore, the basic objective of this unit is to discuss interest rates and their effects on the value of money. Specifically, it covers simple interest, compound interest, annuity and mortgage problems.

4.2 INTERESTS

Interest is the price paid for the use of a sum of money over a period of time. It is a fee paid for the use of another's money, just rent is paid for the use of another's house. A savings institution (Banks) pay interest to depositors on the money in the savings account since the institutions have use of those funds while they are on deposit. On the other hand, a borrower pays interest to a lending agent (bank or individual) for use of the agent's fund over the term of the loan.

Interest is usually computed as percentage of the principal over a given period of time. This is called **interest rate**. Interest rate specifies the rate at which interest accumulates per year through out the term of the loan. The original sum of money that is lent or invested/ borrowed is called the **principal**.

Interests are of two types: simple interest and compound interest. In the first part of this unit we shall explore these two concepts.

4.2.1 Simple Interest

If interest is paid on the initial amount of money invested or borrowed only and not on subsequently accrued interest, it is called **simple interest**. The sum of the original amount (principal) and the total interest is the **future amount or maturity value or in short amount**. Simple interest generally used only on short-term loans or investments – often of duration less than one year. Simple interest is given by the following formula.

$$I = prt \quad \dots\dots\dots (1)$$

Where: I = Simple interest
P = principal amount

r = Annual simple interest rate

t = time in years, for which the interest is paid

If any three of the four variables are given, you can solve for the fourth (unknown variable) and their relationship is as follows:

$$\text{Amount (A)} = P + I$$

$$= P + \text{Prt. factor out the common term } P$$

$$= P (1 + rt) \dots\dots\dots (2)$$
$$\text{A}$$

$$P = I/rt \text{ or } P = \frac{A}{1+rt} \dots\dots\dots (3)$$

$$r = I/pt \dots\dots\dots (4)$$

$$t = I/pr \dots\dots\dots (5)$$

Example 1

Ato Kassahun wanted to buy TV which costs Br. 10, 000. He was short of cash and went to Commercial Bank of Ethiopia (CBE) and borrowed the required sum of money for 9 months at an annual interest rate of 6%. Find the total simple interest and the maturity value of the loan.

Solution:

$$p = \text{Br. } 10,000$$

$$t^{**} = 9 \text{ months} = 9/12 = \frac{3}{4} \text{ year}$$

$$r = 6\% \text{ per year} = 0.06$$

$$I = ? \quad A = ?$$

$$\text{Interest (I)} = \text{Prt}$$

$$= 10,000 \times 0.06 \times \frac{3}{4}$$

$$= \underline{\underline{\text{Br. } 450}}$$

$$A = P + I$$

$$= P (1 + rt)$$

$$= 10,000 (1 + 0.06 \times \frac{3}{4})$$

$$= 10,000 \times 1.045$$

$$= \text{Br. } \underline{\underline{10,450}}$$

The total amount which will have to be repaid to CBE at the end of the 9th month is Br. 10, 450 (the original borrowed amount plus Br. 450 Interest).

**** Note:** It is essential that the time period t and r be consistent with each other. That is if r is expressed as a percentage per year, t also should be expressed in number of years (number of months divided by 12 if time is given as a number of months). If

time is given as a number of days, then $t = \frac{\text{No.of days}}{360 \text{ days}}$. this approach is known as

ordinary interest year method which uses a 360 day years, whereas if we use 365 days years the approach is called **exact time method**.

Example 2

How long will it take if Br. 10, 000 is invested at 5% simple interest to double in value?

Solution

Given: $p = \text{Br.}10,000$

$r = 10\% = 0.10$

$A = \text{Br.}20,000 (2 \times 10,000)$

$t = ?$

$$I = prt \quad \text{Divide both sides of the equation by } pr \text{ and solve for } t.$$

$$t = I^* / pr$$

$$= \frac{10,000}{10,000(0.10)}$$

$$= \underline{10 \text{ Years}}$$

$$I^* = \text{Amount (A)} - \text{principal (p)}$$

$$= 20,000 - 10,000$$

$$= 10,000$$

Therefore it will take 20 years for the principal (Br. 10, 000) to double itself in value if it is invested at 10% annual interest rate.

Example 3

How much money you have to deposit in an account today at 3% simple interest rate if you are to receive Br. 5, 000 as an amount in 10 years?

Solution

$A = \text{Br.}5,000$

$t = 10 \text{ Years}$

$r = 3\% = 0.03$

$P = ?$

$$P = \frac{A}{1+rt}$$

$$= \frac{5,000}{1 + (0.03 \times 10)}$$

$$= \text{Br. } \underline{3,846.15}$$

In order to have Br. 5, 000 at the end of the 10th year, you have to deposit Br. 3846.15 in an account that pays 3% per year.

Example 4.

At what interest rate will Br. 5, 000 yield Br. 2, 000 in 8 years time.

Solution:

$$\begin{aligned}
 P &= \text{Br. } 5,000 & r &= I/pt \\
 & & &= \frac{2,000}{5,000 \times 8} \\
 I &= \text{Br. } 2,000 & &= 0.05 = \underline{\underline{5\%}} \\
 t &= 8 \text{ years} & & \\
 r &= ? & &
 \end{aligned}$$

Example 5.

Find the Interest on Br. 5, 000 at 10% for 45 days.

Solution:

$$\begin{aligned}
 P &= \text{Br. } 5,000 & I &= Prt \\
 t &= 45 \text{ days} = 45/360 \text{ years} & &= 5,000 \times 0.1 \times 45/360 \\
 r &= 10\% & &= \underline{\underline{62.50 \text{ Br.}}} \\
 I &= ? & &
 \end{aligned}$$

Try Your Self:

At what interest rate you should invest Br. 5000, if you want to receive an amount of Br. 7,000 in 8 years time.

Answer = 5%

4.2.2 Compound Interest

If the interest, which is due, is added to the principal at the end of each interest period (such as a month, quarter, and year), then this interest as well as the principal will earn interest during the next period. In such a case, the interest is said to be **compounded**. The result of compounding interest is that starting with the second compounding period, the account earns interest on interest in addition to earning interest on principal during the next payment period. Interest paid on interest reinvested is called **compound interest**.

The sum of the original principal and all the interest earned is the **compound amount**. The difference between the compound amount and the original principal is the compound interest.

The compound interest method is generally used in long-term borrowing unlike that of the simple interest used only for short-term borrowings. The time interval between successive conversions of interest into principal is called the **interest period**, or **conversion period**, or **Compounding period**, and may be any convenient length of time. The interest rate is usually quoted as an annual rate and must be converted to appropriate rate per conversion period for computational purposes. Hence, the rate per compound period (i) is found by dividing the annual nominal rate (r) by the number of compounding periods per year (m):

$$i = r/m$$

Example if $r = 12\%$, i is calculated as follows:

<u>Conversion period (m)</u>	<u>Rate per compound period (i)</u>
1. Annually (once a year) -----	$i = r/1 = 0.12/1 = 0.12$
2. Semi annually (every 6 months) -----	$i = r/2 = 0.12/2 = 0.06$
3. Quarterly (every 3 months) -----	$i = r/4 = 0.12/4 = 0.03$
4. Monthly -----	$i = r/12 = 0.12/12 = 0.01$

Example 1

Assume that Br. 10, 000 is deposited in an account that pays interest of 12% per year, compounded quarterly. What are the compound amount and compound interest at the end of one year?

Solution

$$P = \text{Br. } 10,000$$

$$r = 12\%$$

$$t = 1 \text{ year}$$

m = No. of conversion periods = 4 times per quarter. This means interest will be computed at the end of each three month period and added in to the principal.

$$i = r/m \quad 12\%/4 = 3\%$$

In general, if **p** is the principal earning interest compounded **m** times a year at an annual rate of **r**, then (by repeated use of the simple interest formula, using $i = r/m$, the rate per period, the amount **A** at the end of each period is:

(1) $A = p (1 + i)$compound amount at the end of first period.

If we are interested in determining the compound amount after two periods, it may be computed using the equation:

(2) Compound amount = Compound amount + Interest earned during
 after two periods after one period the 2nd period
 $A = p (1 + i) + [P (1 + i)] (i)$

Factoring P and (1 + i) from both terms of the right side of the equation gives us:

$$\begin{aligned} A &= P (1 + i) (1 + i) \\ &= P (1 + i)^2 \end{aligned}$$

(3) Compound amount = Compound amount + Interest earned
 after three periods after two period during the 3rd period
 $A = P(1 + i)^2 + [P (1 + i)^2] (i)$

Factor out p and (1 + i)² from the terms on the right side of the equation and it gives you:

$$\begin{aligned} A &= P (1 + i)^2 (1 + i) \\ &= P (1 + i)^3 \end{aligned}$$

(4) Compound amount
 after nth period $= P (1 + i)^n$

The compound amount formulas developed so far are summarized below:

1. Compound amount after one period = $p (1 + i)^1$
2. Compound amount after two periods = $p (1 + i)^2$
3. Compound amount after three periods = $p (1 + i)^3$
4. Compound amount after nth periods = $p (1 + i)^n$

$A = P (1 + i)^n$	* Compound amount formula.
-------------------	---------------------------------

Where: A = amount (future value) at the end of n periods.

P = Principal (present value)

i = r/m = Rate per compounding period.

n = mt = total number of conversion periods

t = total number of years

m = number of compounding/ conversion periods

per

Year

r = annual nominal rate of interest

Now let us solve the above problem.

$$A = 10,000 (1.03)^1 = \text{Br. } 10,300 \dots\dots\dots 1^{\text{st}} \text{ quarter}$$

$$A = [10,000 (1.03)] (1.03) = 10,000 (1.03)^2 = 10,609 \dots\dots\dots 2^{\text{nd}} \text{ quarter.}$$

$$A = [10,000 (1.03)^2] (1.03) = 10,000 (1.03)^3 = 10,927.27 \dots\dots\dots 3^{\text{rd}} \text{ quarter.}$$

$$A = [(10,000) (1.03)^3] (1.03) = 10,000 (1.03)^4 = 11,255.088 \dots\dots\dots 4^{\text{th}} \text{ quarter.}$$

In general, the compound amount can be found by multiplying the principal by $(1 + i)^n$.

So for the above problem the amount at the end of the year, using the general formula, is equal to:

$$\begin{aligned} A &= P (1 + i)^n \\ &= 10,000 (1.03)^4 \\ &= \text{Br. } \underline{11,255.088} \end{aligned}$$

$$\begin{aligned} n &= mt = 4 \times 1 = 4 \\ i &= r/m = 12\%/4 = 3\% \end{aligned}$$

Compound Interest = Compound amount – original principal

$$= 11,255.088 - 10,000$$

$$= \text{Br. } \underline{1,255.088}$$

We may evaluate “A” in several different ways. Among the possible alternatives are:

1. Use a hand-held calculator with a Y^x function key. This is the procedure most often used.
2. Using Logarithms

Restate the equation by finding for $\log A$ (or $\ln A$) and then finding the antilog, using either a hand-held calculator with logarithmic functions or a table of logarithms. Let us illustrate this alternative.

$$\begin{aligned}
 A &= 10,000 (1.03)^4 \\
 \log A &= \log 10,000 + \log (1.03)^4 \\
 &= \log 10,000 + 4 \log 1.03 \\
 &= \log 10^4 + 4 \log 1.03 \\
 &= 4 + 4(0.01284) \\
 &= 4 + 0.05135 \\
 \log A &= 4.05135 \\
 A &= \text{Antilog } 4.05135 \\
 &= \text{Br. } \underline{11255.117}
 \end{aligned}$$

Basic rules of logarithm

1. $a^x = b$	Eg. $2^x = 5$
$\log a^x = \log b$	$\log 2^x = \log 5$
$x \log a = \log b$	$x \log 2 = \log 5$
$x = \log b / \log a$	$x = \log 5 / \log 2$
	$x = \underline{2.322}$
2. $ab^x + c = d$	
$ab^x = d - c$	
$b^x = d - c/a$	
$\log b^x = \log \left(\frac{d-c}{a} \right)$ let: $\frac{d-c}{a} = K$	
$X \log b = \log K$	divide both side of the equality by $\log b$.
$X = \frac{\log K}{\log b}$	
Eg. $4(3^x) + 10 = 18$ find the value of X	
$4(3^x) = 18 - 10$	

$$4(3^x) = 8 \quad (\text{Divide by 4})$$

$$3^x = 2$$

$$\log 3^x = \log 2$$

$$X \log 3 = \log 2$$

$$X = \frac{\log 2}{\log 3}$$

$$= \underline{\underline{0.63093}}$$

$$3. \quad X^3 = a$$

$$\log X^3 = \log a$$

$$3 \log x = \log a$$

$$\log x = \frac{\log a}{3} ; \text{ Let } \frac{\log a}{3} = K$$

$$X = \underline{\text{Antilog } K}$$

$$\text{Eg. } X^3 = 1,000$$

$$\log X^3 = \log 1,000$$

$$3 \log X = \log 10^3$$

$$\log X = 3/3$$

$$\log X = 1$$

$$X = \text{Antilog } 1$$

$$X = \underline{10}$$

$$4. \quad a = b(c + x)^d \quad \text{find the value of } X$$

$$(c + x)^d = a/b$$

$$\log (c + X)^d = \log a - \log b$$

$$d \log c + X = \log a - \log b$$

$$\log c + X = \frac{\log a - \log b}{d} \quad \text{Let: } \frac{\log a - \log b}{d} = K$$

$$\log c + x = K$$

$$c + x = \text{Antilog } K$$

$$X = \text{Antilog } K - C$$

$$\text{Eg. } 1,000 = 250(5 + X)^5$$

$$(5 + X)^5 = \frac{1,000}{250} = 4$$

$$\log (5 + x)^5 = \log 4$$

$$\log 5 + X = \frac{\log 4}{5}$$

$$\begin{aligned}
 \log 5 + X &= 0.120412 \\
 X &= \text{antilog } 0.120412 - 5 \\
 &= 1.319508 - 5 \\
 &= \underline{-3.6805}
 \end{aligned}$$

3. The third way of finding the compound amount is using specially prepared tables which provide values of $(1 + i)^n$ for selected values of i and n .

Therefore, to calculate the value of “A” or other variables in the compound interest formula, you can use any of these three approaches which ever convenient to you.

Example 2. Find the compound amount and compound interest after 10 years if Br. 15, 000 were invested at 8% interest;

a) If compounded annually

Compounding annually means that there is one interest payment period per year. Thus

$$\begin{aligned}
 t &= 10 \text{ years} \\
 m &= 1 \\
 n &= mt = 1 \times 10 = 10 \\
 i &= r/m = 8\%/1 = 8\% = 0.08
 \end{aligned}$$

The compound amount will be:

$$\begin{aligned}
 A &= 15,000 (1.08)^{10} \\
 &= 15,000 (2.158925) \\
 &= \text{Br. } \underline{32,383.875}
 \end{aligned}$$

- Compound Interest = compound amount (A) – Principal (P)

$$\begin{aligned}
 &= 32,383.875 - 15,000 \\
 &= \text{Br. } \underline{17,383.875}
 \end{aligned}$$

b) If compounded semiannually

Compounding semiannually means that there are two interest payment periods per year. Thus, the number of payment periods in 10 years $n = 2 \times 10 = 20$ and the interest rate per conversion period will be $i = r/m = 8\%/2 = 4\%$. The compound amount then will be:

$$A = P (1 + i)^n$$

$$\begin{aligned}
&= 15,000 (1.04)^{20} \\
&= 15,000 (2.191123) \\
&= \text{Br. } \underline{32,866.85}
\end{aligned}$$

$$\begin{aligned}
\text{Compound Interest} &= A - P \\
&= 32,866.85 - 15,000 \\
&= \text{Br. } \underline{17,866.85}
\end{aligned}$$

c) If Compounded quarterly

If compounding takes place quarterly (four times a year), then an 8% annual interest rate, the interest rate per conversion period will be $i = 0.08/4 = 0.02$, there will be a total of $n = 4 \times 10 = 40$ conversion periods over the 10 years. The compound amount will be:

$$\begin{aligned}
A &= 15,000 (1.02)^{40} \\
&= 15,000 (2.208039) \\
&= \text{Br. } \underline{33,120.60}
\end{aligned}$$

d) If compound monthly

$$\begin{aligned}
p &= 15,000 \\
t &= 10 \text{ years} \\
m &= 12 \text{ (12 payment periods per year)} \\
n &= 12 \times 10 = 120 \text{ payment periods over the 10 years} \\
i &= r/m = 8\%/12 = 0.667\% = 0.00667
\end{aligned}$$

Under these conditions:

$$\begin{aligned}
A &= 15,000 (1.00667)^{120} \\
&= 15,000 (2.220522) \\
&= \text{Br. } \underline{33,307.84}
\end{aligned}$$

$$\text{Interest} = \text{Br. } \underline{18,307.84} \text{ (33,307.84 - 15,000)}$$

e) If compounded weekly

$$\begin{aligned}
m &= 52 \\
n &= 10 \times 52 = 520
\end{aligned}$$

$$i = 8\%/52 = 0.154\% = 0.00154, \text{ then}$$

$$A = 15,000 (1.00154)^{520}$$

$$= \text{Br. } \underline{33,362.60}$$

$$\text{Interest} = 33,362.60 - 15,000$$

$$= 18362.60$$

f) **Try your self: if Compounded daily, and hourly**, what will be the compound amount respectively? Answer = Br. 33,380.19 and Br. 33,382.99 respectively.

g) **If compounded continuously (Instantaneously)**, what happens to the compound amount if interest is compounded continuously? To derive a formula for continuous compound interest, we begin by writing:

$$(1 + i)^n = (1 + r/m)^{mt}$$

Then, by inserting $1 = r/r$ in the exponent, we obtain

$$(1 + r/m)^{mt (r/r)} = (1 + r/m)^{(m/r) \cdot (rt)}$$

Then, letting $m/r = X$, we have

$$[(1 + 1/X)^X]^{rt}$$

As X increases indefinitely, the term $(1 + 1/X)^X$ approaches the value of the familiar mathematical constant $e = 2.7182818\dots$. This means that the factor

$$(1 + i)^n = [(1 + 1/X)^X]^{rt} \text{ approaches}$$

e^{rt} as n increases indefinitely. The resulting formula for the amount under continuous compounding of interest is given by:

$A = P e^{rt}$

.....**

Where: A = amount at the end of time t under continuous compounding

p = principal

r = annual rate, compounded continuously

t = time, in years

Note: the value of e^{rt} may be found using a calculator.

Solution:

$$p = \text{br. } 15,000$$

$$t = 10 \text{ years}$$

$$A = P e^{rt}$$

$$= 15,000 (e^{0.08 \times 10})$$

$$\begin{aligned}
 A &= ? & & = 15,000 \times e^{0.8} \\
 & & & = 15,000 \times 2.22554 \\
 & & & = \text{Br. } \underline{33,383.11}
 \end{aligned}$$

$$\begin{aligned}
 \text{Compound Interest} &= 33,383.11 - 15,000 \\
 &= \underline{\text{Br. } 18,383.11}
 \end{aligned}$$

What can you observe from the above discussion? When a number of conversion period within a year increases, the interest earned also increases continuously toward an upper limit. The limiting case occurs where interest is compounded continuously.

Example 3.

How long it take to accumulate Br. 8, 000 if you invest Br. 6, 000 at 12% compounded monthly?

Solution:

$$P = \text{Br. } 6,000$$

$$A = \text{Br. } 8,000$$

$$r = 12\%$$

$$i = r/m = 1\% = 0.01$$

$$t = ? \text{ n?}$$

$$A = P(1 + i)^n$$

$$8000 = 6000(1.01)^n$$

we can use logarithm to solve this problem

$$\frac{8000}{6000} = (1.01)^n$$

$$\log 1.33 = (1.01)^n$$

$$n = \frac{\log 1.33}{\log 1.01}$$

$$n = 28.92 \approx 29 \text{ months}$$

It takes 29 months for Br. 6000 invested at 12% to grow to Br. 8000

The present value

Frequently it is necessary to determine the principal P which must be invested now at a given rate of interest per conversion period in order that the compound amount A be accumulated at the end of n conversion periods. Under these conditions, p is called the present value of A. This process is called discounting and the principal is now a discounted value of future income A. If:

$A = P(1 + i)^n$ then dividing both sides by $(1 + i)^n$ leads to

$$P = \frac{A}{(1+i)^n} = A(1+i)^{-n}$$

$P = A(1+i)^{-n}$

..... * Present value of compound amount.

Example 4. How much should you invest now at 8% compounded semiannually to have Br. 10, 000 toward your brother's college education in 10 years?

Solution

$$A = \text{Br. } 10,000$$

$$t = 10 \text{ years}$$

$$m = 2$$

$$n = mt = 20$$

$$r = 8\%$$

$$i = r/m = 4\% = 0.04$$

$$p = ?$$

$$P = A(1+i)^{-n}$$

$$= 10,000(1.04)^{-20}$$

$$= 10,000(0.456387)$$

$$= \text{Br. } \underline{4563.87}$$

4.3 EFFECTIVE RATE

An effective rate is the simple interest rates that would produce the same return in one year had the same principal been invested at simple interest without compounding. In other words, the effective rate r converted m times a year is the simple interest rate that would produce an equivalent amount of interest in one year. It is denoted by r_e .

If principal p is invested at an annual rate r , compounded m times a year, then in one year,

$$A = P(1 + r/m)^m$$

What simple interest rate will produce the same amount A in one year? We call this simple interest rate the effective rate. To find r_e we proceed as follows:

(Amount at simple interest after 1 year) = (Amount at compound interest after 1 year)

$$P(1 + r_e) = p(1 + r/m)^m \text{Divide both sides by } p$$

$$1 + r_e = (1 + r/m)^m \text{isolate } r_e \text{ on the left side and gives you:}$$

$r_e = (1 + r/m)^m - 1$

.....* effective interest rate formula.

Where: r = nominal annual rate of interest

m = no. of conversion periods per year.

$r_e = e^r - 1$	** effective interest rate in continuous
compounding.	

Example 5. An investor has an opportunity to invest in two investment alternatives A and B which pays 15% compounded monthly, and 15.2% compounded semi-annually respectively. Which investment is better investment, assuming all else equal?

Solution

Nominal rate with different compounding periods cannot be compared directly. We must find the effective rate of each nominal rate and then compare the effective rates to determine which investment will yield the larger return.

Effective rate for investment A

$$\begin{aligned}
 r &= 15\% \\
 m &= 12 \\
 i &= 1.25\% = 0.0125 \\
 r_{eA} &= (1 + i)^m - 1 \\
 &= (1.0125)^{12} - 1 \\
 &= \underline{16.076\%}
 \end{aligned}$$

Effective rate for Investment B

$$\begin{aligned}
 r &= 15.2\% \\
 m &= 2 \\
 i &= 7.6\% = 0.076 \\
 r_{eB} &= (1.076)^2 - 1 \\
 &= \underline{15.778\%}
 \end{aligned}$$

Since the effective rate for investment A (16.076%) is greater than the effective rate for investment B (15.778%), A is the preferred investment alternative.

Example 6. What is the effective rate corresponding to a nominal rate of 16% compounded quarterly?

$$\begin{aligned}
 r_e &= (1 + 0.16/4)^4 - 1 \\
 &= (1.04)^4 - 1 \\
 &= 1.169859 - 1 \\
 &= \underline{16.99\%}
 \end{aligned}$$

4.4 ANNUITIES

An annuity is any sequence of equal periodic payments. The payments may be made weekly, monthly, quarterly, annually, semiannually or for any fixed period of time. The time between successive payments is called payment period for the annuity. If payments are made at the end of each payment period, the annuity is called an **ordinary annuity**. If payment is made at the beginning of the payment period, it is called **annuity due**. In this course we will discuss only ordinary annuities. The amount, or future value, of an annuity is the sum of all payments plus the interest earned during the term of the annuity.

The term of an annuity refers to the time from the beginning of the first payment period to the end of the last payment period.

4.4.1 Ordinary Annuity

An ordinary annuity is a series of equal periodic payments in which each payment is made at the end of the period. In an ordinary annuity the first payment is not considered in interest calculation for the first period because it is paid at the end of the first period for which interest is calculated. Similarly, the last payment does not qualify for interest at all since the value of the annuity is computed immediately after the last payment is received.

Future value (Amount) of an ordinary annuity.

Example1. What is the amount of an annuity if the size of each payment is Br. 100 payable at the end of each quarter for one year at an interest rate of 4% compounded quarterly?

Solution

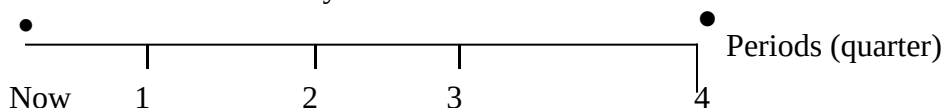
Periodic payment (R) = Br. 100

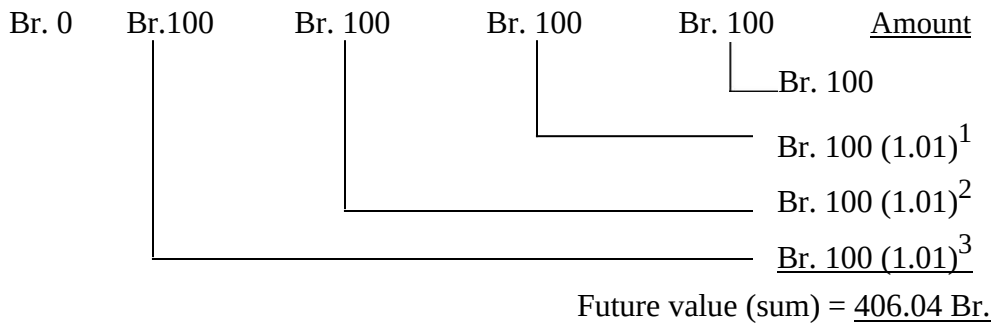
Payment interval (Conversion period) = quarter

Nominal (annual rate) = $r = 4\%$

Interest per conversion period (i) = $r/m = 4\%/4 = 1\%$

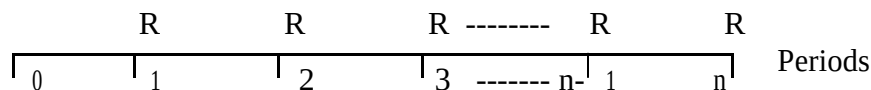
Future value of an annuity = ?





$$\begin{aligned}
 \text{Compound interest} &= \text{Amount} - R(n) \\
 &= 406.04 - 100 (4) \\
 &= \underline{\text{Br. 6.04}}
 \end{aligned}$$

If R represents the amount of the periodic payment, i represents the interest rate per payment period, and n represents the number of payment periods, then



The first payment of R accumulates interest for n-1 periods, the second payment R for n – 2 periods etc. The last payment accumulate no interest, the next to last payment accumulates one period for interest. So using the future value for compound interest we see the future value of the annuity:

$$A = R (1 + i)^{n-1} + R (1 + i)^{n-2} + \dots + R (1 + i)^1 + R \dots \text{Equation 1}$$

Multiplying each side of the equation by (1 + i), we obtain

$$A (1 + i) = R (1 + i)^n + R (1 + i)^{n-1} + \dots + R(1 + i)^2 + R (1 + i) \dots \text{Eg. 2.}$$

Then subtracting the first equation (eq. 1) from the second equation (eq. 2), gives you:

$$\begin{aligned}
 &\left\{ \begin{aligned} A (1 + i) &= R (1 + i)^n + R (1 + i)^{n-1} + \dots + R (1 + i)^2 + R (1 + i) \\ A &= R (1 + i)^{n-1} + \dots + R (1 + i)^2 + R(1 + i) + R \end{aligned} \right. \\
 &\hline
 &A (1 + i) - A = R (1 + i)^n - R \\
 &A [(1 + i)] = R [(1 + i)^n - 1] \\
 &A (i) = R[(1 + i)^n - 1] \qquad \text{Dividing both sides by i, we have}
 \end{aligned}$$

$$A = \left[\frac{R [(1 + i)^n - 1]}{i} \right]$$

.....* Amount of an ordinary annuity

Where: A = Amount (future value) of an ordinary annuity at the end of its term

R = Amount of periodic payment

i = interest rate per payment
period

n = (mt) total no. of payment
periods

Solve the above problem (example 1) using the annuity formula:

$$A = \left[\frac{100 (1.04)^4 - 1}{0.04} \right]$$
$$= \underline{\text{Br. 406.04}}$$

Example 2. Mr X. Deposits Br. 100 in a special savings account at the end of each month. If the account pays 12%, compounded monthly, how much money, will Mr. X have accumulated just after 15th deposit?

Solution:

$$R = \text{Br. 100}$$

$$n = 15$$

$$r = 12\%$$

$$m = 12$$

$$i = r/m = 12\%/12 =$$

$$1\% \quad A = ?$$

$$A = R \left[\frac{(1+i)^n - 1}{i} \right]$$
$$= 100 \left[\frac{(1.01)^{15} - 1}{0.01} \right]$$
$$= 100 (16.096896)$$
$$= \underline{\text{Br. 1609. 69}}$$

Example 3. A person deposits Br. 200 a month for four years into an account that pays 7% compounded monthly. After the four years, the person leaves the account untouched for an additional six years. What is the balance after the 10 year period?

Solution:

$$R = \text{Br. 200}$$

$$t = 4 \text{ years}$$

$$m = 12$$

$$n = mt = 4 \times 12 = 48$$

$$r = 7\%$$

Amount after

$$4 \text{ years } (A_4) = 200 \left[\frac{(1 + 0.07/12)^{48} - 1}{0.07/12} \right]$$
$$= 200 (55.20924)$$
$$= \underline{\text{Br. 11, 041.85}}$$

$$i = 7\%/12 = 0.07/12$$

After the end of the fourth year, we calculate compound interest rate taking Br. 11, 041.85 as principal compounded monthly for the coming 6 years.

$$\begin{aligned} p &= 11, 041.85 & A_{10} &= 11, 041.85 (1 + 0.07/12)^{72} \\ t &= 6 \text{ years} & &= 11, 041.85 (1.5201) \\ m &= 12 & &= \underline{\underline{\text{Br. 16, 784.77}}} \\ n &= 6 \times 12 = 72 \\ r &= 7\% \\ i &= r/m = 7\%/12 = 0.07/12 \\ A_{10} &= ? \end{aligned}$$

Therefore, the balance after 10 years is Br. 16, 784.77.

4.4.2 SINKING FUND

A sinking fund is a fund into which equal periodic payments are made in order to accumulate a definite amount of money up on a specific date. Sinking funds are generally established in order to satisfy some financial obligations or to reach some financial goal.

If the payments are to be made in the form of an ordinary annuity, then the required periodic payment into the sinking fund can be determined by reference to the formula for the amount of an ordinary annuity. That is, if:

$$\begin{aligned} A &= R \left[\frac{(1+i)^n - 1}{i} \right] \quad \text{then,} \\ R &= \frac{A}{\frac{(1+i)^n - 1}{i}} \\ &= A \left[\frac{i}{(1+i)^n - 1} \right] \end{aligned}$$

Example 4. How much will have to be deposited in a fund at the end of each year at 8% compounded annually, to pay off a debt of Br. 50, 000 in five years?

Solution:

$$A = \text{Br. } 50,000$$

$$t = 5 \text{ years}$$

$$m = 1, n = 5 (5 \times 1)$$

$$r = 8\%$$

$$i = r = 8\%$$

$$R = ?$$

$$R = A \left[\frac{i}{(1+i)^n - 1} \right]$$

$$= 50,000 \left[\frac{(0.08)}{(1.08)^5 - 1} \right]$$

$$= 50,000 (0.174056)$$

$$= \text{Br. } \underline{\underline{8,522.80}}$$

The total amount of deposit over the 5 year period is equal to $5 \times 8,522.80 = \text{Br. } 42,614$

Example 5. Ato Ayalkebet has a savings goal of Br. 100,000 which he would like to reach 15 years from now. During the first 5 years he is financially able to deposit only Br. 1000 each quarter into the savings account. What must his quarterly deposit over the last 10 (ten) years be if he is to reach his goal? The account pays 10% interest, compounded quarterly.

Solution:

For the first 5 years

$$R = \text{Br. } 1,000$$

$$t = 5 \text{ years}$$

$$m = 4$$

$$n = 20$$

$$r = 10\%$$

$$i = 2.5\%$$

$$A_5 = ?$$

$$A_5 = 1,000 \left[\frac{(1.025)^{20} - 1}{0.025} \right]$$

$$= 1000 (22.019)$$

$$= \text{Br. } \underline{\underline{22019}}$$

This sum (Br. 22,019) will continue to draw interest at the rate of 10%; compounded quarterly, over the next 10 years; and the amount at the end of the 10th year will be:

$$t = 10 \text{ years}$$

$$m = 4$$

$$n = 40$$

$$i = 2.5\%$$

$$A_{10} = 22019 (1.025)^{40}$$

$$= 22019 (1.103813)$$

$$= \text{Br. } \underline{\underline{24304.86}}$$

To determine the periodic payment for the remaining 10 years, we subtract Br. 24, 304.86 from Br. 100, 000 to obtain the amount of an ordinary annuity for the last 10 years which is equal to Br. 75695.14 (100, 000 – 24, 304.86)

$$\begin{aligned} R &= 75695.14 \left[\frac{0.025}{(1.025)^{40} - 1} \right] \\ &= 75, 695.14 (0.014836) \\ &= \underline{\text{Br. 1, 123.03}} \end{aligned}$$

Thus, if Ayalkebate makes quarterly payments of Br. 1000 into a savings account over the first five years and then quarterly payments of Br. 1, 123.03 over the next 10 years, he will reach his savings goal of Br. 100, 000 at the end of 15 years.

4.4.2 Present value of an ordinary annuity

The present value of an ordinary annuity is the sum of the present values of all the payments, each discounted to the beginning of the term of the annuity. It represents the amount that must be invested now to purchase the payments due in the future.

The present value of an annuity can be computed in two ways:

- Discounting all periodic payments to the present (beginning of the term individually) or
- Discounting the future value (amount) of an annuity to the beginning of the term

Example 6. What is the present value of an annuity if the size of each payment is Br. 200 payable at the end of each quarter for one year and the interest rate is 8% compounded quarterly?

Solution:

$$R = \text{Br. 200}$$

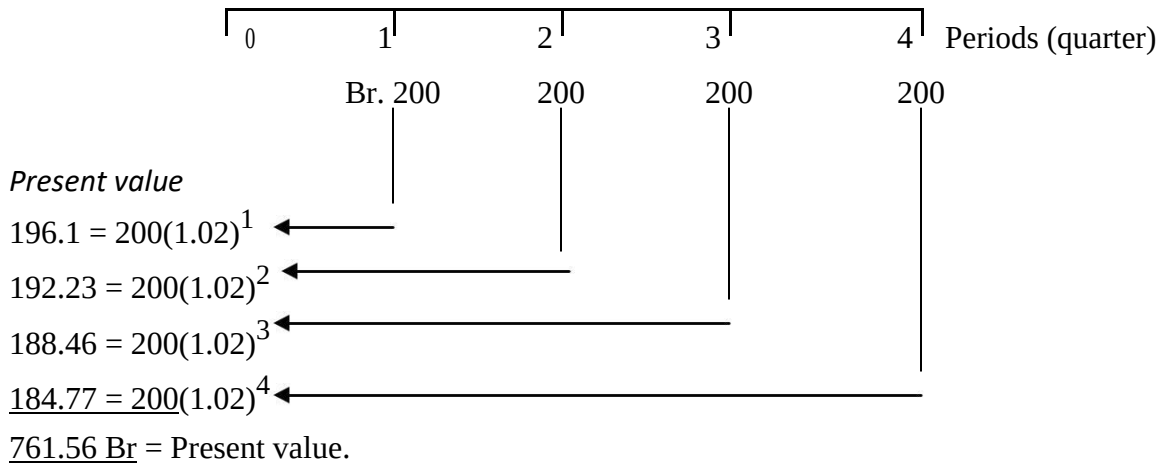
$$r = 8\%, \quad i = 2\%$$

$$m = 4$$

$$n = 4$$

$$p = ?$$

Using the first approach (discounting each payment individually), the present value will be:



Equivalently we may find the future value of the ordinary annuity using the formula and then discount it to the present taking it as a single future value.

$$A = R \left[\frac{(1+i)^n - 1}{i} \right]$$

$$= 200 \left[\frac{(1.02)^4 - 1}{0.02} \right]$$

$$= \text{Br. } \underline{824.32}$$

$$P = A (1 + i)^{-n}$$

$$= 824.32 (1.02)^{-4}$$

$$\text{Br} = \underline{761.56}$$

If we multiply the future value of an ordinary annuity by the compounding discounting factor, we get the present value of an annuity as follows:

$$P = R \left[\frac{(1+i)^n - 1}{i} \right] ((1 + i)^{-n})$$

$$= R \left[\frac{(1+i)^n (1+i)^{-n} - (1+i)^{-n}}{i} \right]$$

$$= \left[\frac{(1+i)^0 - (1+i)^n}{i} \right]$$

$$\boxed{P = R \left[\frac{1-(1+i)^n}{i} \right]} \dots\dots\dots * \text{ Present value of an ordinary annuity.}$$

Using the formula, the present value of the above example is computed as:

$$R = \text{Br. } 200$$

$$r = 8\%, i = 2\%$$

$$m = 4$$

$$t = 4$$

$$\begin{aligned} P &= R \left[\frac{1-(1+i)^n}{i} \right] \\ &= 200 \left[\frac{1-(1.02)^4}{0.02} \right] \\ &= 200 (3.80773) \\ &= \underline{\text{Br. } 761.55} \end{aligned}$$

Example 7. What is the present value of an annuity that pays Br. 400 a month for the next five years if money is worth 12% compounded monthly?

Solution:

$$R = \text{Br. } 400$$

$$t = 5 \text{ years}$$

$$m = 12$$

$$n = 12 \times 5 = 60$$

$$r = 12\%$$

$$i = 12\%/12 = 1\% = 0.01$$

$$p = ?$$

$$\begin{aligned} P &= R \left[\frac{1-(1+i)^n}{i} \right] \\ &= 400 \left[\frac{1-(1.01)^{60}}{0.01} \right] \\ &= 400 (44.955037) \\ &= \text{Br. } \underline{17,982.01} \end{aligned}$$

4.4.3 Amortization

Amortization means retiring a debt in a given length of time by equal periodic payments that include compound interest. After the last payment, the obligation ceases to exist it is dead and it is side to have been amortized by the periodic payments. Prominent examples of amortization are loans taken to buy a car or a home amortized over periods such as 5, 10, 20 or 30 years.

In amortization the interest is to determine the periodic payment, **R**, so as to amortize (retire) a debt at the end of the last payment. Solving the present value of ordinary annuity formula for R in terms of the other variable, we obtain the following amortization formula:

$$R = P \left[\frac{i}{1 - (1+i)^{-n}} \right] \dots\dots\dots ** \text{ Amortization formula}$$

Where: R = Periodic payment

P = Present value of a loan

i = Rate per period

n = Number of payment periods

Example:1

1. Ato Elias borrowed Br. 15, 000 from Commercial Bank of Ethiopia and agree to repay the loan in 10 equal installments including all interests due. The bank's interest charges are 6% compounded Quarterly. How much should each annual payment be in order to retire the debt including the interest in 10 years.

Solution

Pv = Br. 15, 000

t = 10 years

n = 10 x 4 = 40

m = 4

r = 6% i = 6%/4 = 1.5% R = ?

$$R = 15,000 \left[\frac{0.015}{1 - (1.015)^{-40}} \right]$$

$$(1.015)^{40} = 1.814$$

$$(0.033427) = \text{Br. } 501.4$$

$$\text{Interest} = [501.4 \times 40] - 15,000$$

$$= 20056 - 15,000$$

$$= \text{Br. } 5056$$

Check your progress:

If you have Br. 100,000 in an account that pays 6% compounded monthly and if you decide to withdraw equal monthly payments for 10 years at the end of which time the account will have a zero balance, how much should be withdrawn each month?

Answer: Br. 1,110,205

2. An employee has contributed with her employer to a retirement plan for 20 years a certain amount twice a year. The contribution earns an interest rate of 10% compounded semiannually. At the date of her retirement the total retirement

benefit is Br. 300, 000. The retirement program provides for investment of this amount at an interest rate of 10% compounded semiannually. Semiannual payments will be made for 10 years to the employee of her family in the event of her death.

1. What semi annual payment should she made?
2. What semi annual payment should be made for her or her family?
3. How much interest will be earned on Br. 300, 000 over the 40 years?

Solution:

Retirement plan

Employment period		Retirement period		Time
0	Pay	20	Receive	40 yrs
	m = 2			

$A_{20} = 300,000$ Br.

$R_2 = ?$

$t = 20$ yrs

$P_v = 300,000$

$m = 2, n = 40$

$t = 20$ yrs,

$r = 10\%, i = 5\%$

$m = 2, n = 40$

$R_1 = ?$

$r = 10\%, I = 5\%$

$$R_1 = 300,000 \left[\frac{0.05}{1 - (1.05)^{-40}} \right]$$

$$= \text{Br. } \underline{\underline{2483.45}}$$

$$R_2 = \left[\frac{0.05}{1 - (1.05)^{-40}} \right] \times 300,000$$

$$= \underline{\underline{17483.45}}$$

4.4.4 Mortgage Payments

In a typical home purchase transaction, the home buyer pays part of the cost in cash and borrows the remaining needed, usually from a bank or a savings and loan associations. The buyer amortizes the indebtedness by periodic payments over a period of time. Typically **payments** are **monthly** and the time period is long such as 30 years, 25 years and 20 years. Mortgage payment and amortization are similar. The only differences are:

- the time period in which the debt/ loan is amortized
- /repaid/ the amount borrowed.

In mortgage payments m is equal to 12 because the loan is repaid from monthly salary or Income, but in amortization money take other values. Similarly stated mortgage

payments are of amortization in nature involving the repayment of loan monthly over an extended period of time.

Therefore, in mortgage payments we are interested in the determination of monthly payments.

Taking:

A = total debt

R = monthly mortgage payments

r = stated nominal rate per annum

n = 12 x number of years (period of the loan)

R can be determined as follows:

$$R = A \left[\frac{r/12}{1 - (1 + r/12)^{-n}} \right] \quad \text{or} \quad R = A \left[\frac{i}{1 - (1 + i)^{-n}} \right]$$

$$A = R \left[\frac{1 - (1 + r/12)^{-n}}{r/12} \right]$$

Example: 1

Ato Assefa purchased a house for Br. 115, 000. He made a 20% down payment with the balance amortized by a 30 year mortgage at an annual interest of 12% compounded monthly so as to amortize/ retire the debt at the end of the 30th year.

Required:

1. Find the periodic payment
2. Find the interest charged.

Find the interest charged.

Solution:

Selling price = Br. 115, 000

r = 12%, i = 0.01

Less: Down payment = 23, 000 (20% x 115,000)

m = 12, n = 360

Mortgage (A) = Br. 92, 000

t = 30 years

R = ?

$$R = A \left[\frac{r/12}{1 - (1 + r/12)^{-n}} \right] = 92, 000 \left[\frac{0.01}{1 - (1.01)^{-360}} \right]$$

$$= 92, 000 (0.010286125)$$

$$= \text{Br. } 946.32$$

Ato Assefa should pay Br. 946.32 every month

to

retire the debt within 30 years or 360 monthly payments.

Interest = Actual payment – mortgage (loan)

$$= (946.32 \times 360) - 92,000$$

$$= \text{Br. } 340,675.20 - 92,000$$

$$= \text{Br. } \underline{248,675.20}$$

Throughout the 30 years period the loan earns an interest of Br, 284,675.20 **Example: 2**

Ato Amare purchased a house for Br. 50,000. He made an amount of down payment and pay monthly Br. 600 to retire the mortgage for 20 years at an annual interest rate of 24% compounded monthly.

Required.

Find the mortgage, down payment, interest charged and percentage of the down payment to the selling price.

Solution:

Selling price = Br. 50,000

Down payment = ?

Mortgage (A) = ?

$$R = \text{Br. } 600$$

$$r = 24\%, i = 2\%$$

$$m = 12, n = 240$$

$$t = 20 \text{ years}$$

$$\begin{aligned} * \text{ Mortgage (A)} &= R \left[\frac{1 - (1+i)^{-n}}{i} \right] \\ &= 600 \left[\frac{1 - (1.02)^{-240}}{0.02} \right] \\ &= 600 (49.56855) \end{aligned}$$

$$= \text{Br. } \underline{29,741.13}$$

* Down payment = Selling price – mortgage.

$$= 50,000 - 29,741.13$$

$$= \text{Br. } \underline{20,258.87}$$

* Interest charged = Actual payment – mortgage]

$$= 600 \times 240 - 29,741.13$$

$$= 144,000 - 29,741.13$$

$$= \text{Br. } \underline{114,258.87}$$

$$\text{* Percentage of down payment} = \frac{\text{Down payment}}{100\% \text{ Selling price}} \times$$

$$= \frac{20,258.87}{50,000} \times 100\%$$

$$= \underline{40.52\%}$$

Try your self (check your progress)

Ato Liku purchases a house for Br. 250,000. He makes a 20% down payment, with a balance amortized by a 30 year mortgage at an annual interest rate of 12% compounded monthly.

- a) Determine the amount of the monthly mortgage payment.
- b) What is the total amount of interest Ato Liku will pay over the life of the mortgage?
- c) Determine the amount of the mortgage Ato Liku will have paid after 10 years?

Answer:

- a) R = Br. 2,507.20
- b) Interest = Br. 540,602.80
- c) Payment after 10 years = Br. 13,163.57

4.5 EXERCISES/ PROBLEMS

1. If you borrow Br. 1, 000 from Commercial Bank of Ethiopia for 1 year to pay at 6% interest rate your tuition fee. Find the simple interest and the maturity value of the loan.
2. What is the present value of a loan that will amount to Br. 5, 000 in 5 years if money is worth 3% compounded semi-annually?
3. How much should be deposited in an account paying 10% compounded quarterly in order to have a balance of Br. 10, 000 ten (10) years from now? What would be the amount of compound interest after 10 years?
4. A small boy at the age of 10 drops 0.25 cents into a Jar each day. At the end of each month (30 days months) he deposits this amount at Dashen Bank that pays 5% interest compounded quarterly. If he makes the deposit without interruption, how much will the boy have at the age of 20.
5. Hiwot deposits Br. 1, 000 at the end of every 3 months period in to an account for 5 years which earn 10% interest compounded quarterly and then her deposits are changed to Br. 500 monthly for the next 5 years which earn 12% interest compounded monthly. How much is the account by the end of the time period considered?
6. NTT Company purchased a delivery truck on credit from AMCE which requires a payment of Br. 400, 000 plus 5% interest compounded annually at the end of 5 years. The Company plans to set up a sinking fund to accumulate the amount required to settle the debt.

Required:

- A. find the total debt at the end of the 5 year.
 - B. what should be the monthly deposit into the fund be if the account pays 15% interest, compounded monthly?
7. If Br. 10, 000 is invested at 8% compounded:
 - A. Annually

B. Semi Annually

C. Quarterly

D. What can you observe from your answers in A, B and

C What is the amount after 5 years?

8. Assume you won a lottery and you want to deposit/ invest your money in the following to investment alternatives. Investment A which pays 15% compounded monthly and B that pays 14% compounded semi annually, which is the better investment, assuming other things are the same.

4.6 ANSWER TO THE EXERCISE / PROBLEMS

1. Given:

$$P = 1,000 \text{ Br.}$$

$$T = 1 \text{ year}$$

$$R = 6\% / \text{year}$$

$$I = ?$$

$$A = ?$$

Solution

$$I = PRT$$

$$= 1,000 \times 0.06 \times 1$$

$$= \text{Br. } 60$$

$$A = P + I$$

$$= 1,000 + 60$$

$$= \underline{\underline{\text{Br. } 1060}}$$

$$\text{or } A = P(1 + rt)$$

$$= 1,000(1 + 0.06 \times 1)$$

$$= 1,000(1.06)$$

$$= \text{Br. } \underline{\underline{1,060}}$$

2. Given:

$$A = \text{Br. } 5,000$$

$$t = 5 \text{ years}$$

$$r = 3\% / \text{year}$$

$$m = 2 \text{ time}$$

$$n = mt = 2 \times 5 = 10$$

$$i = r/m = 3\%/2 = 1.5\%$$

$$p = ?$$

Solution:

$$p = A(1 + i)^{-n}$$

$$= 5,000(1.0015)^{-10}$$

$$= 5,000(0.985123)$$

$$= \underline{\underline{\text{Br. } 4925.62}}$$

3. Given:

$$p = 10,000$$

$$t = 10 \text{ years}$$

$$m = 4 \text{ times}$$

Solution:

$$A = P(1 + i)^n$$

$$= 10,000(1.0025)^{40}$$

$$= 10,000(1.105033)$$

$$\begin{aligned}
 r &= 10\% & &= \underline{\underline{11.050.33 \text{ Br.}}} \\
 i &= 10\%/4 = 2.5\% & \text{Compound Interest} &= A - P \\
 n &= mt = 10 \times 4 & &= 11,050.33 - 10,000 \\
 &= 40 & &= \underline{\underline{1050.33 \text{ Br.}}} \\
 A &= ?
 \end{aligned}$$

5) Solution

Amount of money at the end of every month the boy will have Br. 7.50 (0.25 x 30) to be deposited at Dashen Bank.

Therefore, $R = 7.50$

$$\begin{aligned}
 t &= 10 \text{ years} \\
 r &= 5\% \\
 m &= 4, n = 40 \\
 i &= 5\%/4 = 0.00125
 \end{aligned}
 \quad
 \begin{aligned}
 A &= 7.5 \left[\frac{(1.00125)^{40} - 1}{0.00125} \right] \\
 &= 7.5 \times 514.89557 \\
 &= \underline{\underline{\text{Br. } 3861.72}}
 \end{aligned}$$

If the small boy saves 0.25 cents every day and deposits it monthly in a bank account that pays 5% compounded quarterly, he will have Br. 3,861.72 at after 10 years.

6. Solution First

$$\begin{aligned}
 &5 \text{ years} \\
 R_1 &= 1000 \\
 r &= 10\% \\
 m &= 4 \\
 i &= 2.5\% \\
 t &= 5 \\
 n &= 20 \\
 A_5 &= ?
 \end{aligned}
 \quad
 \begin{aligned}
 A_1 &= 1,000 \left[\frac{(1.025)^{20} - 1}{0.025} \right] \\
 &= 1,000 (25.54466) \\
 &= \underline{\underline{\text{Br. } 25544.66}}
 \end{aligned}$$

This amount will be compounded quantity at a 12% interest rate. Therefore. The amount at the end of the next 5 years will be calculated as follows.

7. a) $A = ?$

$$\begin{aligned}
 t &= 5 \text{ years} \\
 p &= 400,000 \text{ Br.} \\
 m &= 1, \\
 r &= 5\%, n = 5
 \end{aligned}$$

$$\begin{aligned}
 A &= P (1 + i)^n \\
 &= 400,000 (1.05)^5 \\
 &= \underline{510512.625 \text{ Br.}}
 \end{aligned}$$

b) Given

$$A = 510512.625 \text{ Br.}$$

$$t = 5 \text{ years}$$

$$m = 12$$

$$n = 5 \times 12 = 60$$

$$r = 12\%$$

$$i = 1\%$$

$$R = ?$$

$$\begin{aligned}
 R &= A \left[\frac{i}{(1+i)^n - 1} \right] = 510512.625 \times \frac{0.01}{(1.01)^{60} - 1} \\
 &= 510512.625 \times \frac{0.01}{0.81669669} \\
 &= 510512.625 \times 0.0122444 \\
 &= \text{Br. } \underline{6250.95}
 \end{aligned}$$

The company should deposit Br. 6250.95 every month for 5 years or 60 periods to a sinking fund in order to settle its debt of 510512.625

8. Check your progress

- Answer**
- A) A = Br. 14693.28
 - B) A = Br. 14862.44
 - C) A = Br. 14859.47

9. **Solution:** Nominal rates with amount compounding periods in this case (quarterly and monthly) cannot be compared directly. You have to find the effective rate of each nominal rate and then compare the effective rates to determine which investment will yield the return

Effective rate for A =

Effective rate for Investment A:

Effective rate for Investment B:

$$\begin{aligned}
 r_e A &= (1 + i)^M - 1 \\
 &= (1 + 0.0125)^{12} - 1 \\
 &= 0.16075 = 16.075\%
 \end{aligned}$$

$$\begin{aligned}
 r_e B &= (1 + 0.07)^2 - 1 \\
 &= (1.07)^2 - 1 \\
 &= \underline{\underline{0.1449 = 14.5\%}}
 \end{aligned}$$

Therefore, investment A is a better alternative

$$\begin{aligned}
 P &= 25,544.66 & A_{10} &= (25544.66 (1.01)^{60} \\
 r &= 12\% & &= \underline{\underline{\text{Br. } 46406.895}}
 \end{aligned}$$

$$m = 12$$

$$i = 1\%$$

$$t = 5$$

$$n = 5 \times 12 = 60$$

Since Hiwot continues her payment for the second 5 years of Br. 500, the amount will be calculated using an annuity formula as:

$$\begin{aligned}
 A_2 &= 500 \left[\frac{1.01^{60} - 1}{0.01} \right] \\
 &= 500 (81.66966) \\
 &= \underline{\underline{\text{Br. } 40834.845}}
 \end{aligned}$$

At the end of the 10 years Hiwot will have a total amount of Br. 87,241.73 (Br. 46,406.9 + Br. 40834.84).

UNIT 5: ELEMENTS AND APPLICATIONS OF CALCULUS

Contents

- 5.0 Aims and Objectives
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 - The Second Derivative Test
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5.0 OBJECTIVES

After reading this unit students must be able to:

- to explain the concepts of limits and continuity
- to provide an understanding of average rate of change
- to provide an understanding of derivative
- to illustrate a wide variety of applications of optimization procedures
- to understand skills in problem formulation
- to reinforce skills of interpretation of mathematical skills

5.1 INTRODUCTION

This unit examines the calculus and its application to business, economics and other areas of problem solving. The major areas of study within the calculus are differential calculus and integral calculus. Differential calculus focuses on rates of change in analyzing a situation.

Integral calculus involves summation of a special type. Graphically the concepts of area in two dimensions or volume in three dimensions are important in integral calculus. The goal in this chapter is to provide an appreciation for what the calculus is and where it can be applied. Though it would take several semester of intensive study to understand most of the finer points of the calculus, your coverage will enable you to understand the tools for conducting analysis at an elementary level.

Calculus is a mathematical tool used to solve problems in business, Economics and other areas.

- differential calculus and
- integral calculus

- 1) * Differential calculus focuses on rates of change in analyzing a situation. It broadens the concept of slope.

- Limits
- Continuity
- Derivatives

Application areas:

- Optimization problems i.e minimizing cost and /or maximizing profit, revenue---

- 2) Integral calculus involves summation of a special type, total change -----

It is the inverse of Differential calculus or vice versa like that of log and antilog.

Scope of this chapter = Differential calculus, derivative and its application in solving optimization problems.

5.2 THE DERIVATIVE

The process of finding a derivative is called differentiation. A set of rules of differentiation exists for finding the derivatives of many common functions.

5.2.1 The rules of differentiation:

The rules of differentiation have been developed using the limit approach. The mathematics involved in providing these rules can be finally complicated. For our purposes it will suffice the rules without proof.

The rules of differentiation apply to functions, which have specific structural characteristics. A rule will state that if a function has specific characteristics, then the derivative of the function will have resulting form.

⇒ Each function can be graphed and that the derivative is a general expression for the slope of the function.

Notation: $\frac{dy}{dx} \Rightarrow$ “which read the derivative of Y with respect to X”

: This notation can be used interchangeably with the notation “**F**’ (X) –read **f prime** of X” – which represent the derivative of the function **f** at **X**. That is given $f(X)$

$$\frac{dy}{dx} = f' \quad (X)$$

⇒ Differentiation by direct application of the limit formula is time consuming and difficult, therefore, functions should be classified into certain groups according to their behavior so that the process of finding a derivative when applied to these groups follows a definite pattern (formula)

1. Derivative of a constant function.

If $f(x) = K$

Eg. $F(x) = 10$,

$$f'(x) = 0$$

$$f'(x) = 0 \text{ slope} = 0$$

If you consider what the function ($f(x) = 10$) looks like graphically, this result seems reasonable. The function $f(x) = 10$ is horizontal line intersecting the Y axis at $(x, 10)$. The slope at all points along such function equals 0.

2. Power functions

If $f(x) = X^n$, where n is a real number $f'(X) = nx^{n-1}$

Example

1) If $f(X) = X = X^1$ where, $n = 1$

$$f'(X) = nx^{n-1}$$

$$= 1X^{1-1}$$

$$= \underline{1}$$

This implies that for the function $f(X) = X$, the slope equals 1 at all points. You should recognize that $f(X) = X$ is a linear function with slope 1.

$$2) f(X) = X^5$$

$$f'(X) = \underline{\underline{5X^4}}$$

$$4) f(X) = \sqrt[3]{X^2}$$

$$= X^{2/3}$$

$$3) f(X) = \frac{1}{X^3}$$

$$= X^{-3}$$

$$f'(X) = -3X^{-4}$$

$$= \underline{\underline{\frac{-3}{X^4}}}$$

$$f'(X) = \frac{2}{3} X^{-1/3}$$

$$= \frac{2X^{-1/3}}{3}$$

$$= \frac{2}{3X^{1/3}}$$

$$= \underline{\underline{\frac{2}{3\sqrt[3]{X}}}}}$$

3. A constant times a function

If $f(X) = K \cdot g(X)$, where K is a constant and g is a differentiable function,

$$f'(X) = K \cdot g'(X)$$

$$\text{a) } f(X) = 5X^3$$

$$f'(X) = 15X^2$$

$$\text{b) } f(X) = X^2 - 5X$$

$f(X)$ can be expressed as

$$f(X) = g(X) - h(X), \text{ where } g(X) = X^2$$

$$h(X) = 5x$$

$$f'(X) = g'(X) - h'(X)$$

$$= \underline{\underline{2x - 5}}$$

4. Sum or differences of functions.

This implies that the derivative of function formed by the sum (difference) of two or more component functions is the sum (difference) of the derivatives of the component functions.

$$(X) = [g(X) \pm h(X)]$$

$$f'(X) = [g'(X) \pm h'(X)]$$

$$f(X) = 6X^4 - 5X^2$$

$$g(X) = 6X^4$$

$$g'(X) = 24X^3$$

$$f'(X) = 24X^3 - 10x$$

$$h(X) = 5x^2$$

$$h'(X) = 10^x$$

$$g'(X) - h'(X) = 24x^3 - 10^x$$

$$f(X) = 3x^2 + 2x + 1$$

$$f'(X) = \underline{\underline{6x + 2}}$$

5. Product of functions

$$\text{If } h(X) = f(X) \cdot g(X)$$

$$h'(X) = f(X) \cdot g'(X) + g(X) \cdot f'(X)$$

$$h(X) = (2x^2 - 5)(x + 3)$$

$$f(X) = 2x^2 - 5$$

$$= (2x^2 - 5) \times 1 + (x + 3)(4x)$$

$$g'(X) = x + 3$$

$$= \underline{\underline{6x^2 + 12x - 5}}$$

$$f'(X) = 4x$$

$$g'(X) = 1$$

$$h'(X) = f(X) \cdot g'(X) + g(X) \cdot f'(X)$$

$$\underline{\underline{2x^2 - 5 \times 1 + (x + 3)(4x)}}$$

$$\text{Eg. 2. } f(X) = (x^2 - 5)(X - X^3)$$

$$g(X) = x^2 - 5 \quad g'(X) = 2x$$

$$h(X) = X - X^3 \quad h'(X) = 1 - 3X^2$$

$$f'(X) = g(X) \cdot h'(X) + h(X) \cdot g'(X)$$

$$= (x^2 - 5)(1 - 3x^2) + (X - X^3)(2x)$$

$$= \underline{\underline{-5x^4 + 18x^2 - 5}}$$

6. Derivative of the Quotient of function

$$h(X) = \frac{f(X)}{g(X)}$$

$$h'(X) = \frac{g(X) \cdot f'(X) - f(X) \cdot g'(X)}{[g(X)]^2}$$

Eg. 1. $h(X) = \frac{1-X}{X^2}$

$$h'(X) = \frac{(-1)(X^2) - (1-X)(2X)}{(X^2)^2}$$

$$= \frac{-X^2 - 2X + 2X^2}{(X^2)^2} = \frac{X^2 - 2X}{X^4}$$

Eg. 2. $f(X) = \frac{1}{X^3}$ $f'(X) = \frac{-3}{X^4}$

3. $F(X) = (3X^2 - 5) \cdot (1 - X^3)$ $f'(X) = \frac{3x^4 - 15x^2 - 6x}{(1-x^3)^2}$

7. Other rules-----

5.3 APPLICATION OF CALCULUS IN BUSINESS DECISIONS

5.3.1 Revenue, Cost, And Profit Applications

5.3.1.1 Revenue Applications:

5.3.1.2 Marginal analysis

Examines Incremental Effects

$$C(X) = \text{total cost function} \Rightarrow c'(X) = \text{marginal Cost}$$

$$R(X) = \text{total Revenue function} \Rightarrow R'(X) = \text{Marginal revenue}$$

$$P(X) = R(X) - (C(X)) = \text{Profit} \Rightarrow p'(X) = \text{marginal Profit}$$

$$p'(X) = R'(X) - c'(X)$$

Marginal Cost = is the additional cost incurred as a result of producing and selling one more unit of a product or service. Linear cost functions assume that the variable cost per unit is constant for such functions the marginal cost is the same at any level of output.

- A non-linear cost function is characterized by variable marginal costs.
 - For the total cost function $C(X)$, the derivative $C'(X)$ represents
 - i) The instantaneous rate of change in TC given a change in the number of units produced.
 - ii) A general expression for the slope of the graph of the TC function
 - iii) The marginal cost, $MC = c'(X)$
- $\therefore c'(X)$ Can be used to approximate the marginal cost associated with producing the next unit.

→ It is the rate of change in total cost per unit change in production at an out put level of X unit. It is also an optimization to the actual cost of making one more unit at any production level X (non-linear functions).

Example: Suppose the total cost $C(X)$ in thousands of dollars for manufacturing X unit is given by the function

$$C(X) = 575 + 25x \frac{X^2}{4} ; 0 < X \leq 50$$

[0 50]

Required:

- 1) Find the MC at a production level of X units

- 2) Find the MC at a production level of 40 unit and interpret the result
- 3) Find the actual cost of producing the 41st unit and compare this cost with the result found in question number 2

Solution

$$1) c'(X) = 25 - x/2$$

$$2) c'(40) = 25 - 40/2$$

$$= 5 \text{ Br.} = 5,000 \text{ Br.} \Rightarrow \text{marginal cost of 41}^{\text{st}} \text{ unit.}$$

$$3) C(41) - C(40) = ?$$

$$C(41) = 575 + 25(41) - (41)^2/4$$

$$= 1179.75$$

$$C(40) = 575 + 25(40) - (40)^2/4$$

$$= 1175$$

$$C(41) - C(40) = 4750 \text{ Br.} \Rightarrow 1179.75 - 1175 = 4.75 \times 1000$$

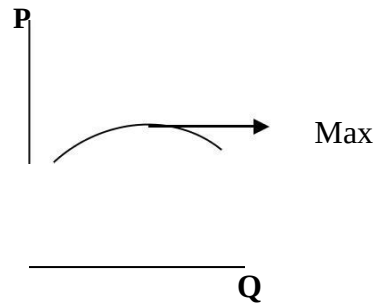
At a production level of 40 unit the rate of change of TC relative to production is Br. 5,000. In Other words, the cost of producing 1 more unit at this level of production (40) is approximately 5,000Br.

Marginal Revenue = Marginal revenue (MR) is the additional revenue derived from selling one more unit of a product or service. If each unit of a product sells at the same price, the MR is always equal to the price. Eg. $R = 10x$ $MR = 10 \text{ Br.}$

Marginal revenue for non-linear total revenue function is not constant.

For a total revenue function $R(X)$, the derivative $R'(X)$ represents the instantaneous rate of change in total revenue given a change in the number of units sold. For the purpose of marginal analysis, the derivative is used to represent the

Marginal revenue or $MR = R'(X)$



It represent / approximately the marginal revenue from selling the next unit

Eg. $R(X) = 500x - 0.005x^2$ $X = \text{number of units}$

$$R'(X) = 500 - 0.01X$$

Marginal Profit = Marginal revenue – Marginal cost

$$P'(X) = R'(X) - C'(X)$$

- Marginal revenue (Profit) analysis is concerned with the effect on profit if one-additional unit of a product is produced and sold. As long as the additional revenue brought in by the next unit exceeds the cost of producing and selling that unit, there is a net profit from producing and selling that unit and total profit increases.

- If $MR > MC$, produce the next unit
- If $MR < MC$, do not produce the next unit.

5.3.2 Profit maximization criterion

If $MR = MC$, for the last unit produced and sold, total profit will be maximized.

$$P(X) = R(X) - C(X) = 0$$

$$P'(X) = R'(X) - C'(X) = 0$$

$$P'(X) = 0$$

$$R'(X) = C'(X)$$

Example: The market research department of a Company recommends that the Company to manufacture and market a new transistor radio after suitable test. The marketing department also presents the following demand equation.

$$X=10,000-1000P \quad \text{i.e. } P=10-X/1000$$

Furthermore, the financial department provides the following cost equation:

$$C(X) = 7,000 + 2x$$

Conduct a marginal analysis for the company.

$$1) C'(X) = 2$$

$$2) TR = P \cdot X = 10x - \frac{X^2}{1,000}$$

$$R'(X) = 10 - \frac{X}{500}$$

$$3) P(X) = R(X) - C(X)$$

$$= (10x - \frac{X^2}{1,000}) - (7,000 + 2x)$$

$$= 8x - \frac{X^2}{1,000} - 7,000$$

$$P'(X) = 8 - \frac{X}{500} \quad \text{or}$$

$$P'(X) = R'(X) - C'(X)$$

$$= 10 - \frac{X}{500} - 2$$

$$= 8 - \frac{X}{500}$$

Average cost, Average revenue, Average profit

$$\rightarrow \text{Average Cost} = \bar{C}(X) = \frac{C(X)}{X} \Rightarrow \text{Cost per unit}$$

$$\text{Marginal Average cost} = \bar{C}'(X) \Rightarrow \text{the rate of change of average cost}$$

$$\rightarrow \text{Average revenue} = \bar{R}(X) = \frac{R(X)}{X} \Rightarrow \text{Revenue/unit}$$

$$\text{Marginal Average revenue} = \bar{R}'(X) \Rightarrow R'(X) \cdot X$$

$$\rightarrow \text{Average Profit} = \bar{P}(X) = \frac{P(X)}{X}$$

$$\text{Marginal Average profit} = \bar{P}'(X)$$

$$\text{Suppose: } C(X) = 1,000 + 25x - \frac{X^2}{10}$$

$$C'(X) = 25 - \frac{X}{5}$$

$$\bar{C}(X) = \frac{1,000}{X} + 25 - \frac{X}{10}$$

$$\bar{C}'(X) = \frac{-1,000}{X^2} - \frac{1}{10}$$

$$C'(10) = \frac{1,000}{100} - \frac{1}{10}$$

= Br. -10.10 Shows that a unit increase in production will decrease the average cost by approximately Br. 10.10 at a production level of 10 units.

If f' set equal to zero

$$\bar{C}'(X) = \frac{1000}{X^2} + 25 - \frac{X}{10} = 0$$

X = represents the minimum value of f , If $f'(X) > 0$ i.e. -Cost

$f'(X) < 0$ i.e. -Profit

5.3.3 Higher-order Derivatives

If a function F has a derivative for each value of X in some specified interval, then the derivative function f' is defined for that interval. If in turn the derivative function itself has a derivative for points in that interval this new derivative function is called the second derivative of the original function f or the first derivative of f' .

→ **The first derivative test**

* Locate all critical values X^*

$$f'(X) = 0$$

* For any critical value X^* , determine the value of X and right

→Second derivative test

For critical points, where $f'(X) = 0$, the most expedient test is the second –derivative test. Intuitively the 2nd derivative test attempts to determine the concavity of the function at a critical point.

Eg. $f(X) = \frac{X^4}{4} - \frac{9X^2}{2}$

$$f'(X) = X^3 - 9X$$

$$f'(X) = 0$$

$$X^3 - 9X = 0$$

$$X(X^2 - 9) = 0$$

$$X(X + 3)(X - 3) = 0$$

$$X = 0, X + 3 = 0 \text{ or } X - 3 = 0$$

$$X = 0, X = -3, \text{ or } X = 3$$

$$\downarrow f'(X) = -ve$$

Concave upward

Concave downward

$$\downarrow f'(X) = +ve$$

$$f''(X) = 3X^2 - 9$$

$f(0, -3, \text{ or } 3) = (0, 0), (-3, -\frac{81}{4}), (3, -\frac{81}{4})$
 $\uparrow \quad \uparrow \quad \uparrow$
 max min min

Rule

→1 find $f'(X)$, set it equal to zero, and solve for candidate values, X.

2. Find $f''(X)$ and evaluate $f''(X)$

- If $f''(X)$ is negative, a local maximum occurs at X
- If $f''(X)$ is positive, a local minimum occurs at X.
- If $f''(X)$ is zero; the test fails to determine what happens at X.

* $f'(X) = 0$ is called stationary points. The slope of the line is tangent to a curve.

g. $f(X) = 3X^3 + 5X^2 + 2X + 3$

$$f'(X) = 9X^2 + 10X + 2$$

$$f''(X) = 18X + 10$$

$$f'''(X) = 18$$

$$f^{iv}(X) = 0$$

Where:

f = The original function

f' = Relates information about the behavior of f

f'' = Relates information about f'

Given f :

If $f'(X) > 0$ f is increasing

If $f'(X) < 0$ f is decreasing

If $f'(X) = 0$ f is constant

If $f''(X) > 0$ f' is increasing

$f''(X) < 0$ f' is decreasing

$f''(X) = 0$ f' is constant

5.3.4 Optimization Problems

Ex. 1. A company manufactures and sales X units of transistor radios per week. If the weekly cost and demand equations are:

$$C(X) = 5,000 + 2x$$

$$P = 10 - \frac{X}{1,000} \quad [0, 8000]$$

Finding for each week

- The production level that leads to maximum revenue and the maximum revenue.
- The production levels that leads to maximum profit and the maximum profit.
- The production level that leads to minimum cost and the minimum cost.

Solution

$$1) R(X) = 10x - \frac{X^2}{1,000} = (10 - \frac{X}{1,000}) \cdot (X)$$

$$R'(X) = 10 - \frac{X}{500} \quad R'(X) = 0$$

$$= 10 - \frac{X}{500} = 0$$

$$X = \underline{5,000 \text{ units}}$$

$$R''(X) = -\frac{1}{500} = -0.02, \text{ so } X = 5,000 \text{ units leads to maximum revenue}$$

$$TR = XP = 5,000 \times 5$$

$$(5,000 \text{ } 25,000)$$

$$= \underline{25,000 \text{ Br}}$$

$$R(5,000) = 10 \times 5,000 - \frac{(5,000)^2}{1,000}$$

$$= \underline{25,000}$$

$$2) P(X) = 10x - \frac{X^2}{1,000} - 5,000 - 2x$$

$$= 8x - \frac{X^2}{1,000} - 500$$

$$P'(X) = 8 - \frac{X}{500}$$

$$P'(X) = 0$$

$$8 - \frac{X}{500} = 0$$

$$\underline{4,000 = X}$$

$$P''(X) = -\frac{1}{500}$$

Negative, so it is an optimal solution

(4,000 unit Br. 21,000)

$$P(4,000) = 4,000 \times 8 - \frac{(4,000)^2}{1,000} - 50,000$$

$$= \underline{21,000}$$

3) Cost is minimum at zero production level. If there is non-linear function, we can use the 2nd derivative test.

$$C(X) = 5,000 + 2x$$

$$C'(X) = 2$$

$$C'(X) = 0 \neq 2$$

Ex. 2. When X gallons of alcohol are produced, the average cost per gallons is given by the following function.

$$C(X) = \frac{200}{0.1X + 5} + 0.05X, X > 0$$

Required:

- 1) Find the gallon of alcohol production level that leads to minimum cost.
- 2) Prove that this value (1) of X occurs at a local minimum of C(X)
- 3) Compute the minimum average cost per gallon.

Solution

1. We can rewrite $C(X)$ as

$$C(X) = 200 (0.1x + 5)^{-1} + 0.05X$$

$$C'(X) = -200 (0.1x + 5)^{-2(0.1)} + 0.05$$

$$= -20 (0.1x + 5)^{-2} + 0.05$$

$$C'(X) = 0$$

$$-20 (0.1x + 5)^{-2} + 0.05 = 0$$

$$\frac{-20}{(0.1x + 5)^2} + 0.05 = 0$$

$$(0.1x + 5)^{-1/2} = 400^{1/2}$$

$$0.1x + 5 = \pm 200$$

$$X = 150 \text{ or } X = -250$$

We discard $X = -250$ since it is negative.

2. To show $X = 150$ yields a minimum cost, we start with,

$$A'(X) = -20 (0.1x + 5)^{-2} + 0.05$$

Find $A''(X)$

$$A''(X) = -20 (-2) (0.1x + 5)^{-3} (0.1)$$

$$= \frac{4}{(0.1x + 5)^3}$$

" $\curvearrowright$ $\curvearrowright$
A X > 0 Minimum point.

$$A'(150) = \text{Positive}$$

3. Minimum average cost / gallon

$$C(X) = \frac{200}{0.1x + 5} + 0.05x$$

$$C(150) = \frac{200}{0.1(150) + 5} + 0.05(150)$$
$$= \underline{17.5 \text{ Br/ gallon}}$$

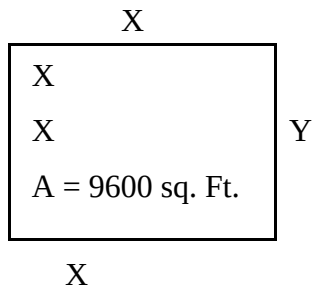
Example: 3

A rectangular warehouse with a flat roof is to have a floor area of 9600 square feet. The interior is to be divided into storeroom and office space by an interior wall parallel to one

pair of the sickles of the building (as shown below). The roof and floor areas will be of 600 square feet for any building, but the total wall length will vary for different dimensions.

Required:

Find the dimensions that minimize the total amount of wall.



$$XY = 9600 \text{ Sq.ft}$$

$$Y = \frac{9600}{X}$$

$$W = 3X + 2Y \text{ i.e total wall}$$

$$W(X) = 3X + 2 \frac{9600}{X}$$

$$= 3X + \frac{19200}{X}$$

$$W(X) = 3 - \frac{19200}{X^2}$$

Set $W'(X)$ equal to zero.

$$3 - \frac{19200}{X^2} = 0$$

$$3X^2 - 19200 = 0$$

$$X^2 = \frac{19200}{3} = 6400$$

$$X = \pm \sqrt{6400}$$

$$= \pm 80$$

$$Y = 120$$

$$X = 80$$

$X = -80$ is discarded because X must be positive in this problem.

$$Y = \frac{9600}{80} = 120$$

⇒ Note that the second derivative

$$W(X) = 3 - 19200 X^{-2}$$

$$W'(X) = -2(-19200) X^{-3}$$

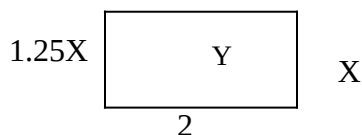
$$= \frac{38400}{X^3}$$

* $W'(80) > 0$ + positive. So we have a minimum value.

* The minimum length is

$$\begin{aligned} W &= 3X + 2Y = 3(80) + 2(120) \\ &= 240 + 240 \\ &= \underline{486} \text{ feet.} \end{aligned}$$

Ex2. Fence is required on three sides of a rectangular plot. Fence for the two ends costs Br. 1.25 per running foot; fence for the third side costs Br. 2 per running foot. Find the maximum area that can be enclosed with Br. 100 worth of fence. Let



$$2(1.25X) + 2Y = 100$$

$$2.50X + 2Y = 100$$

$$A = XY$$

$$A(X) = X \left(\frac{100 - 2.50X}{2} \right)$$

$$Y = \frac{100 - 2.56X}{2}$$

$$2Y = 100 - 2.50X$$

$$A(X) = X \frac{100X - 2.5X^2}{2} = 50X - \frac{25}{20X^2}$$

$$Y = \frac{100 - 2.5X}{2}$$

$$A(X) = 50 - 2 \left(\frac{25}{20} \right) X$$

$$A'(X) = 0$$

$$50 - 2.5X = 0$$

$$50 = 2.5X$$

$$X = \frac{50}{2.5} = 20$$

$$A(20) = 500 \text{ sq.feet}$$

$X = 20 \Rightarrow$ leads to max area.

$$Y = 25$$

$$Y = \frac{100 - 2.5X}{2}$$

$$= \frac{100 - 2.5(20)}{2}$$

$$= \frac{50}{2} = \underline{\underline{25}}$$

$$2.5X + 2Y = 100$$

$$2.5X = 100 - 2Y$$

$$X = \frac{100 - 2Y}{2.5}$$

$$A(Y) = \left(\frac{100 - 2Y}{2.5} \right)$$

$$= \frac{100Y - 2Y^2}{2.5}$$

$$= 40Y - 0.8Y^2$$

$$A'(Y) = 40 - 1.6Y$$

$$A'(Y) = 0$$

$$40 - 1.6Y = 0$$

$$Y = \frac{40}{1.6}$$

$$= \underline{\underline{25}}$$

